AMBA and AXI Connect Example

Temperature–Humidity Sensor Module

[Reference] ▪

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AMBA and AXI Connect Example

Temperature–Humidity Sensor Module

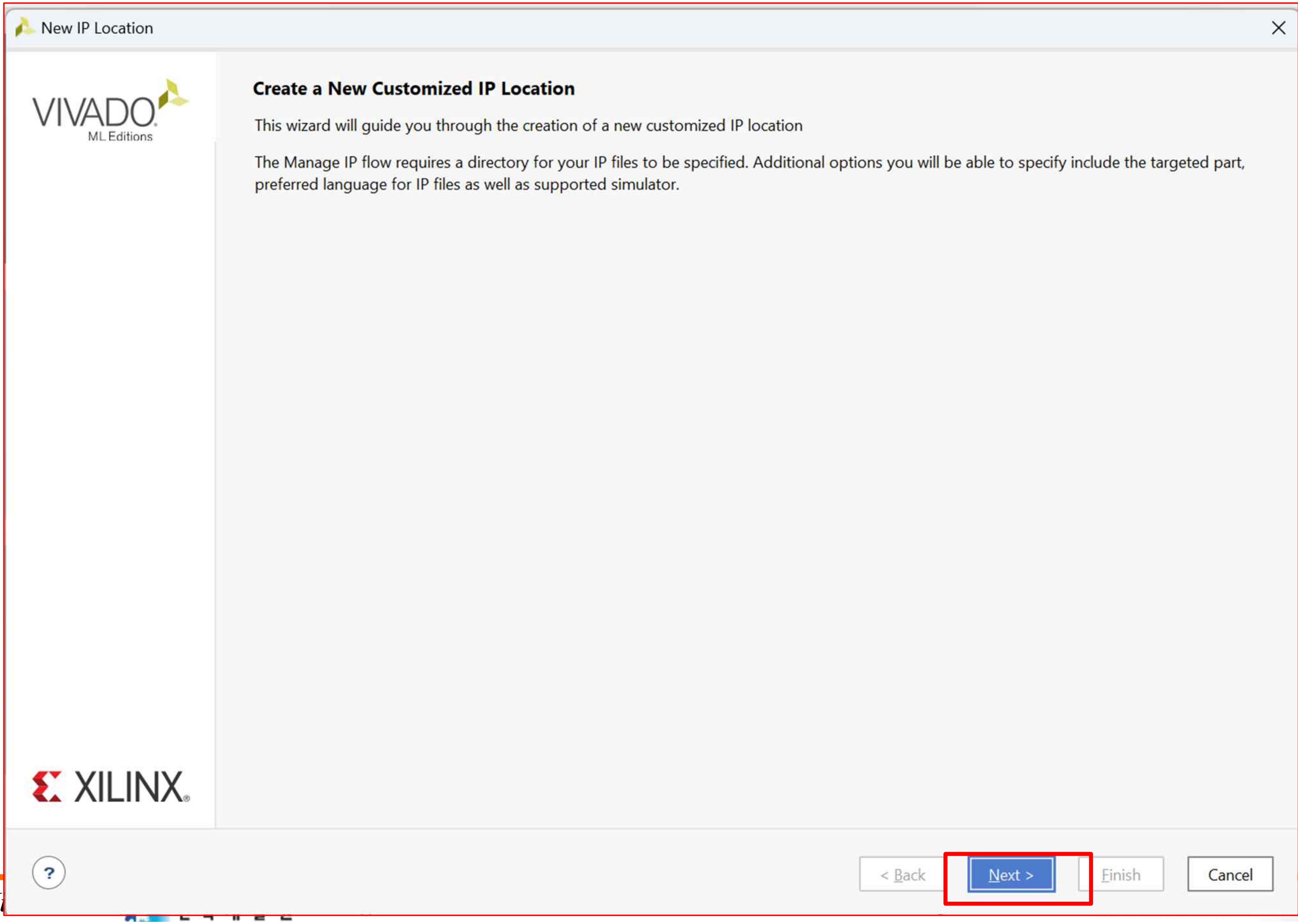
[*Create IP*]

➢ AXI Connect ➔ AXI Template

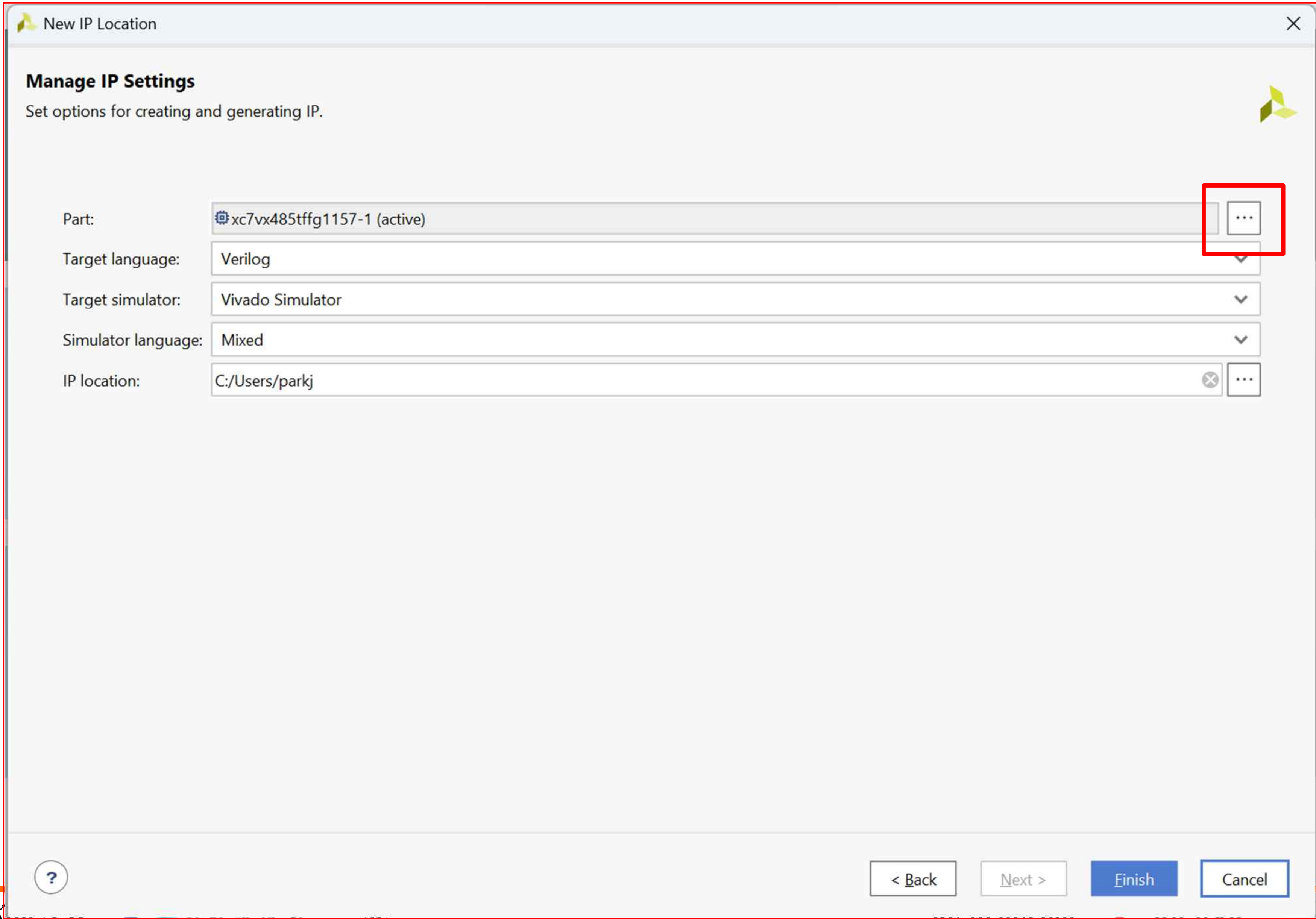
VIVADO - Manage IP – New IP location

(이미 생성한 ***Location***이 있으면 해당 폴더로 지정)

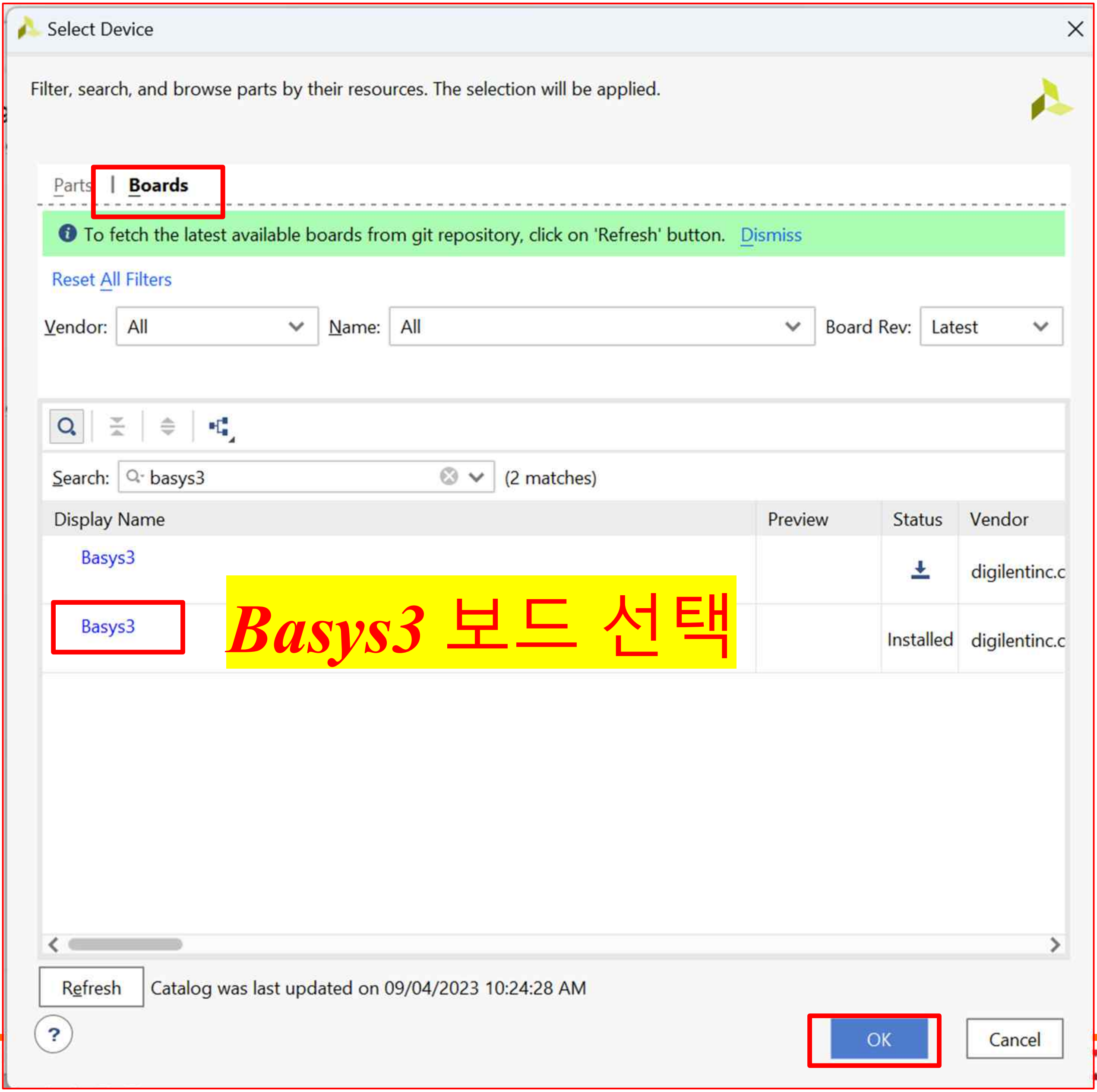
➢ ***AXI Connect ➔ AXI Template***



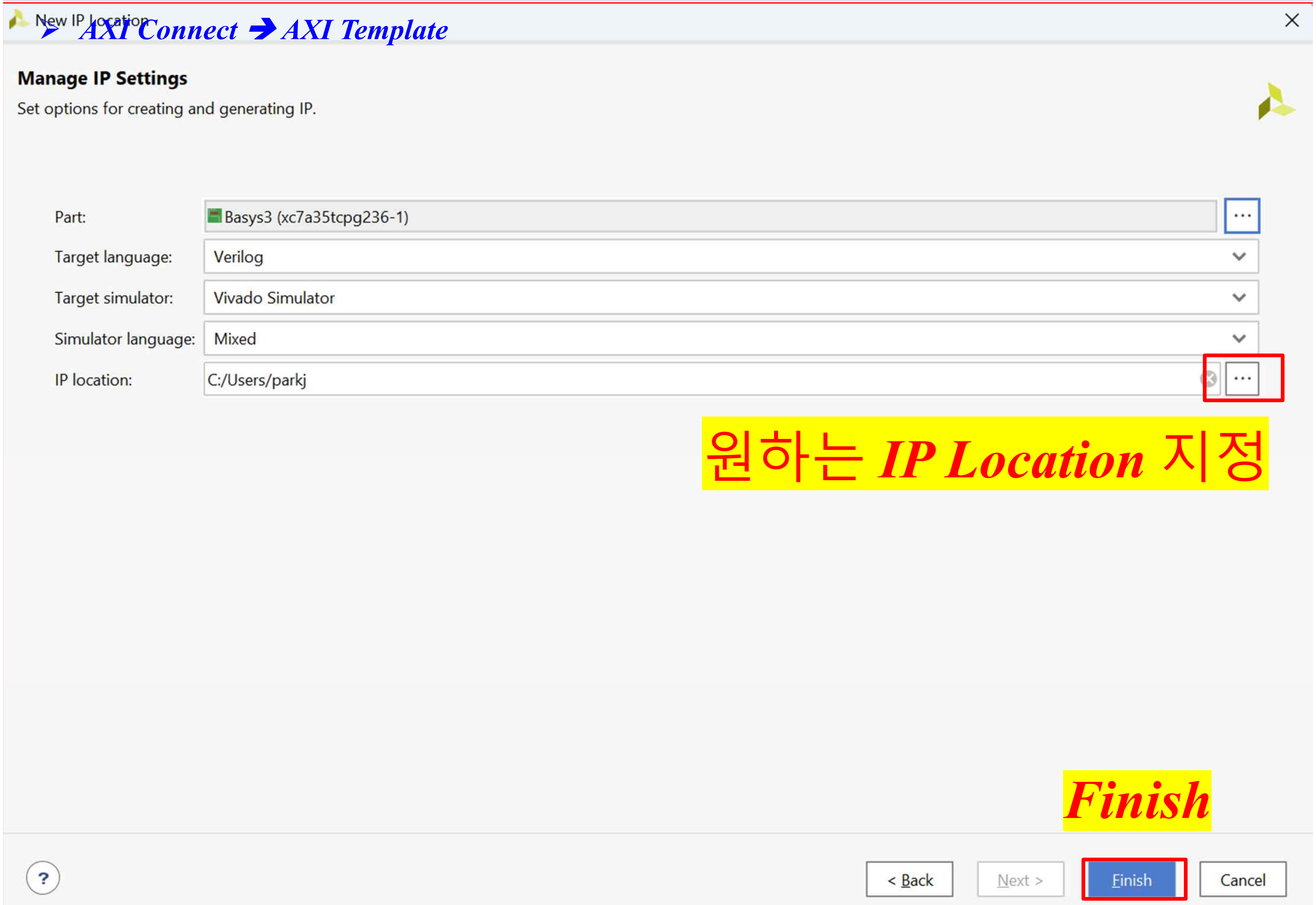
➢ ***AXI Connect ➔ AXI Template***



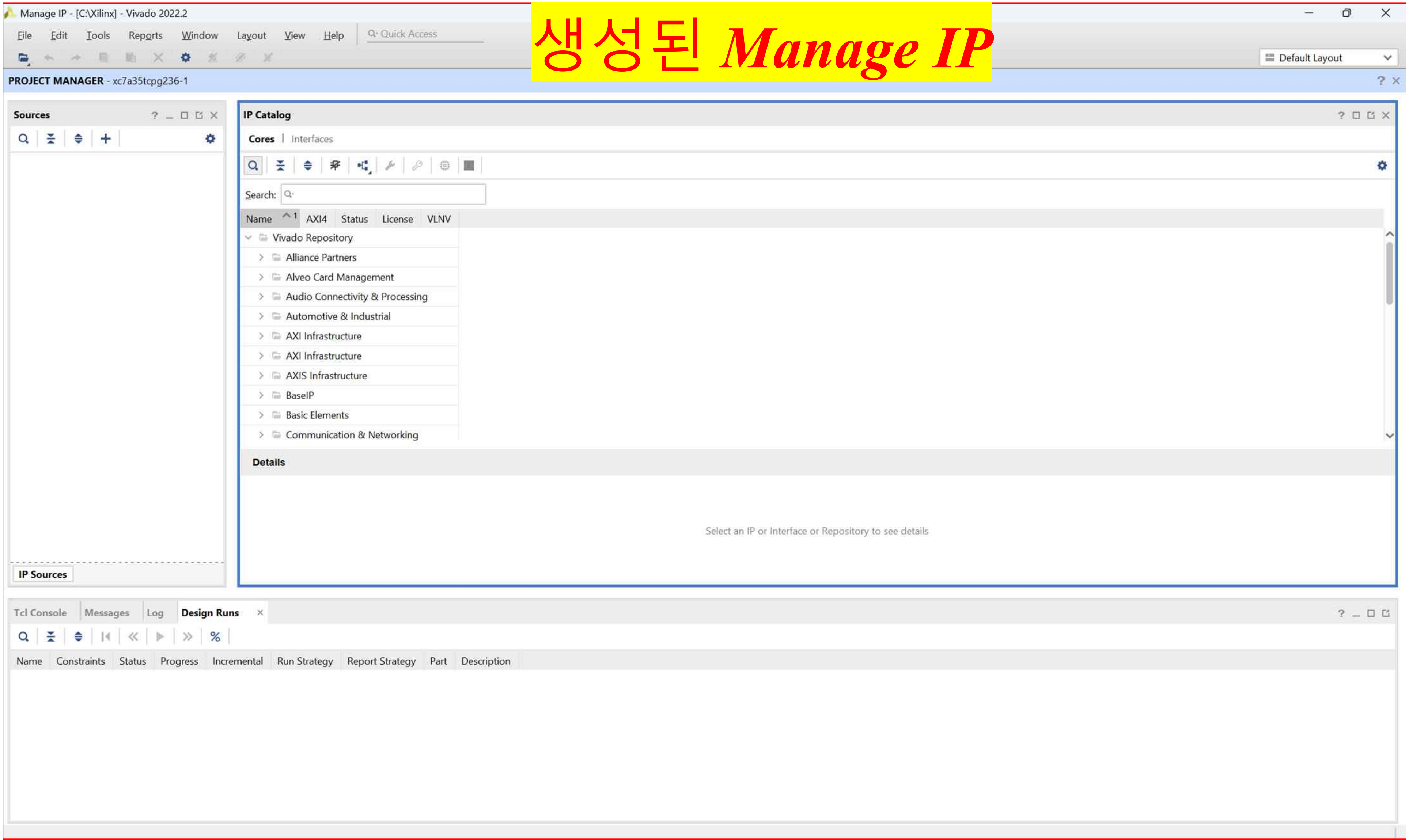
➢ *AXI Connect ➔ AXI Template*



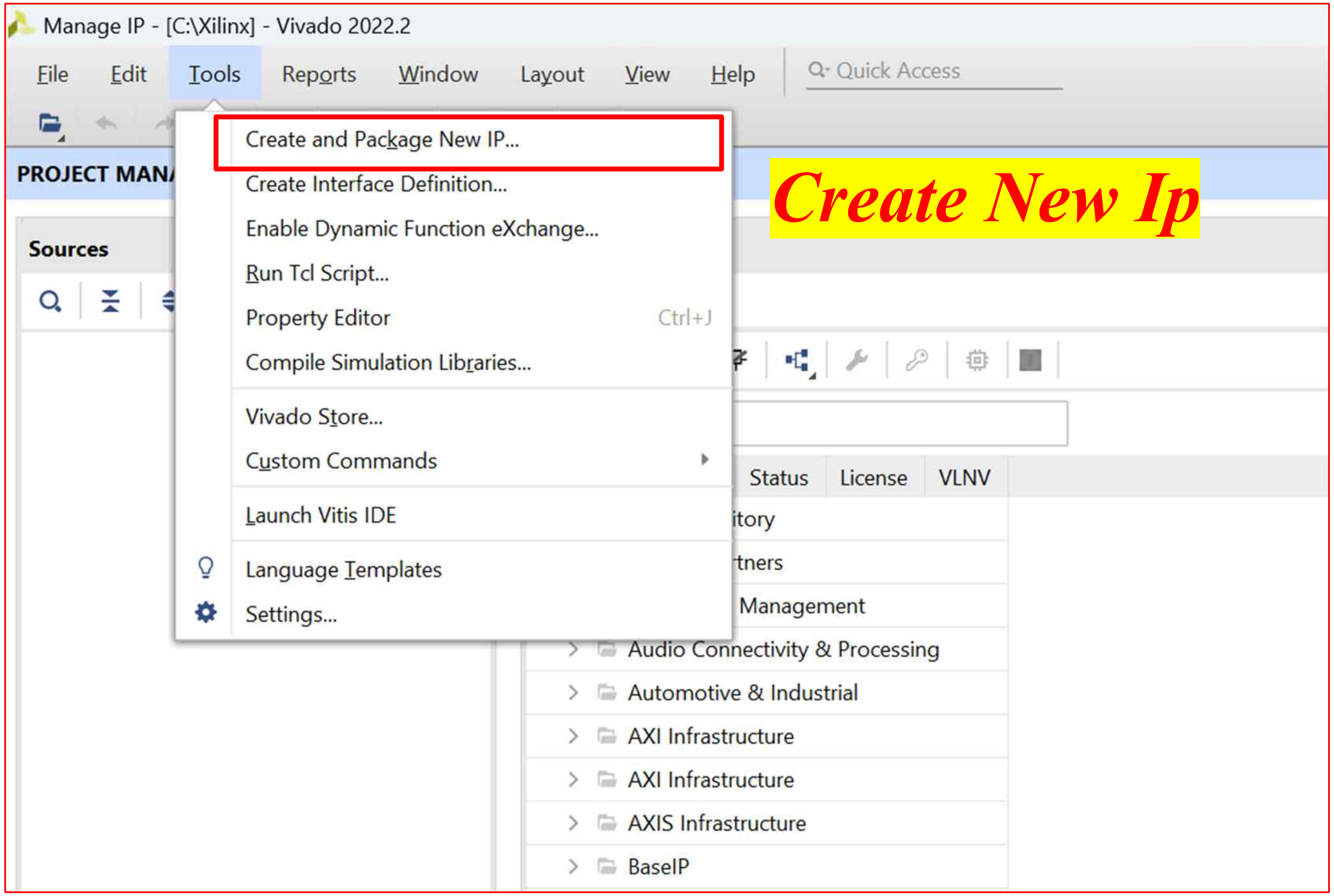
➢ AXI Connect ➔ AXI Template



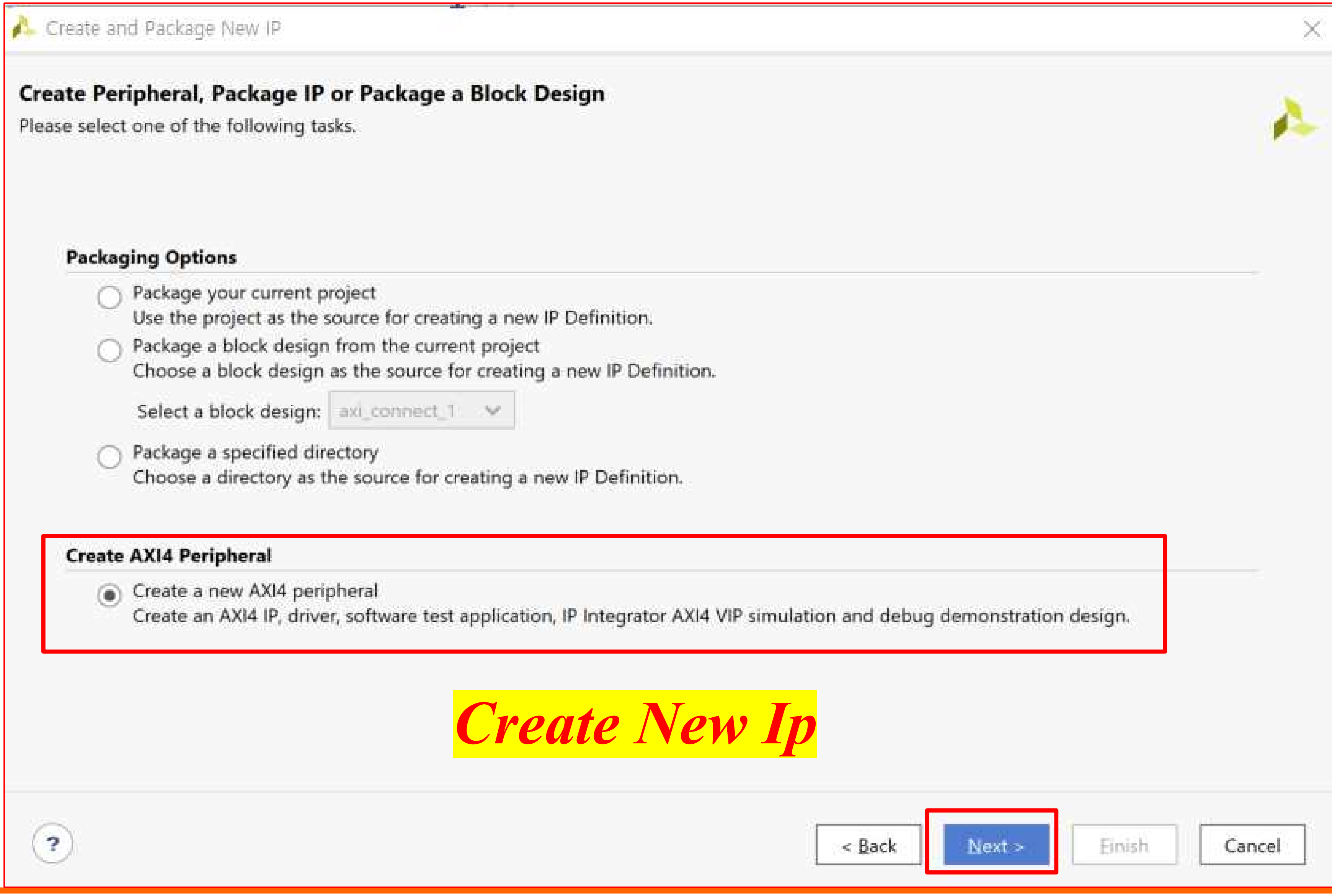
- ➢ *AXI Connect ➔ AXI Template*



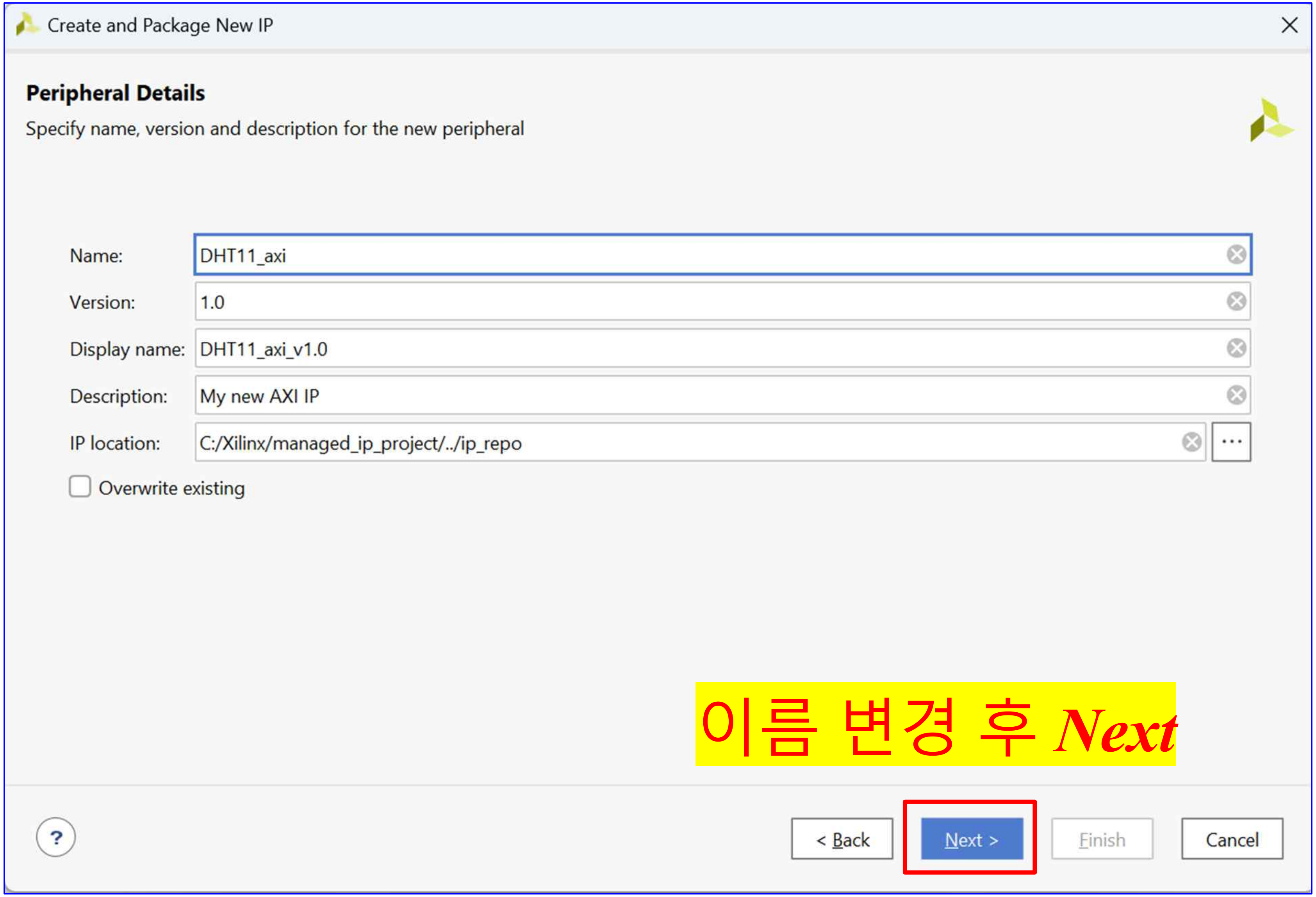
➢ ***AXI Connect → AXI Template***



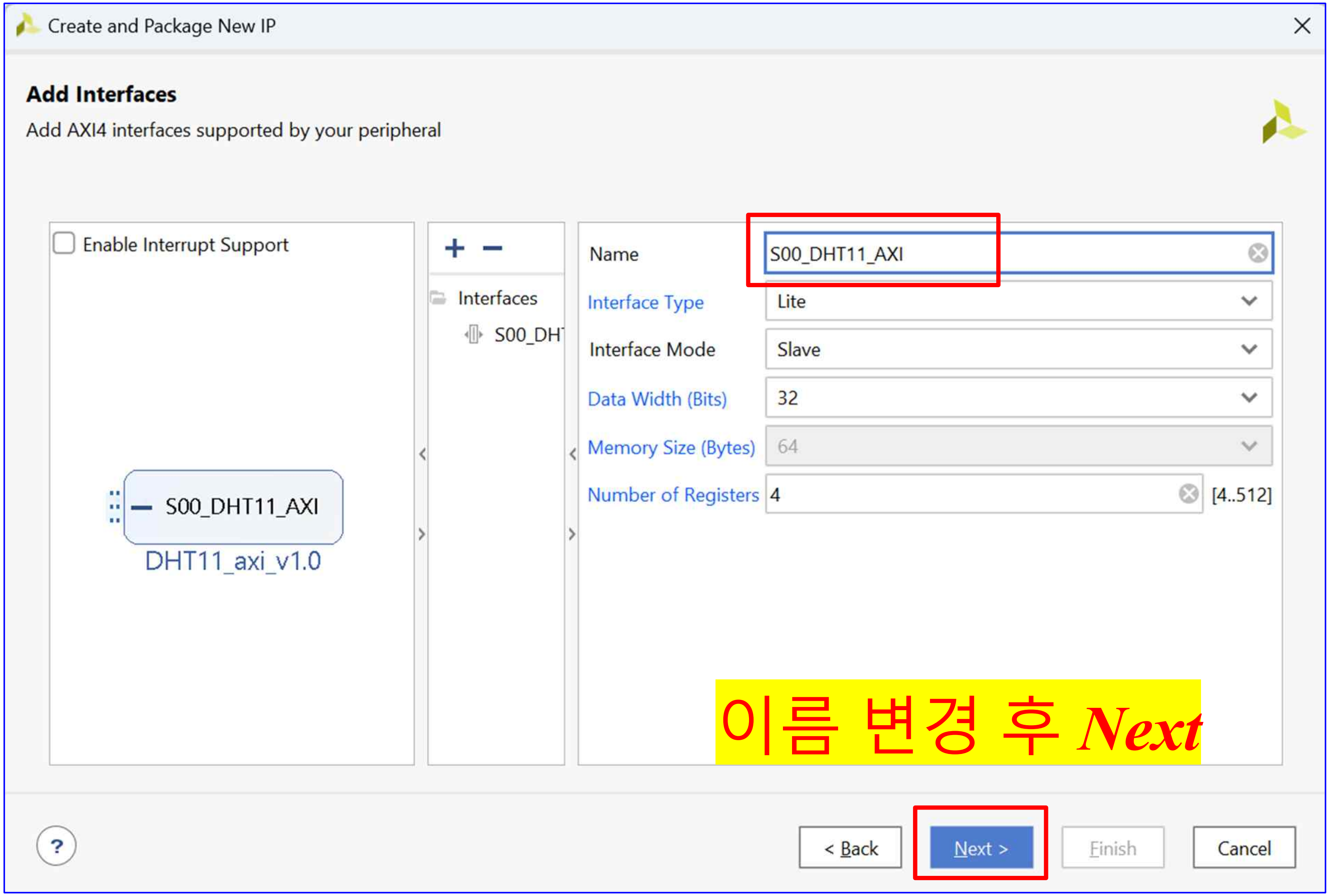
➢ *AXI Connect ➔ AXI Template*



- ***AXI Connect ➔ AXI Template***



➢ ***AXI Connect → AXI Template***



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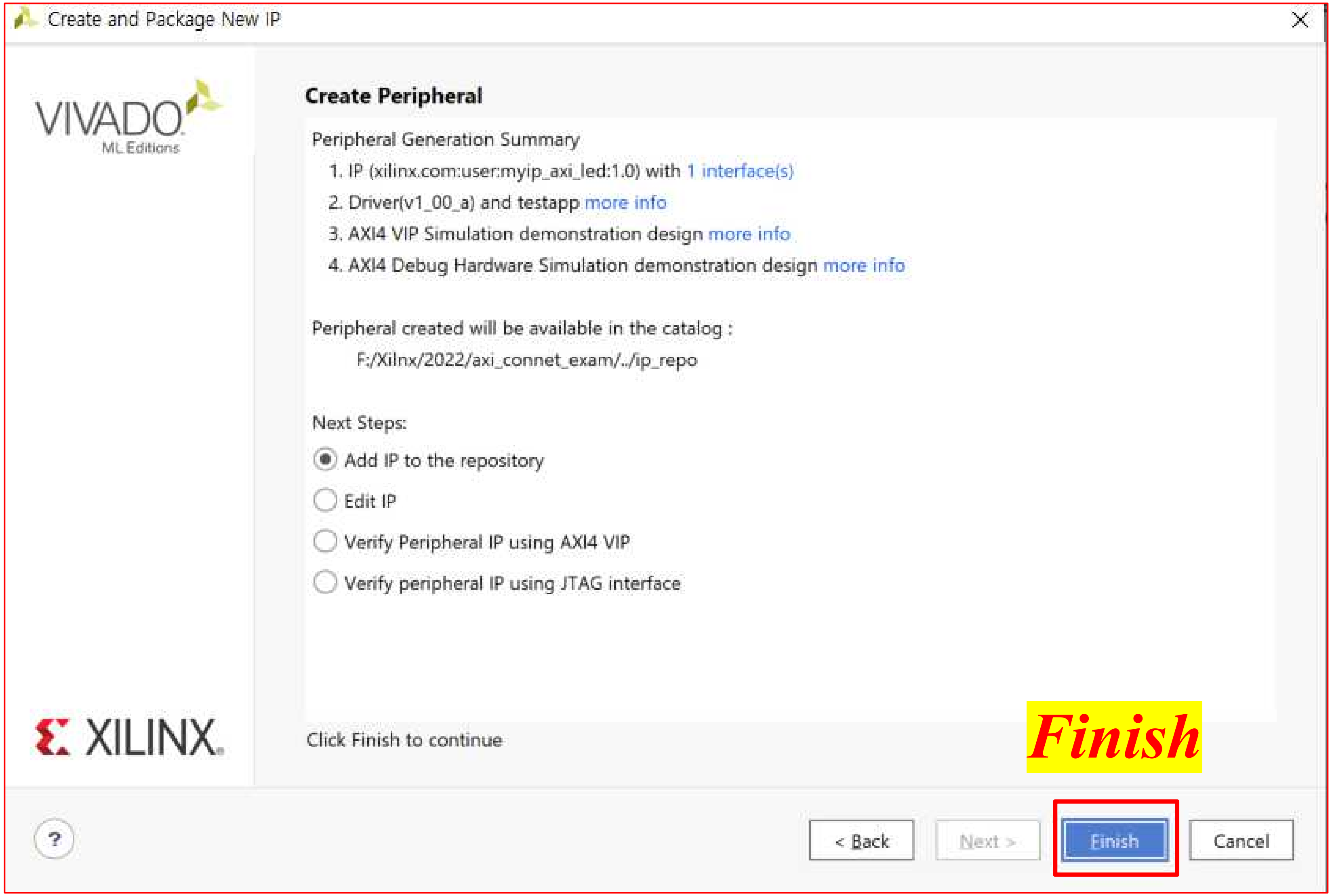
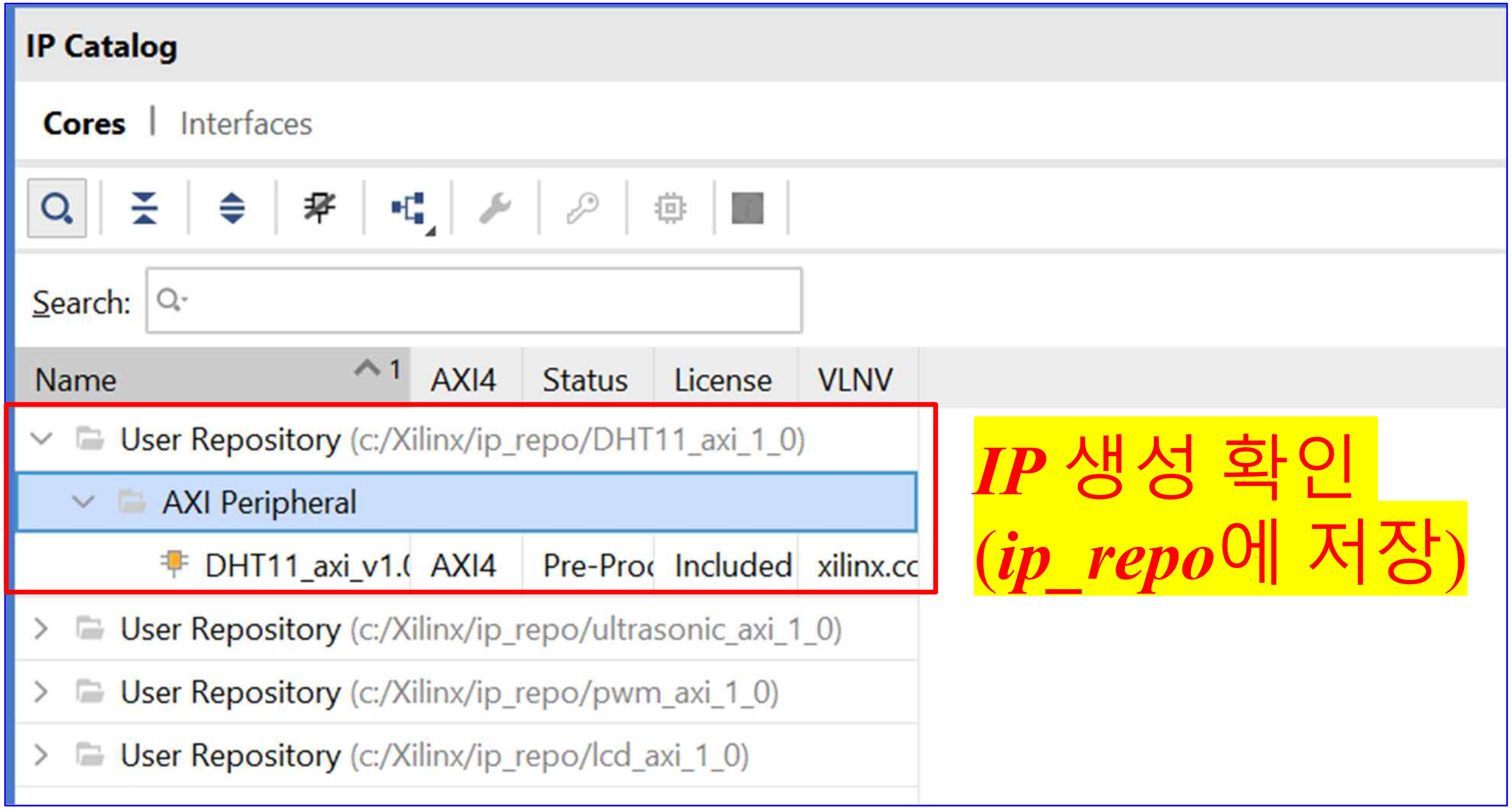
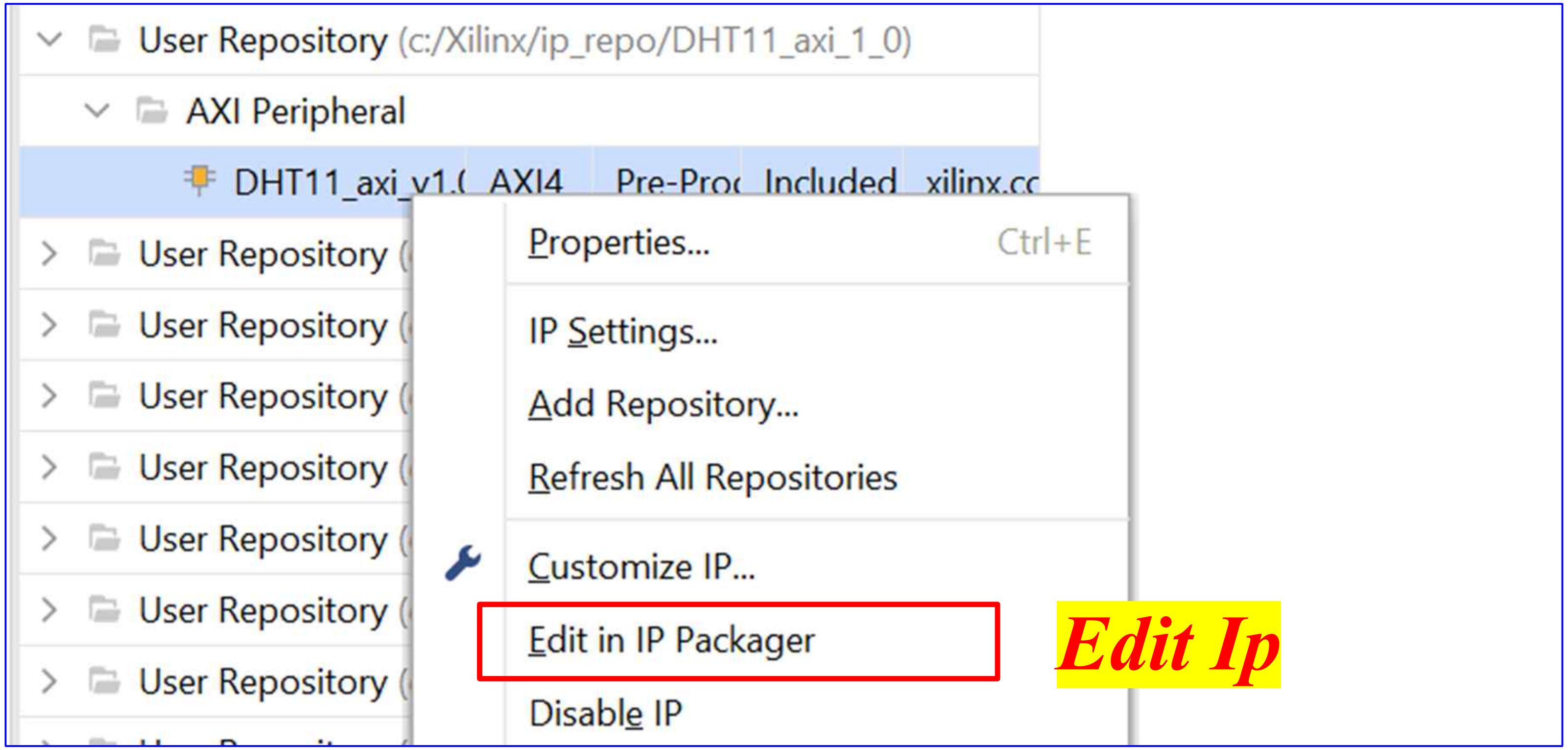


➢ AXI Connect ➔ AXI Template

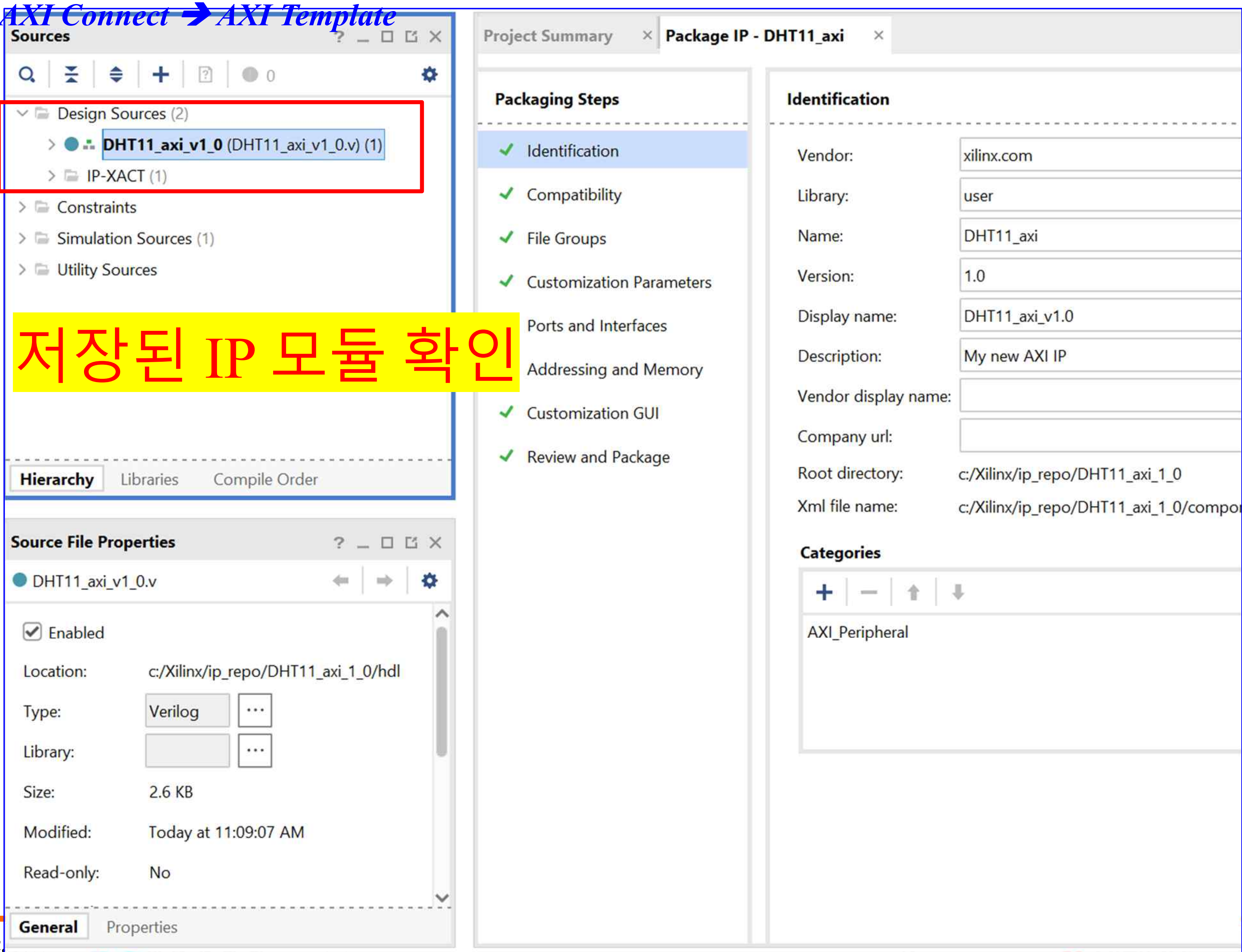


IP 생성 확인
(ip_repo에 저장)

- ***AXI Connect → AXI Template***



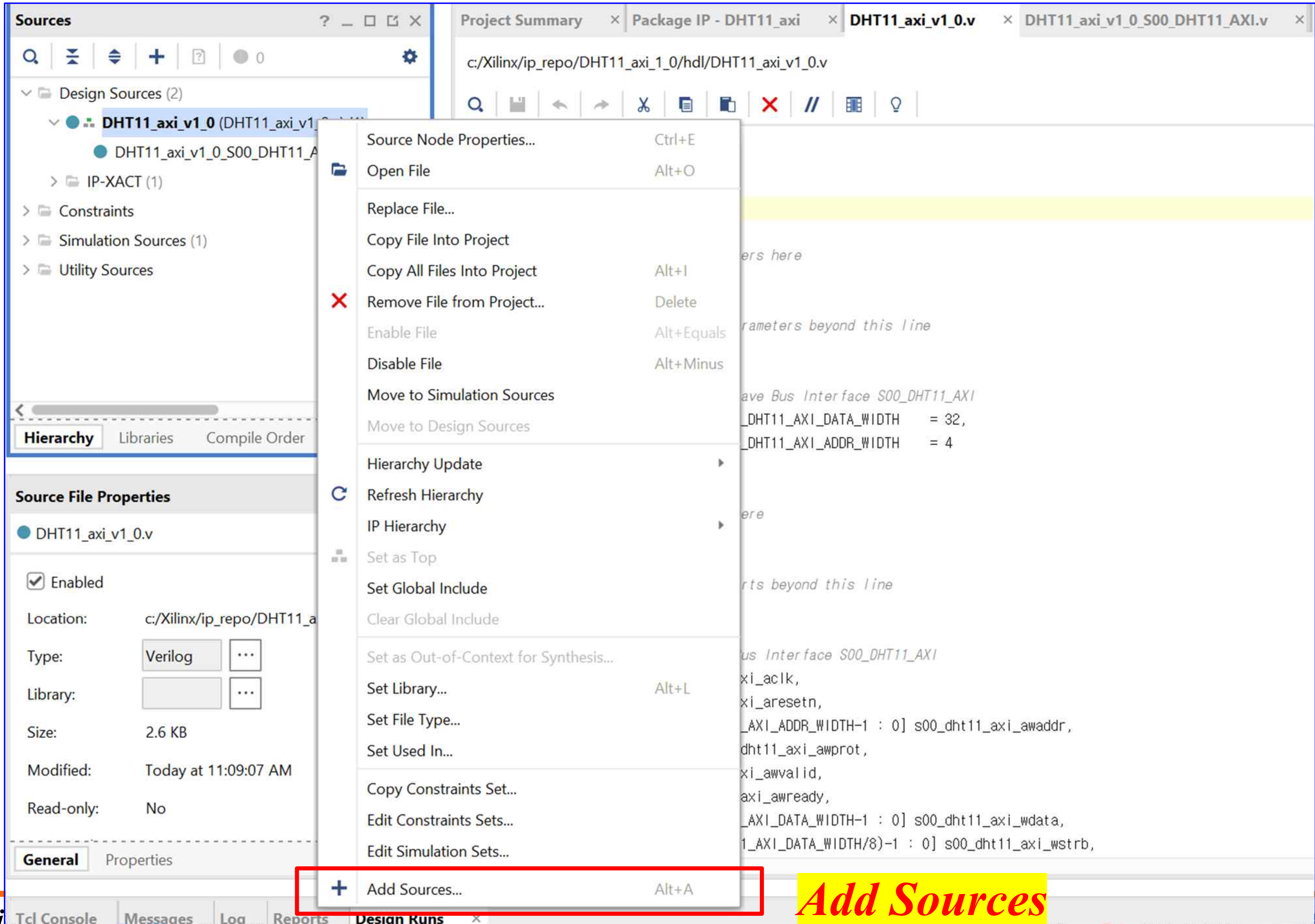
- *AXI Connect ➔ AXI Template*



AMBA and AXI Connect [Temperature–Humidity Sensor Module]

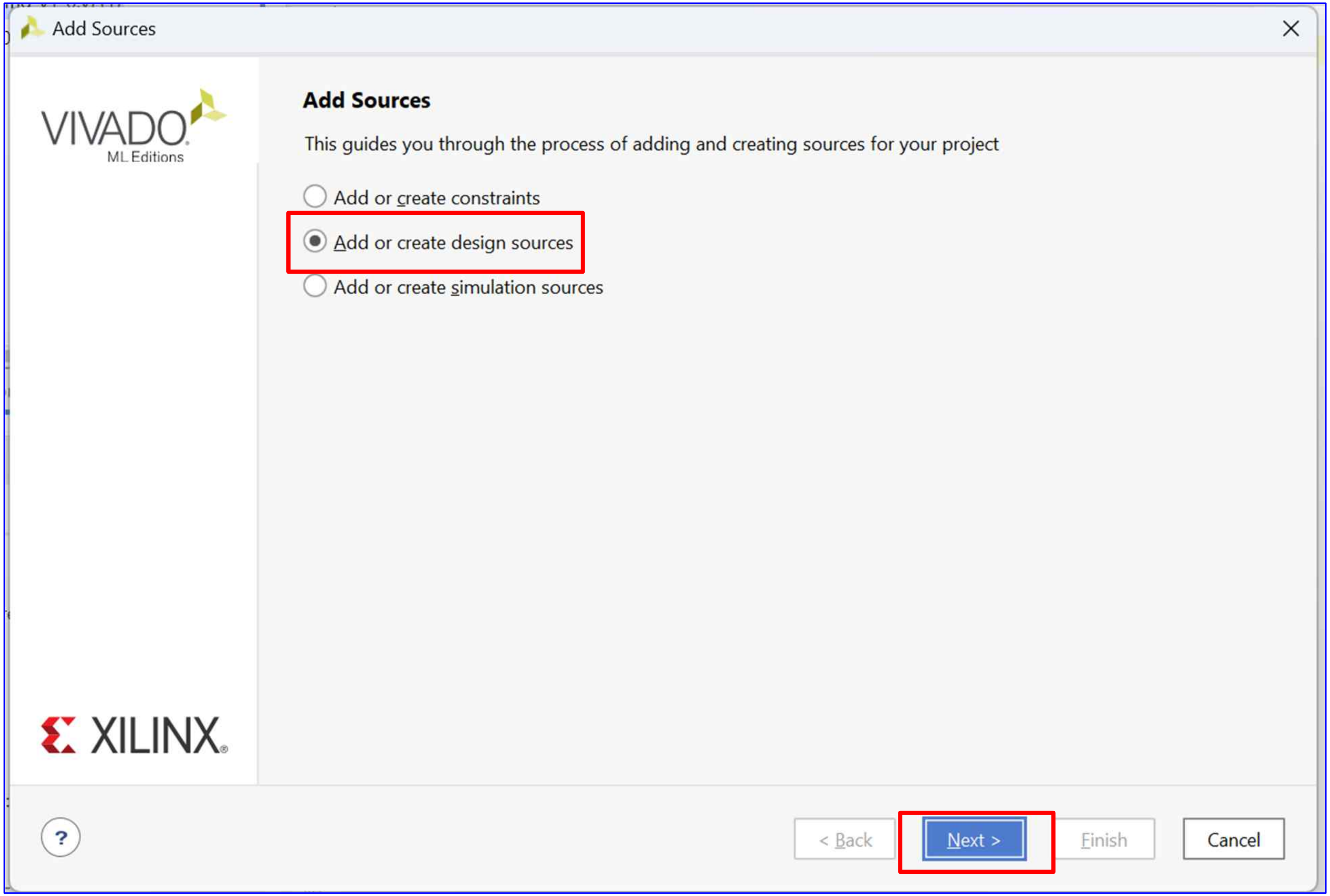
Axi_dht11.xpr

➢ ***AXI Connect ➔ AXI Template***

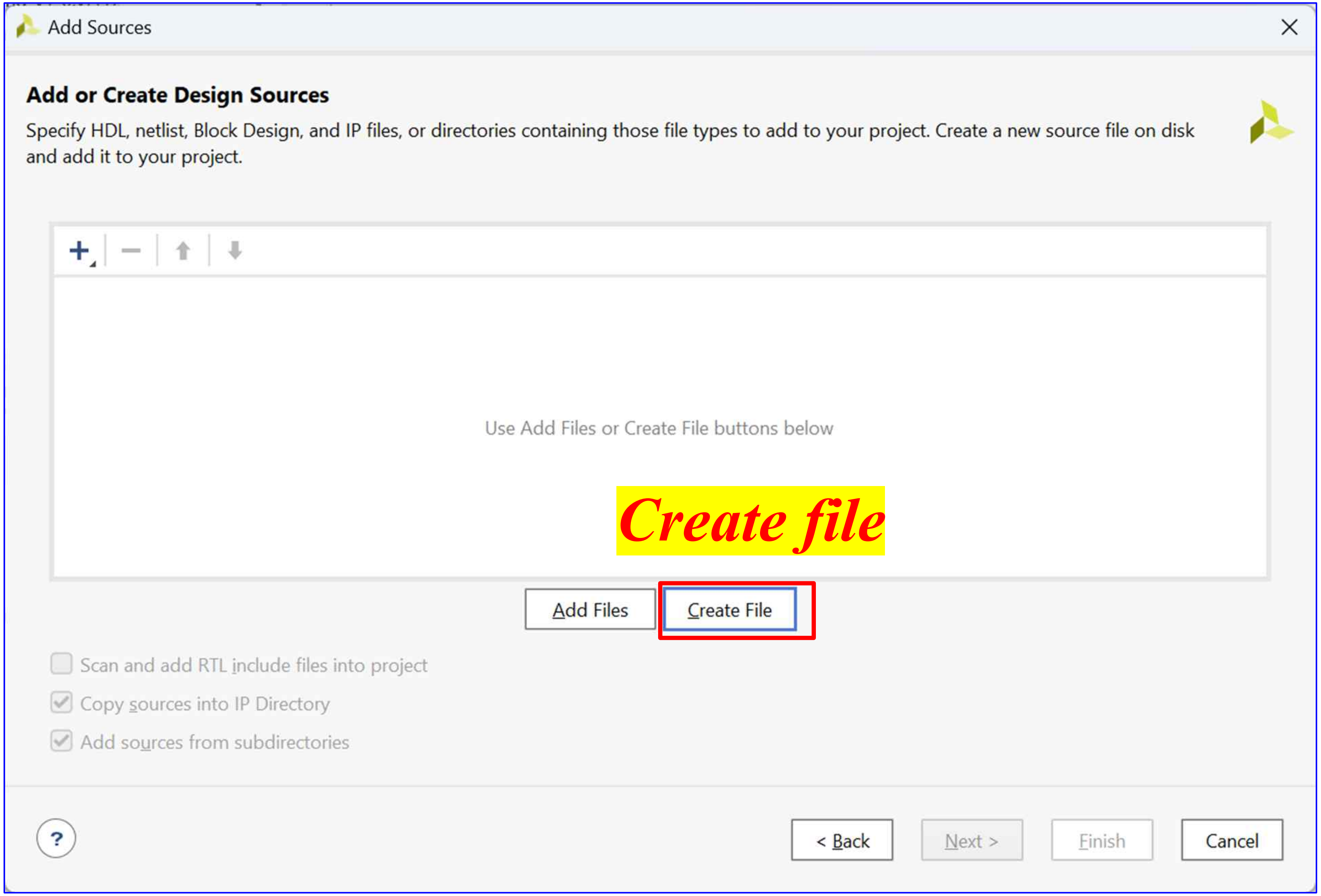


Add Sources

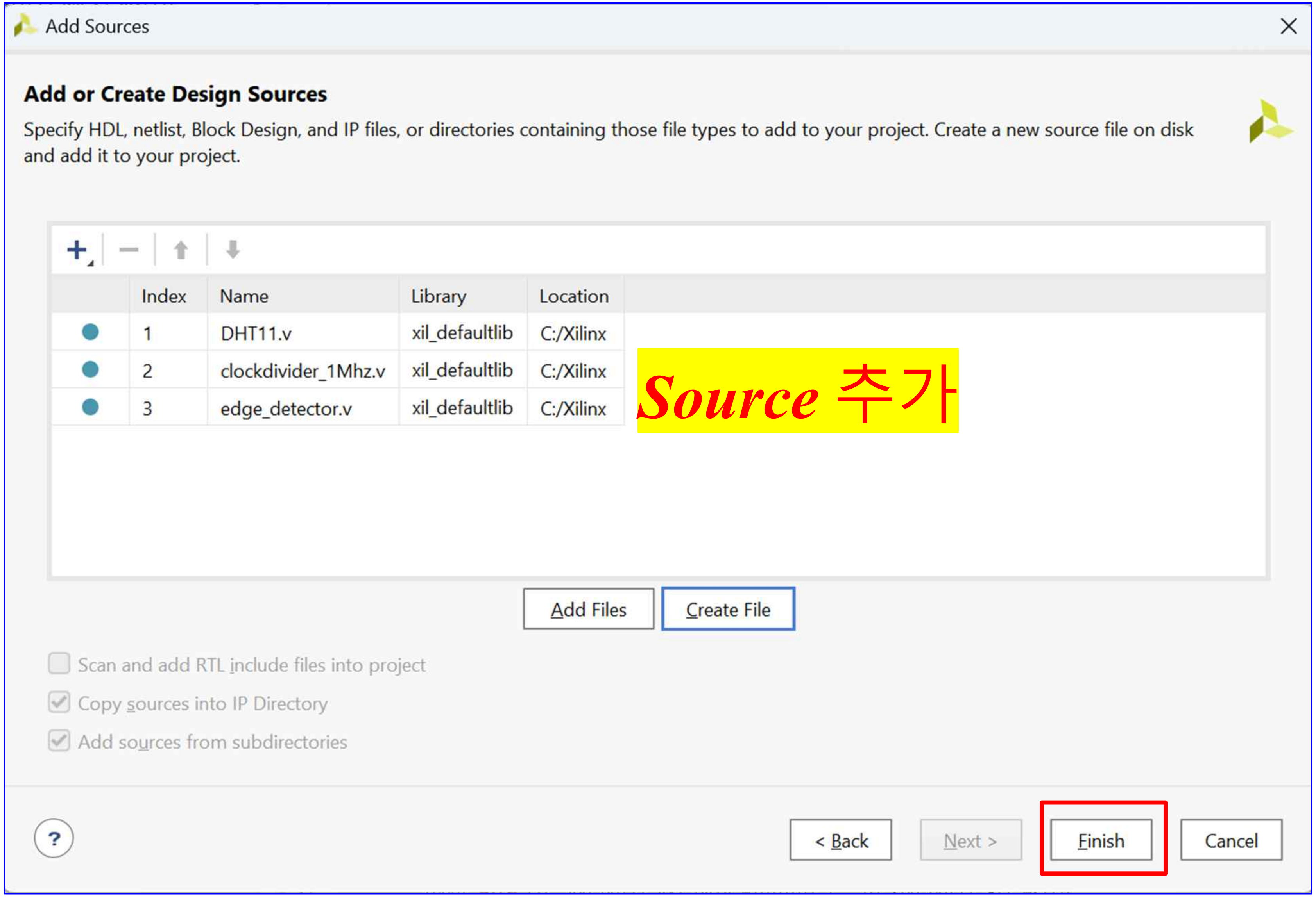
- *AXI Connect ➔ AXI Template*



➤ *AXI Connect ➔ AXI Template*



➢ *AXI Connect → AXI Template*



➢ AXI Connect ➔ AXI Template

```
module DHT11(
input clk, reset,clk1Mhz,
inout dht11_data,
output [3:0] humidity10,humidity0,
output [3:0] temperature10,temperature0

);

    reg [7:0] humidity,temperature;

    assign humidity10 = humidity/10;
    assign humidity0 = humidity%10;
    assign temperature10 = temperature/10;
    assign temperature0 = temperature%10;

    parameter S_IDLE = 3'b000;
    parameter S_LOW_18MS = 3'b001;
    parameter S_HIGH_20US = 3'b010;
    parameter S_LOW_80US = 3'b011;

    parameter S_HIGH_80US = 3'b100;
    parameter S_READ_DATA = 3'b101;
    parameter S_WAIT_PEDGE = 3'b110;
    parameter S_WAIT_NEDGE=3'b111;

wire dht_nedge, dht_pedge;
    reg count_usec_e;
    reg [21:0] count_usec;
    reg [2:0] state, next_state, read_state;
    reg dht11_data_oe, dht_buffer;
    reg [5:0] data_count;
    reg [39:0] temp_data;

    assign dht11_data = dht_buffer;

always @(negedge clk1Mhz or negedge
reset) begin
    if(~reset) count_usec = 0;
    else begin

  if(count_usec_e)
   count_usec = count_usec + 1;

   else if(!count_usec_e) count_usec = 0;
   end

  end

always @(negedge clk1Mhz or negedge
reset)begin
                if(~reset)state = S_IDLE;
                else state = next_state;
    end

    edge_detector ed_dec
(.clk(clk1Mhz), .cp_in(dht11_data), .reset(
reset), .n_edge(dht_nedge), .p_edge(dht_pe
dge));

 always @(posedge clk1Mhz or negedge
  reset)begin

    if(~reset)begin
                count_usec_e <= 0;
                next_state <= S_IDLE;
                dht11_data_oe <= 0;
                dht_buffer <= 1'bz;
                read_state <= S_WAIT_PEDGE;
                data_count <= 0;

                end
    else begin

    case(state)
    S_IDLE : begin

if(count_usec< 22'd3_000_000) begin

                count_usec_e <= 1;
                dht_buffer = 1'bz;
                end
                else begin

                next_state <= S_LOW_18MS;
                count_usec_e <= 0;
                end
        end
```

➢ AXI Connect ➔ AXI Template

```
S_LOW_18MS : begin

    if(count_usec < 22'd20_000) begin
                count_usec_e <= 1;
                dht_buffer <= 0;
                end
    else begin
                count_usec_e <= 0;
                next_state <= S_HIGH_20US;
                 dht_buffer = 1'bz;
                end
    end

S_HIGH_20US : begin

    if(count_usec < 22'd70) begin
                dht_buffer = 1'bz;
                count_usec_e <= 1;
        if(dht_nedge)begin
                next_state <= S_LOW_80US;
                count_usec_e <= 0;
        end
else begin next_state <= S_HIGH_20US;
                count_usec_e <= 1;
                end
    end
  else begin
    next_state <= S_IDLE;
    count_usec_e <= 0;
    end
  end
```

```
S_LOW_80US : begin

    if(count_usec < 22'd90)begin

       if(dht_pedge)begin
         next_state <= S_HIGH_80US;
         count_usec_e <= 0;
         end

    else begin
    next_state <= S_LOW_80US;
    count_usec_e <= 1;
        end

    end   else begin
    next_state <= S_IDLE;
    count_usec_e <= 0;
    end
    end

S_HIGH_80US : begin

    if(count_usec < 22'd90)begin

       if(dht_nedge)begin
    next_state <= S_READ_DATA;
    count_usec_e <= 0;
    end

    else begin
    next_state <= S_HIGH_80US;
    count_usec_e <= 1;
    end
    end

    else begin
    next_state <= S_IDLE;
    count_usec_e <= 0;
    end
    end
```

```
S_READ_DATA : begin

    case(read_state)
      S_WAIT_PEDGE: begin
           if(dht_pedge) begin
                read_state <= S_WAIT_NEDGE;
                count_usec_e <= 1;
           end
           else begin
               count_usec_e = 0;
           end
      end

  S_WAIT_NEDGE: begin
       if(dht_nedge) begin
           data_count <= data_count + 1;
            read_state <= S_WAIT_PEDGE;

            if(count_usec < 50) begin
            temp_data <= {temp_data[38:0],  1'b0};

           end
           else begin
           temp_data <= {temp_data[38:0], 1'b1};
            end
  end
  else begin
  count_usec_e <= 1;
  read_state <= S_WAIT_NEDGE;
  end
  end
  default: read_state = S_WAIT_PEDGE;
  endcase
```

➢ AXI Connect ➔ AXI Template

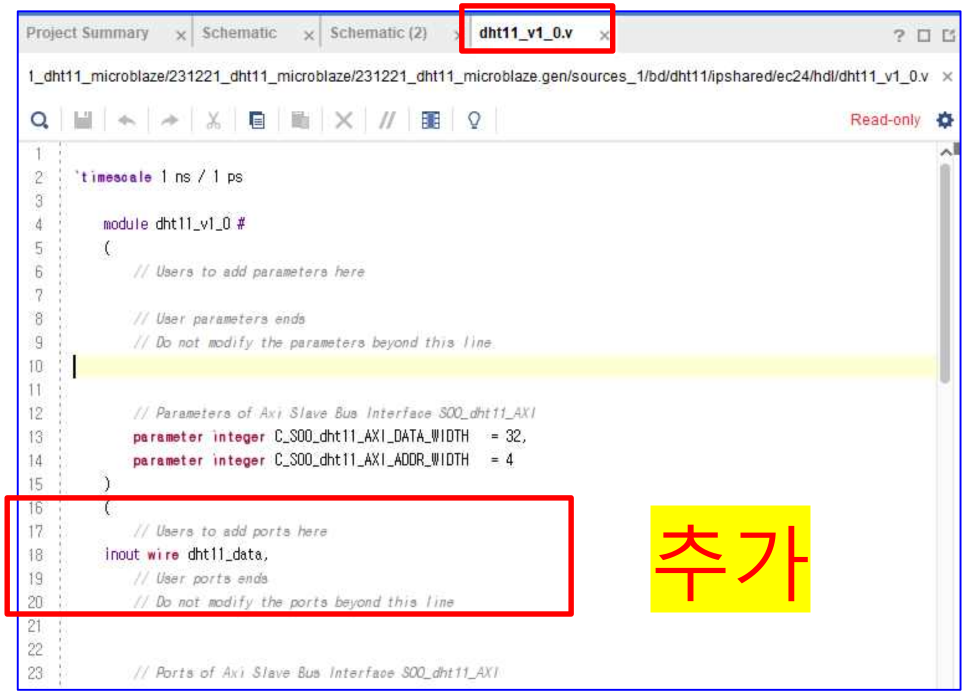
```
if(data_count >= 40)begin

    data_count <= 0;
    next_state <= S_IDLE;

if(temp_data[39:32] +
    temp_data[31:24] + temp_data[23:16]
    + temp_data[15:8] == temp_data[7:0])
    begin

   humidity <= temp_data[39:32];
   temperature <= temp_data[23:16];
      end
    end
   end
    default : next_state = S_IDLE;
   endcase
    end
    end

endmodule
```

AMBA and AXI Connect [Temperature–Humidity Sensor Module]

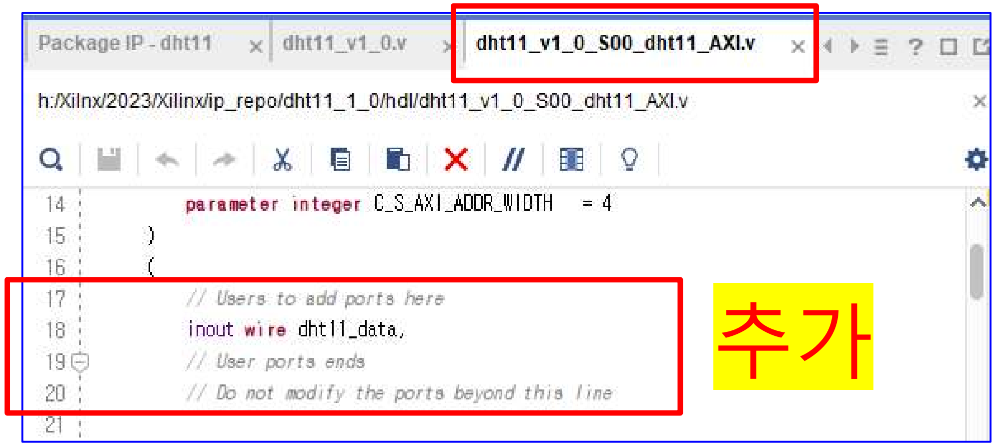

```
`timescale 1ns / 1ps
module clockdivider (
    input  clk,
    input  reset,
    output reg  clk1Mhz
);
  reg [25:0] cnt = 0;
  always @(posedge clk or negedge reset) begin
    if (~reset) begin
      cnt <= 0;
      clk1Mhz <= 0;
    end else begin
      if (cnt == (50 - 1)) begin
        cnt <= 0;
        clk1Mhz <= ~clk1Mhz;
      end else begin
        cnt <=  cnt + 1;
      end
    end
  end
endmodule
```

```
`timescale 1ns / 1ps
module edge_detector(
input clk,
input cp_in,
input reset,
output p_edge,
output n_edge
   );

  reg cp_in_old, cp_in_cur;
  always@(posedge clk or negedge reset) begin
  if(~reset) begin
  cp_in_old=0; cp_in_cur = 0;
  end else begin
  cp_in_old <= cp_in_cur;
  cp_in_cur <= cp_in;
  end
 end

  assign p_edge = ~cp_in_old & cp_in_cur;
  assign n_edge = cp_in_old & ~cp_in_cur;
endmodule
```

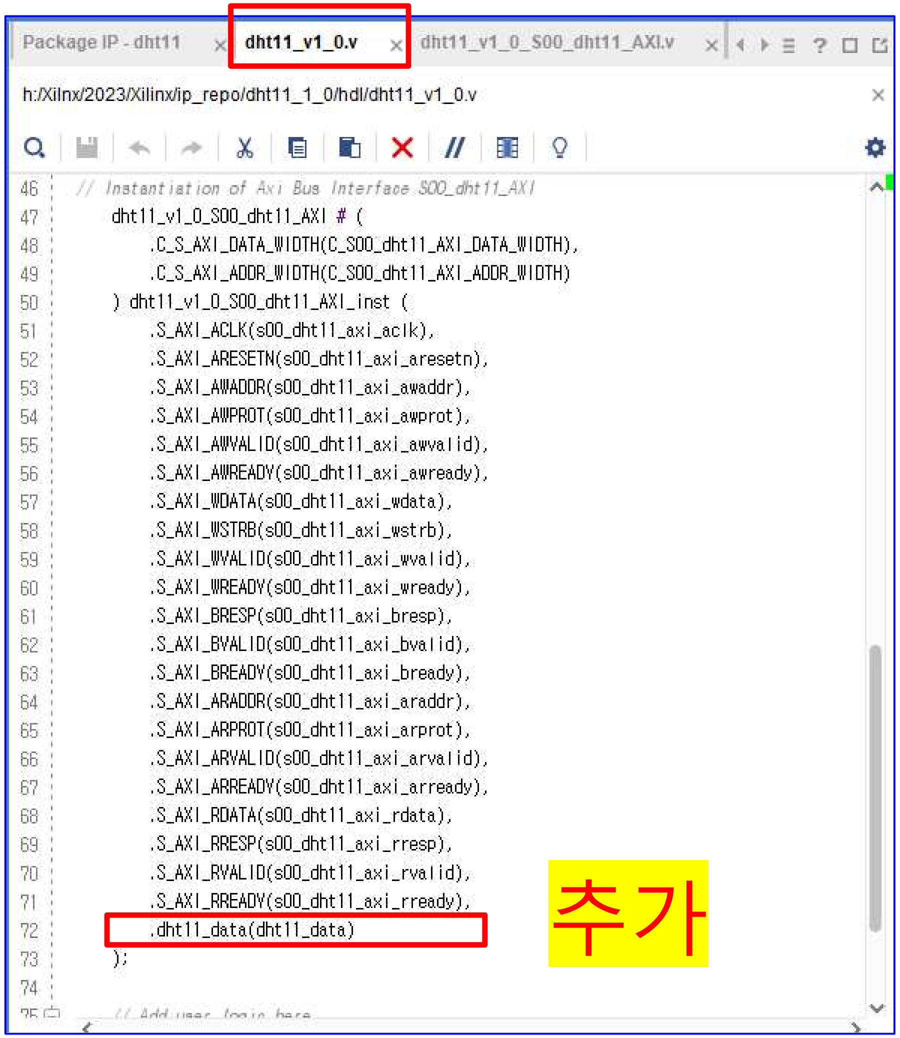
➢ AXI Connect ➔ AXI Template



추가



추가



추가

➢ *AXI Connect ➔ AXI Template*

Package IP - dht11 | dht11_v1_0.v | dht11_v1_0_S00_dht11_AXI.v

h:/Xilnx/2023/Xilinx/ip_repo/dht11_1_0/hdl/dht11_v1_0_S00_dht11_AXI.v

```
	// Address decoding for reading registers
	case ( axi_araddr[ADDR_LSB+OPT_MEM_ADDR_BITS:ADDR_LSB] )
	  2'h0   : reg_data_out <= temperature0;
	  2'h1   : reg_data_out <= temperature10;
	  2'h2   : reg_data_out <= humidity0;
	  2'h3   : reg_data_out <= humidity10;
	  default : reg_data_out <= 0;
	endcase
      end

      // Output register or memory read data
```

변경

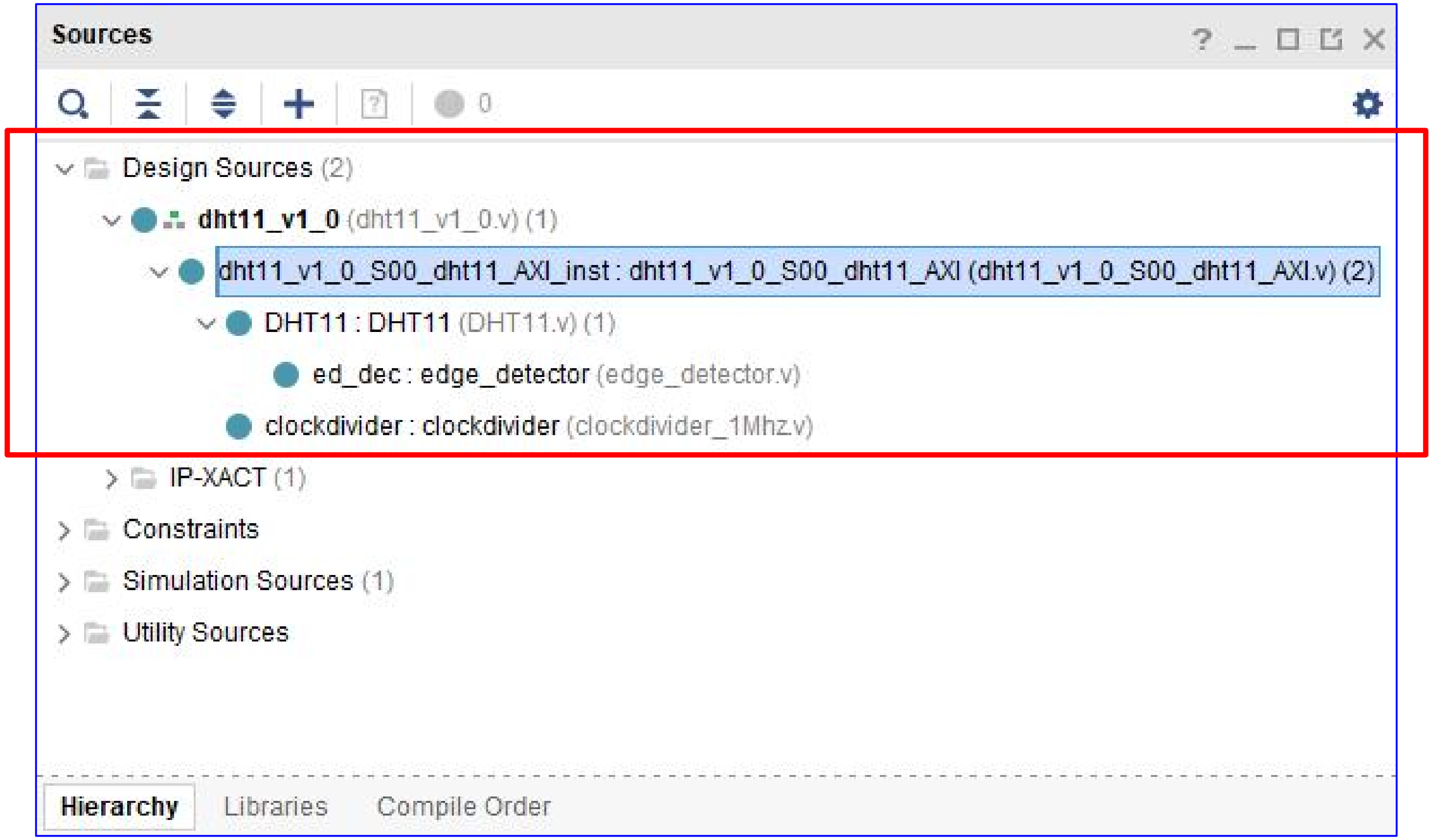
Package IP - dht11 | dht11_v1_0.v | dht11_v1_0_S00_dht11_AXI.v *

h:/Xilnx/2023/Xilinx/ip_repo/dht11_1_0/hdl/dht11_v1_0_S00_dht11_AXI.v

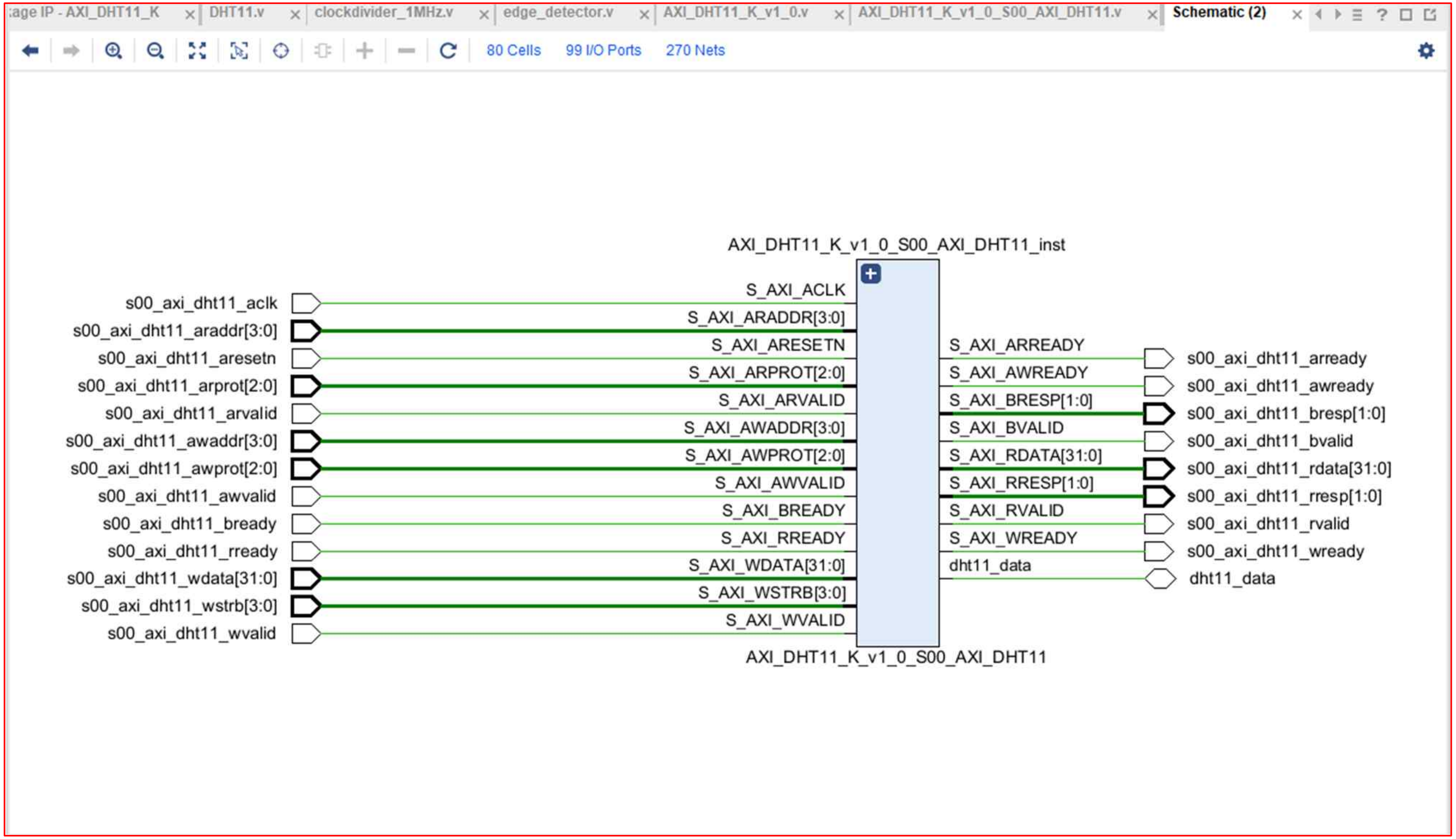
```
	// Add user logic here
DHT11 DHT11(
.clk(S_AXI_ACLK),
.reset(S_AXI_ARESETN),
.clk1Mhz(clk1Mhz),
.dht11_data(dht11_data),
.humidity10(humidity10),
.humidity0(humidity0),
.temperature10(temperature10),
.temperature0(temperature0)
 );

 wire [3:0] humidity10,humidity0, temperature10, temperature0;

clockdivider clockdivider(
.clk(S_AXI_ACLK),
.reset(S_AXI_ARESETN),
.clk1Mhz(clk1Mhz)
);
	// User logic ends

	endmodule
```

추가

➢ *AXI Connect ➔ AXI Template*



- *AXI Connect → AXI Template → RTL*



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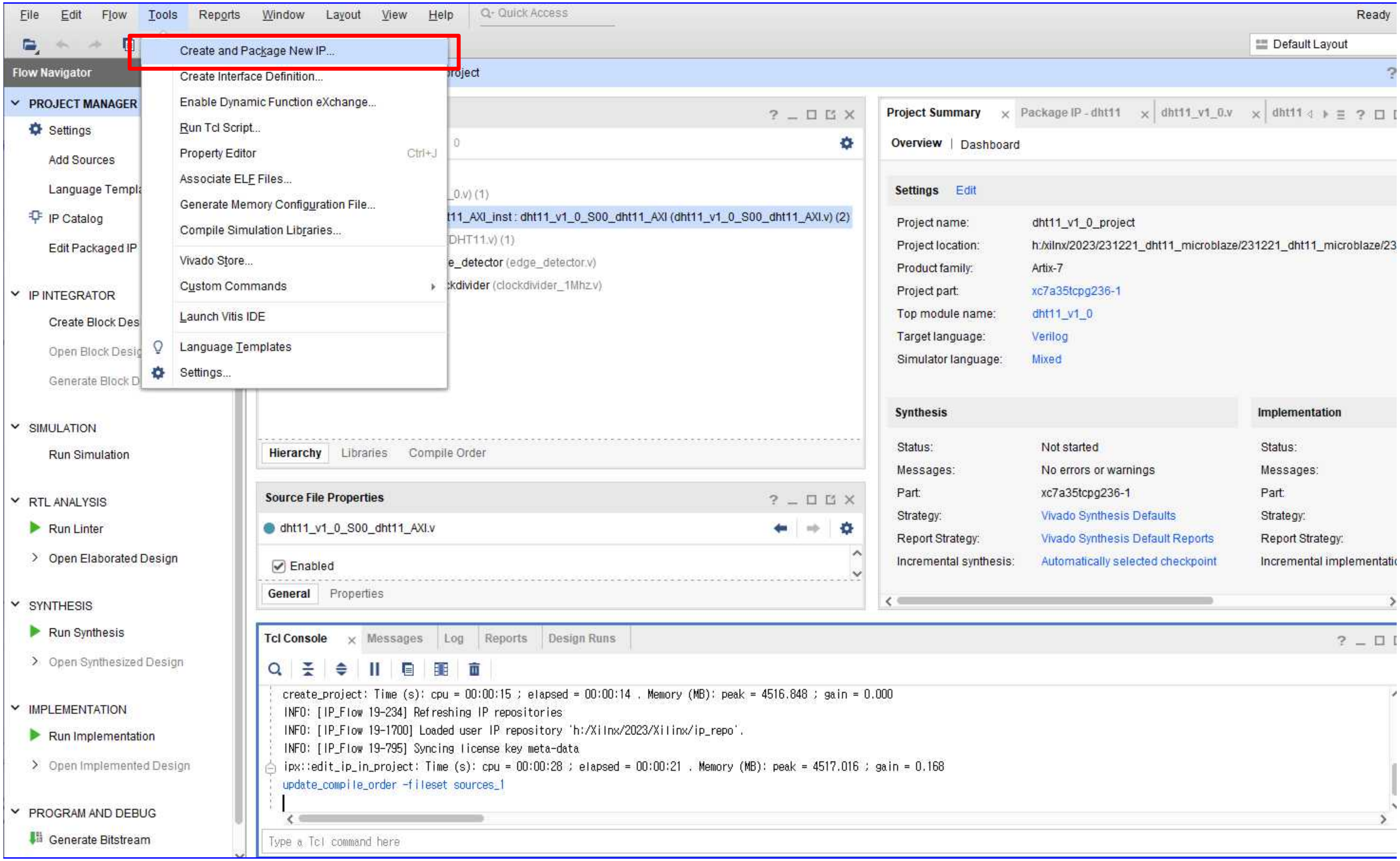
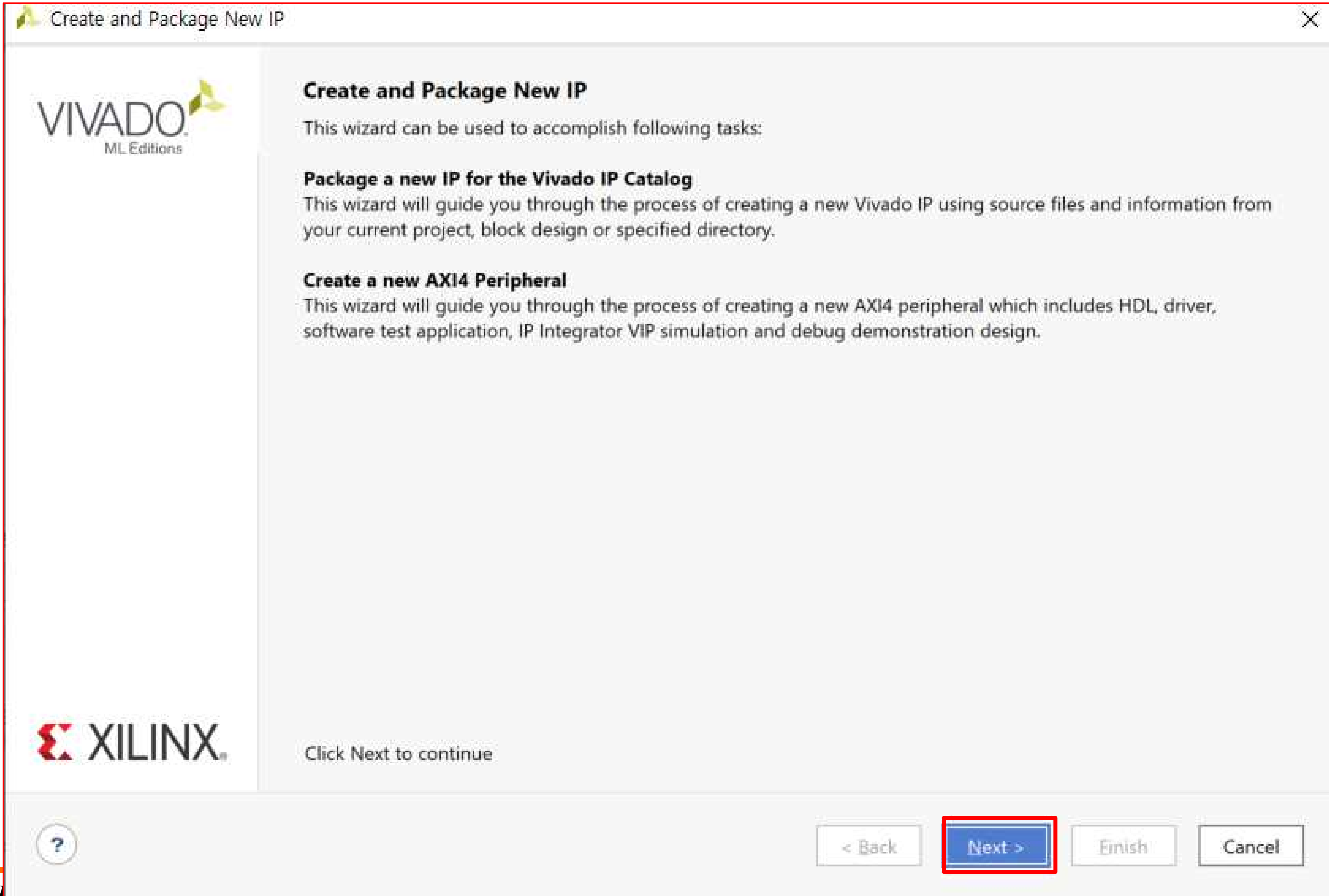
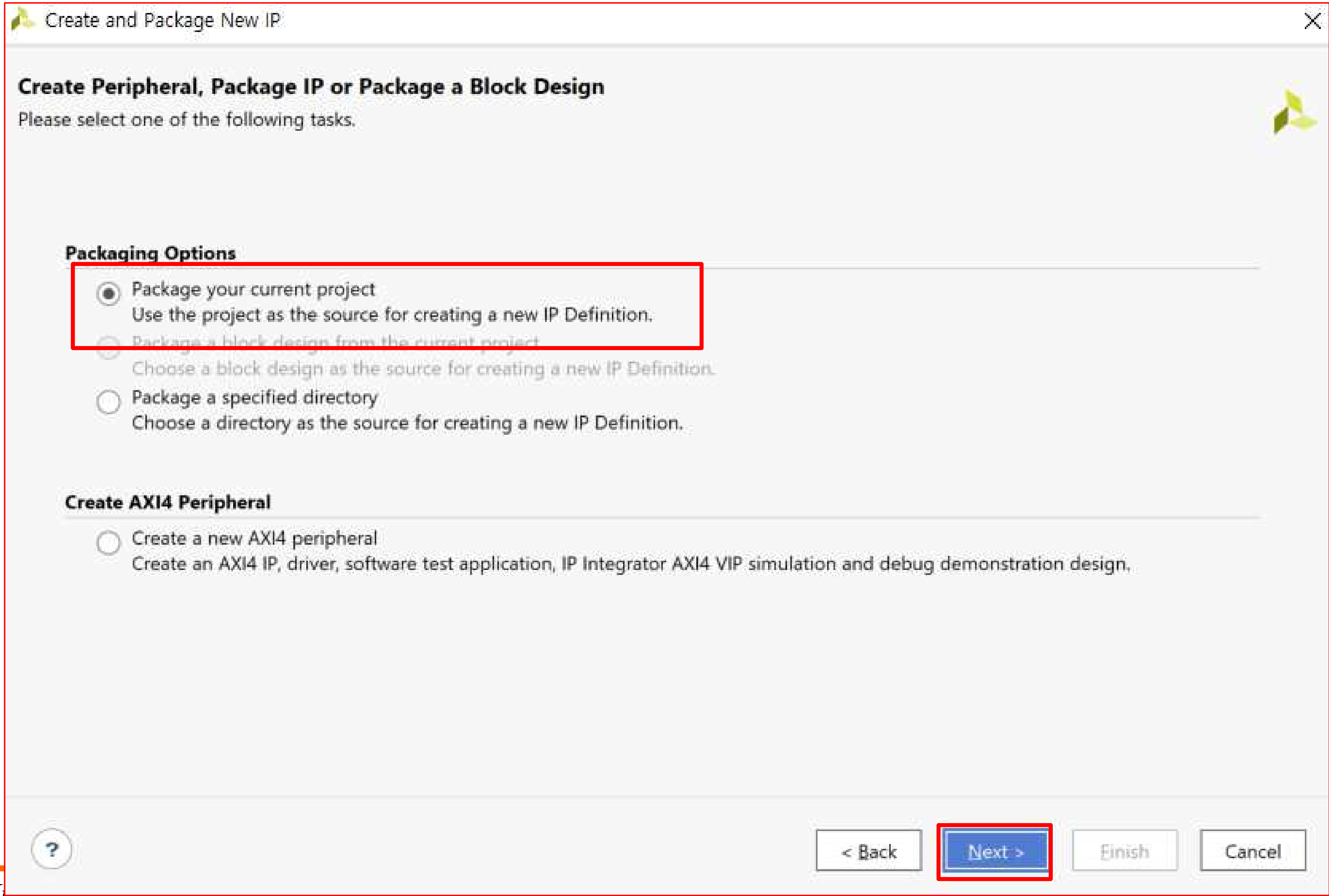


➢ *AXI Connect ➔ AXI Template*



- ***AXI Connect → AXI Template***



Create and Package New IP

➢ *AXI Connect ➔ AXI Template*

VIVADO ML Editions

New IP Creation

The following pieces of information will be gathered:

- Identification information based on top module name
- Family compatibility based on part in the project
- File(s) from Synthesis and Simulation file sets
- Ports from the file containing the top module
- Parameters from the file containing the top module
- Bus Interfaces based on port names
- Address Spaces and Memory Maps based on inferred bus interfaces

Following file will be created on disk along with corresponding customization files:
f:/xilnx/2022/amba_axi_connect_test/amba_axi_connect_test.srcs/component.xml

XILINX

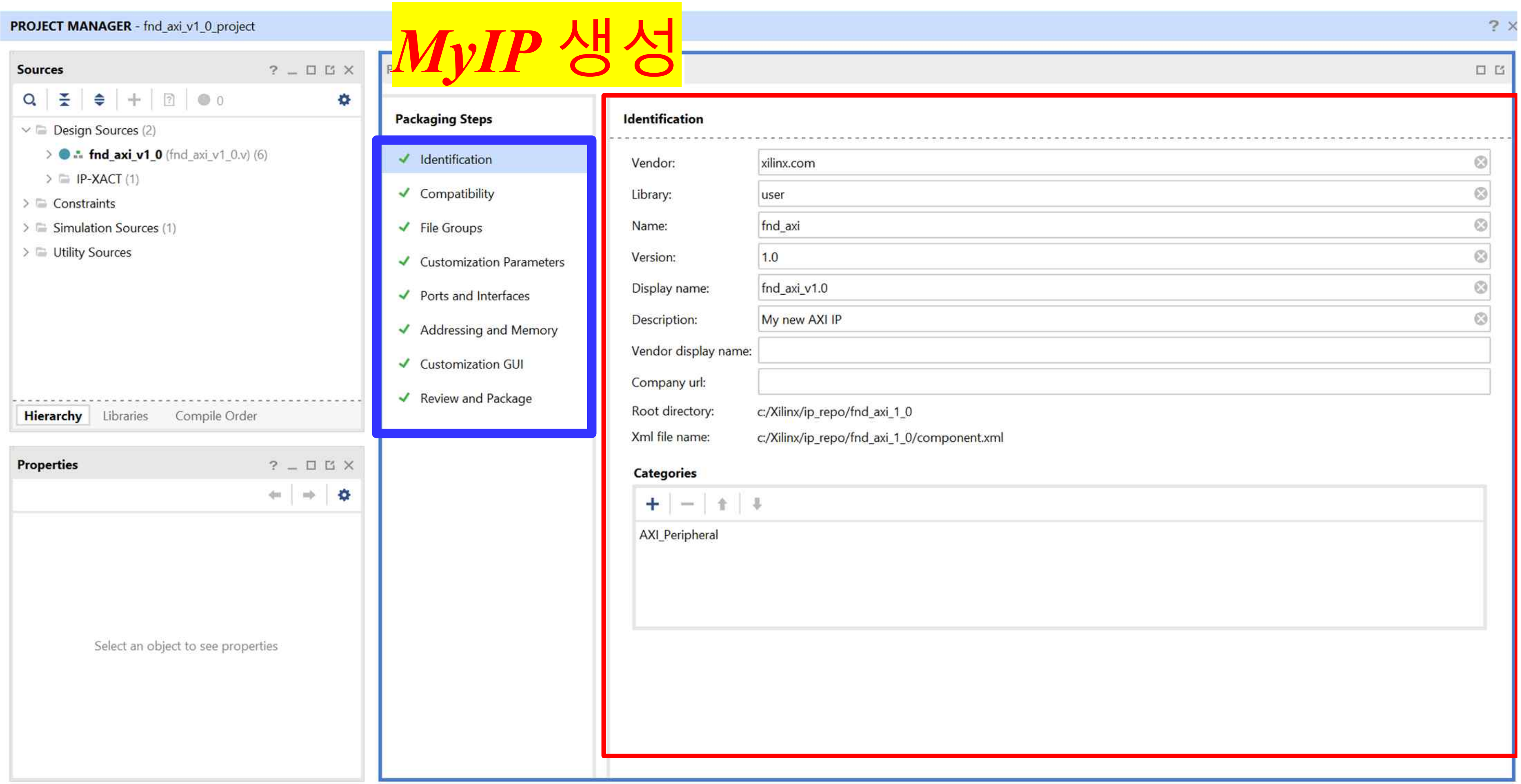
Click Finish to continue

? < Back Next > Finish Cancel

AMBA and AXI Connect [Temperature–Humidity Sensor Module]

Axi_dht11.xpr

➢ *AXI Connect ➔ AXI Template*

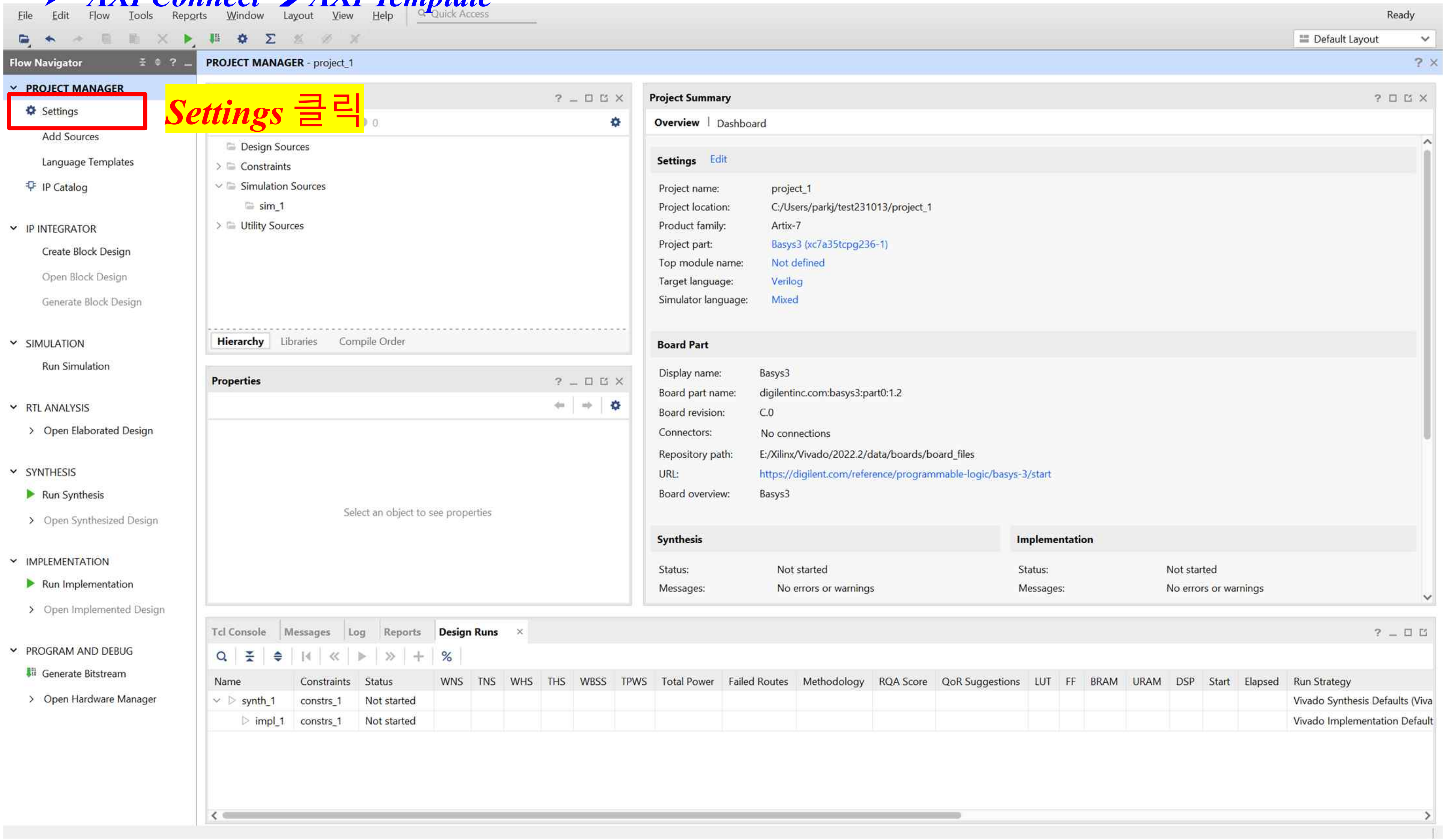


AMBA and AXI Connect Example

Temperature-Humidity Sensor Module

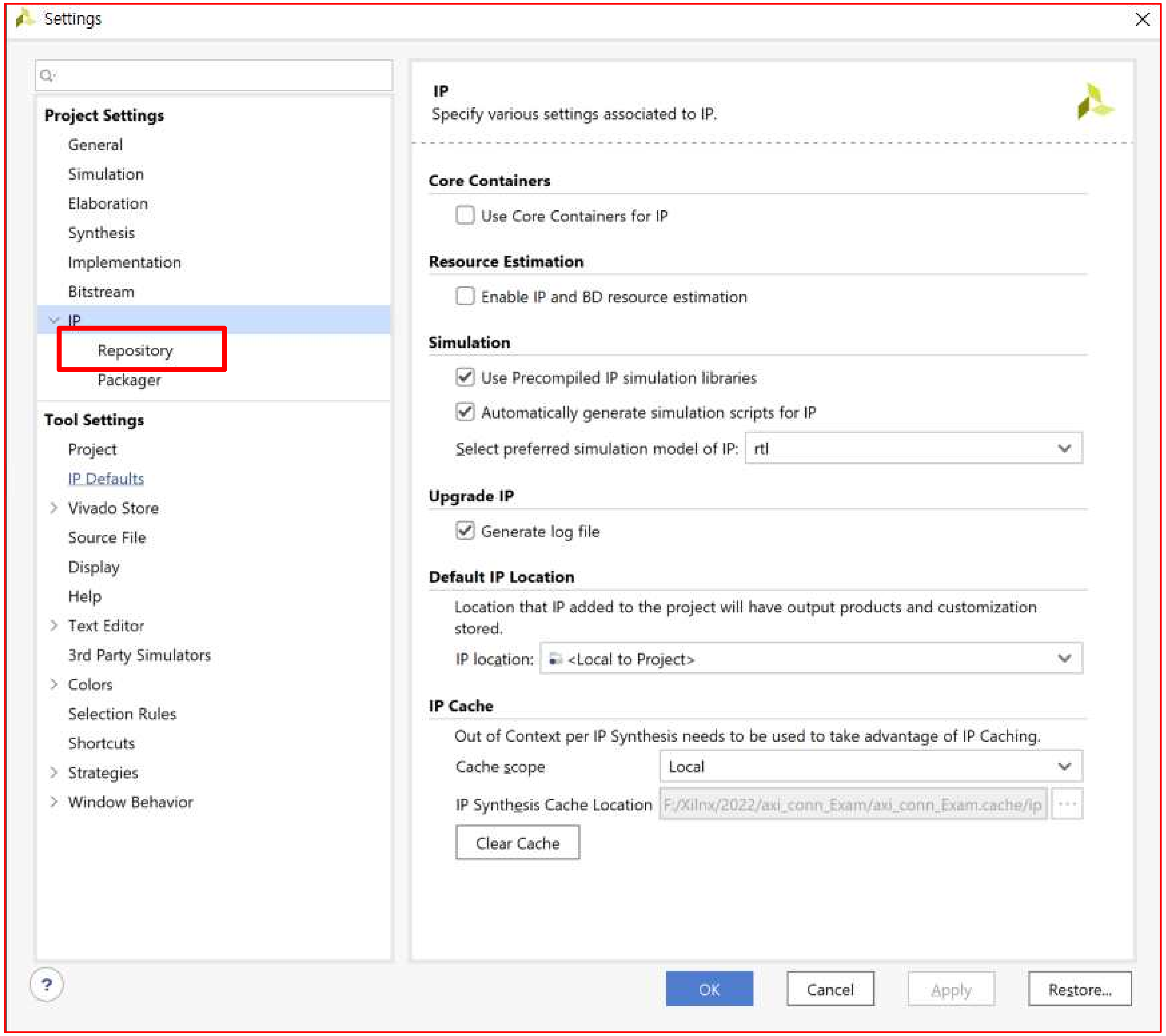
[*Create Block Design*]

➢ *AXI Connect* ➔ *AXI Template*

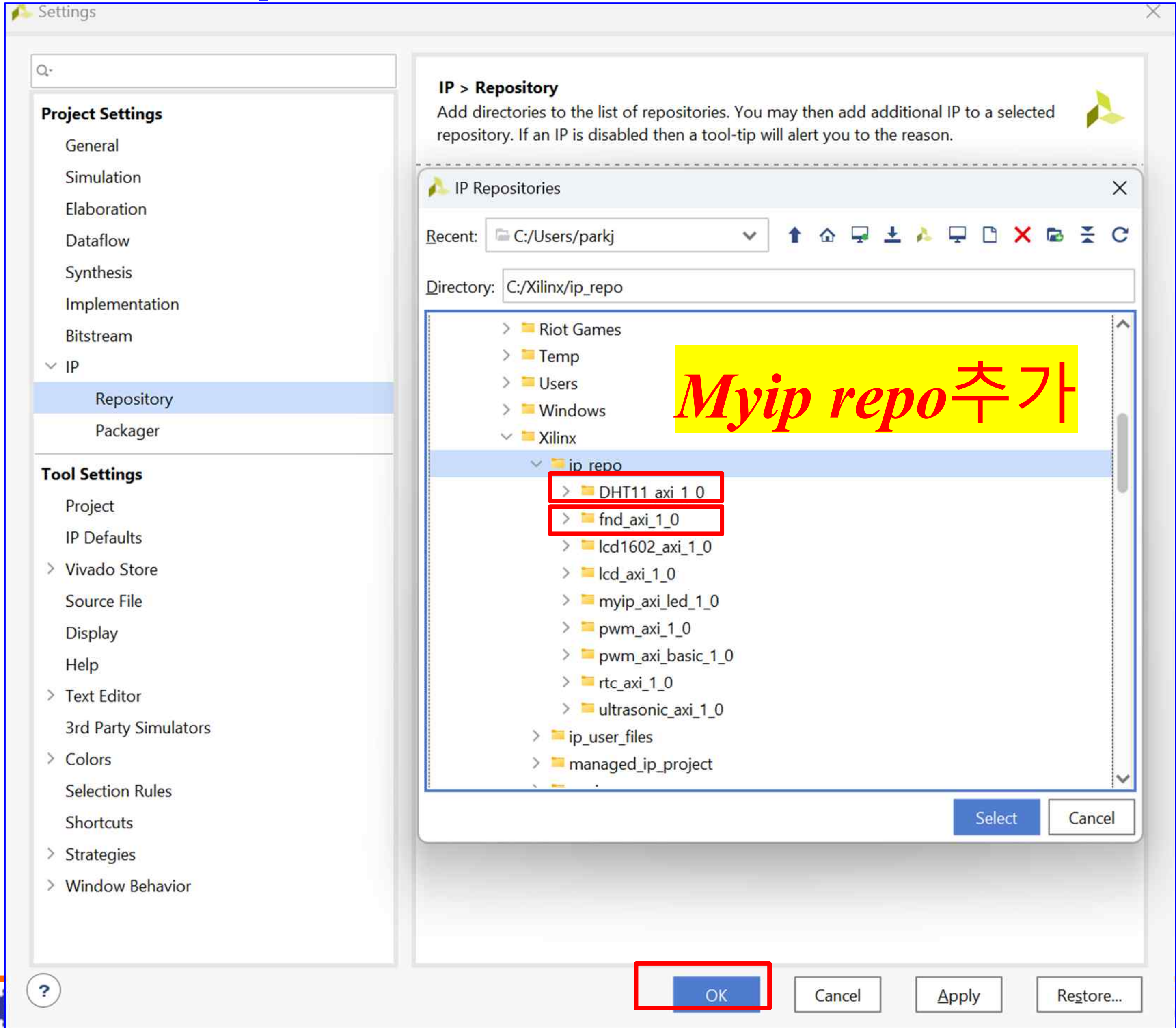


➢ *AXI Connect ➔ AXI Template*

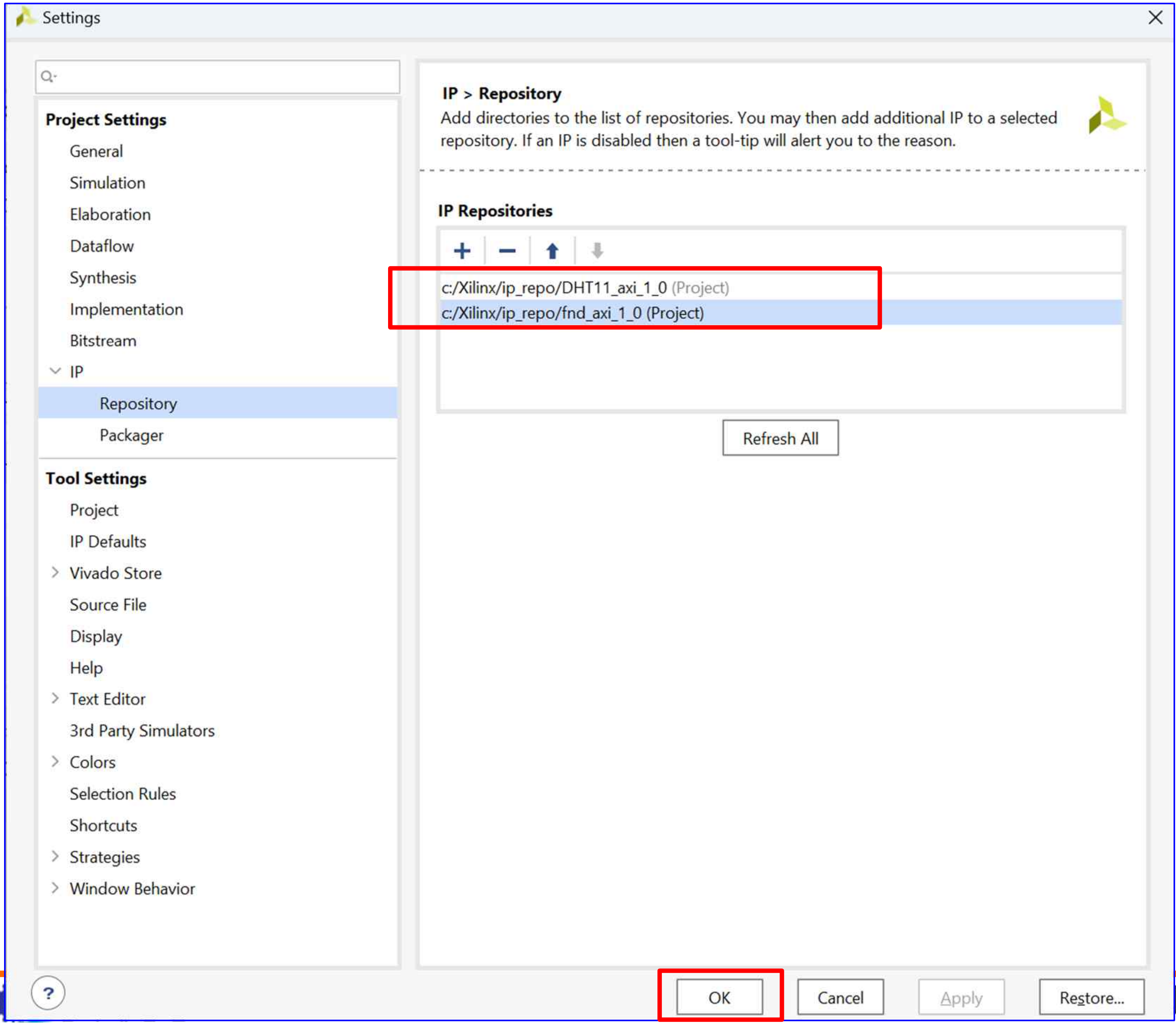
Repository



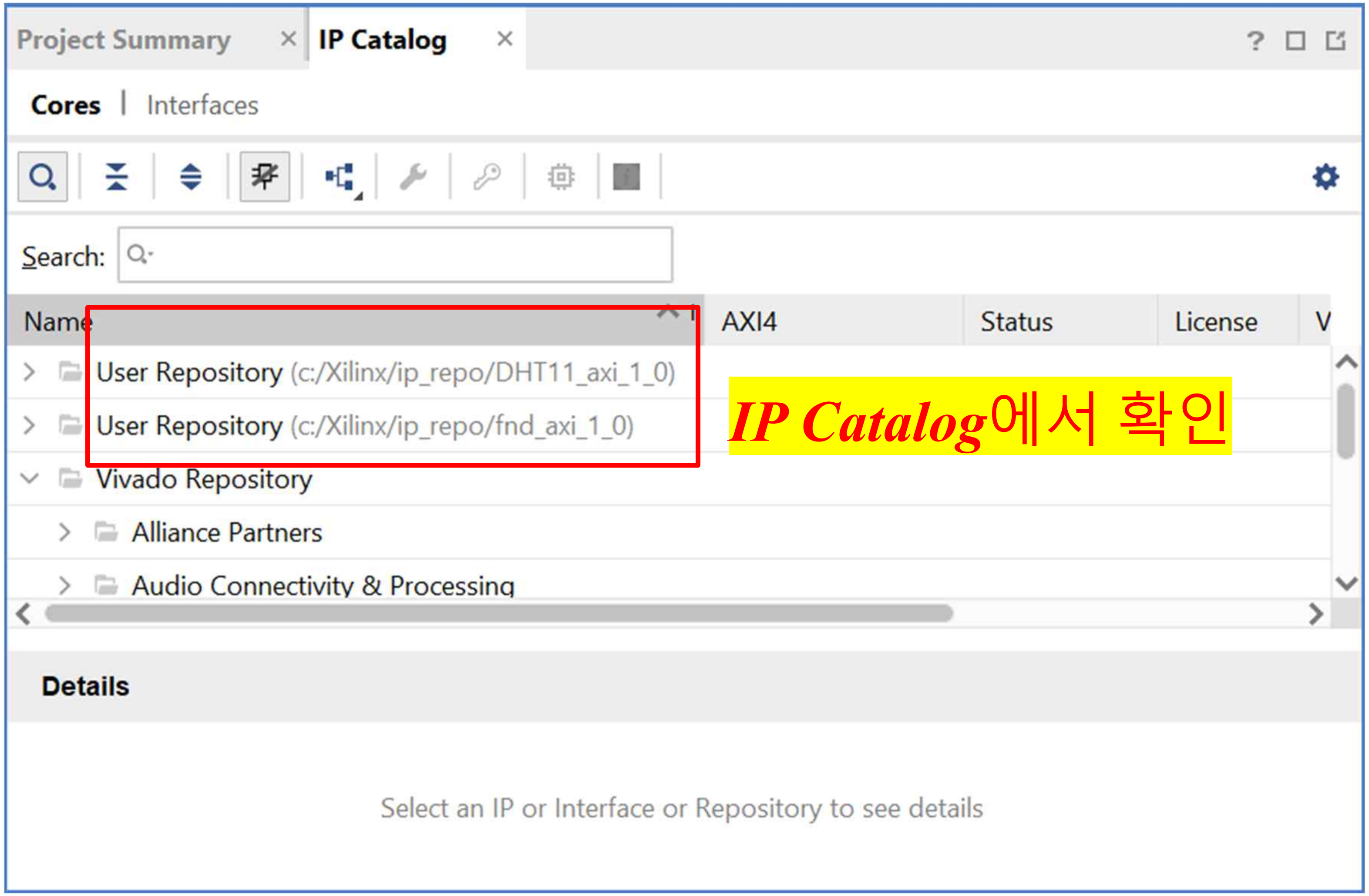
➤ *AXI Connect ➔ AXI Template*



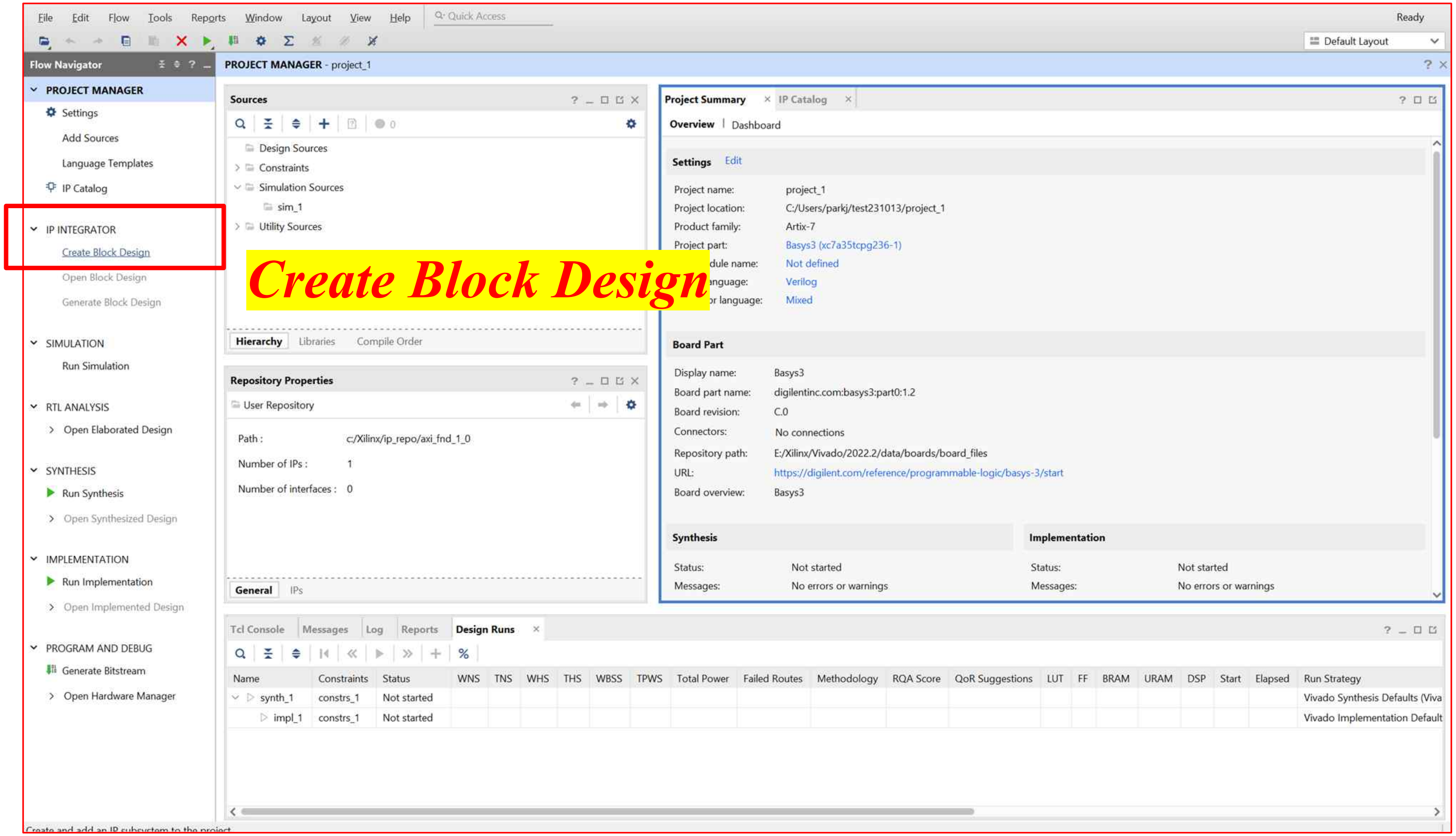
➢ AXI Connect → AXI Template



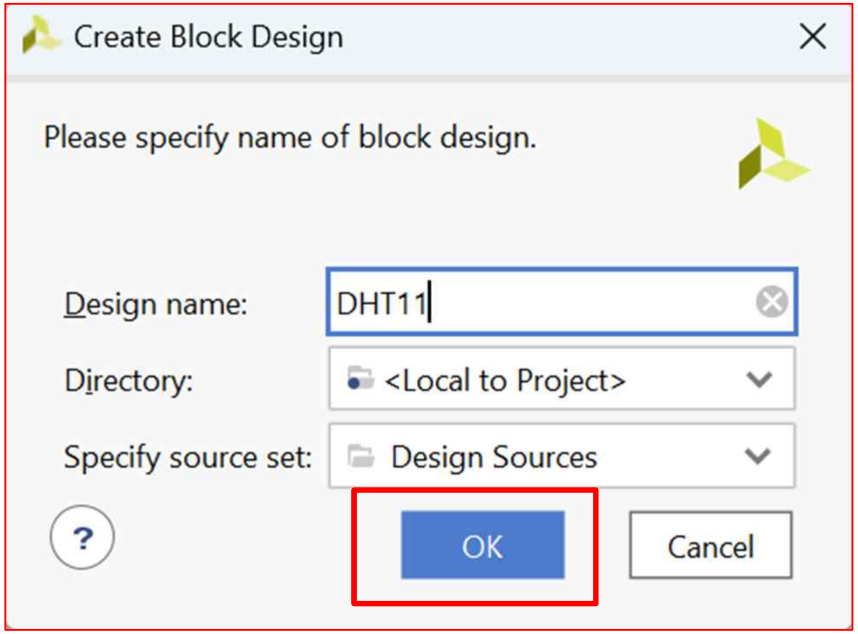
➢ *AXI Connect ➔ AXI Template*



➢ *AXI Connect ➔ AXI Template*

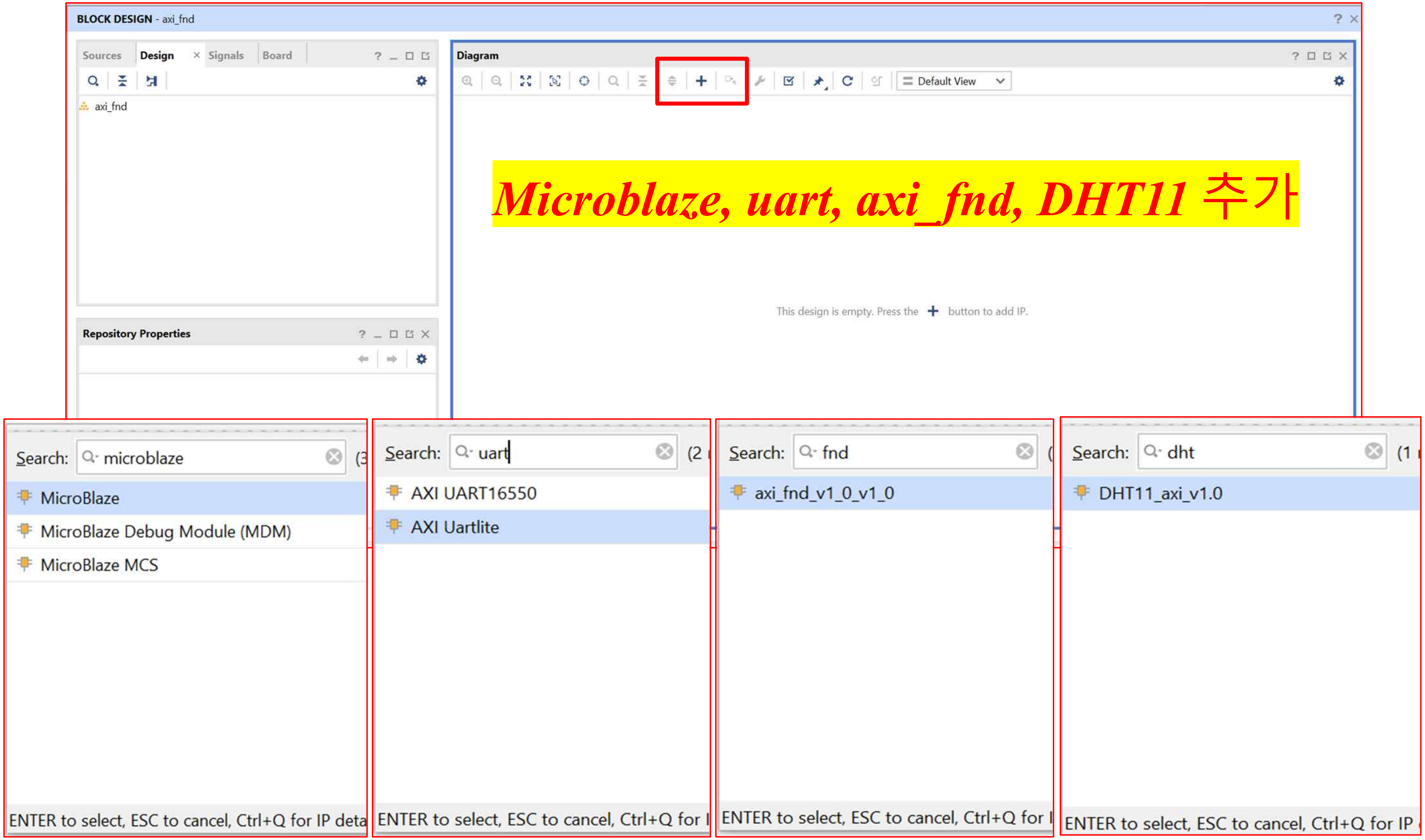


- *AXI Connect ➔ AXI Template*

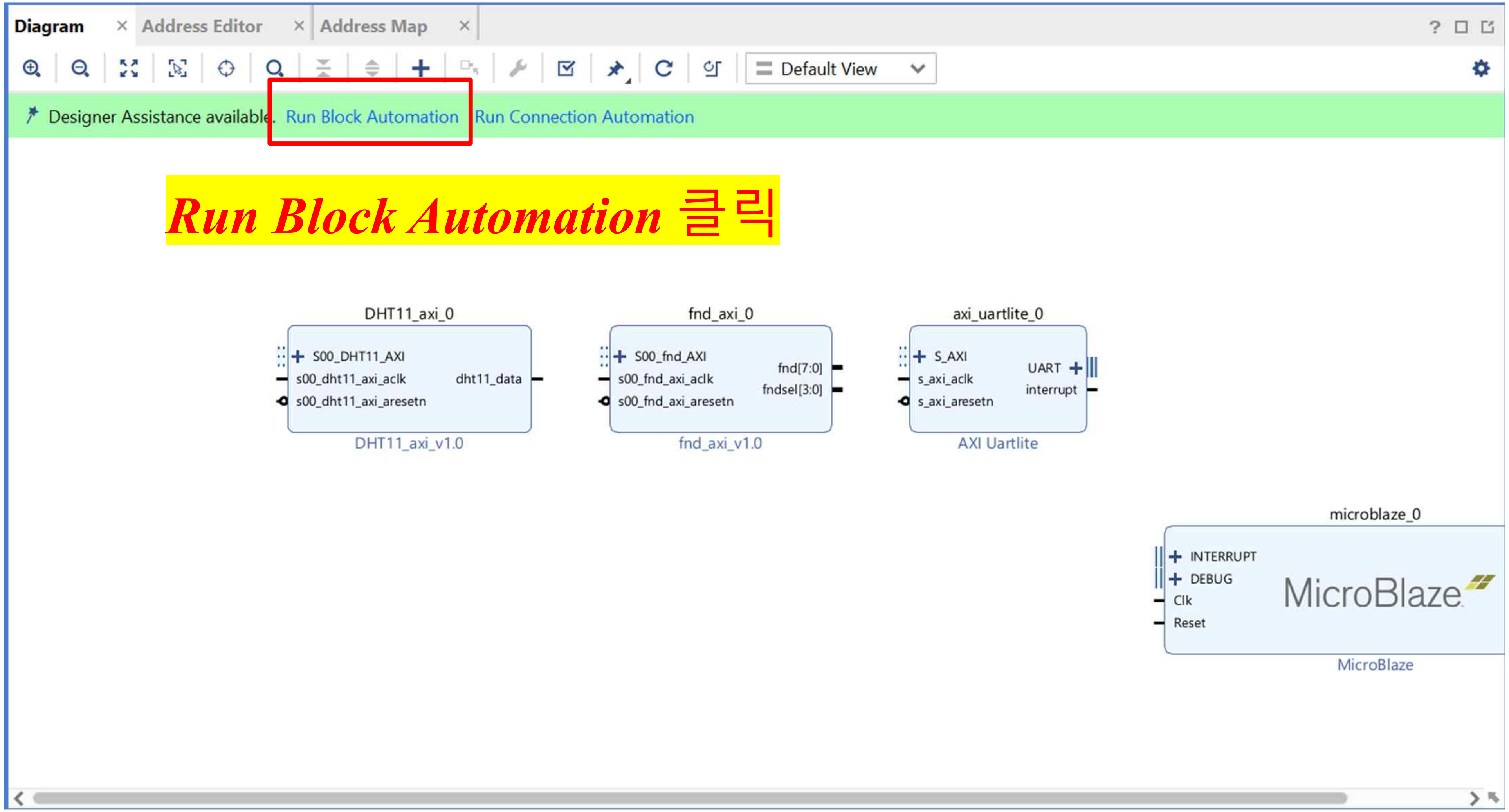


이름 설정 후 OK

➢ AXI Connect ➔ AXI Template



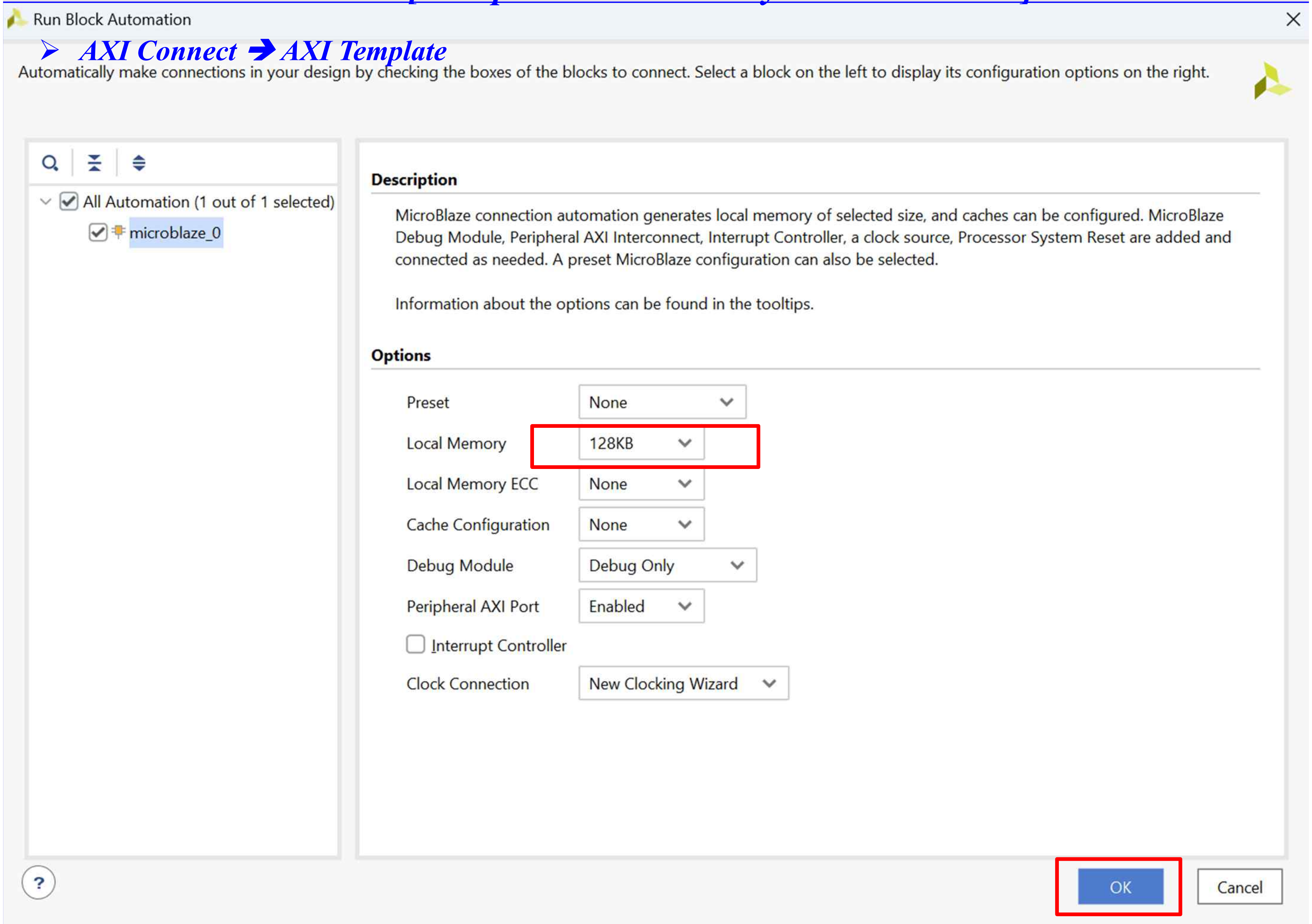
➢ *AXI Connect ➔ AXI Template*



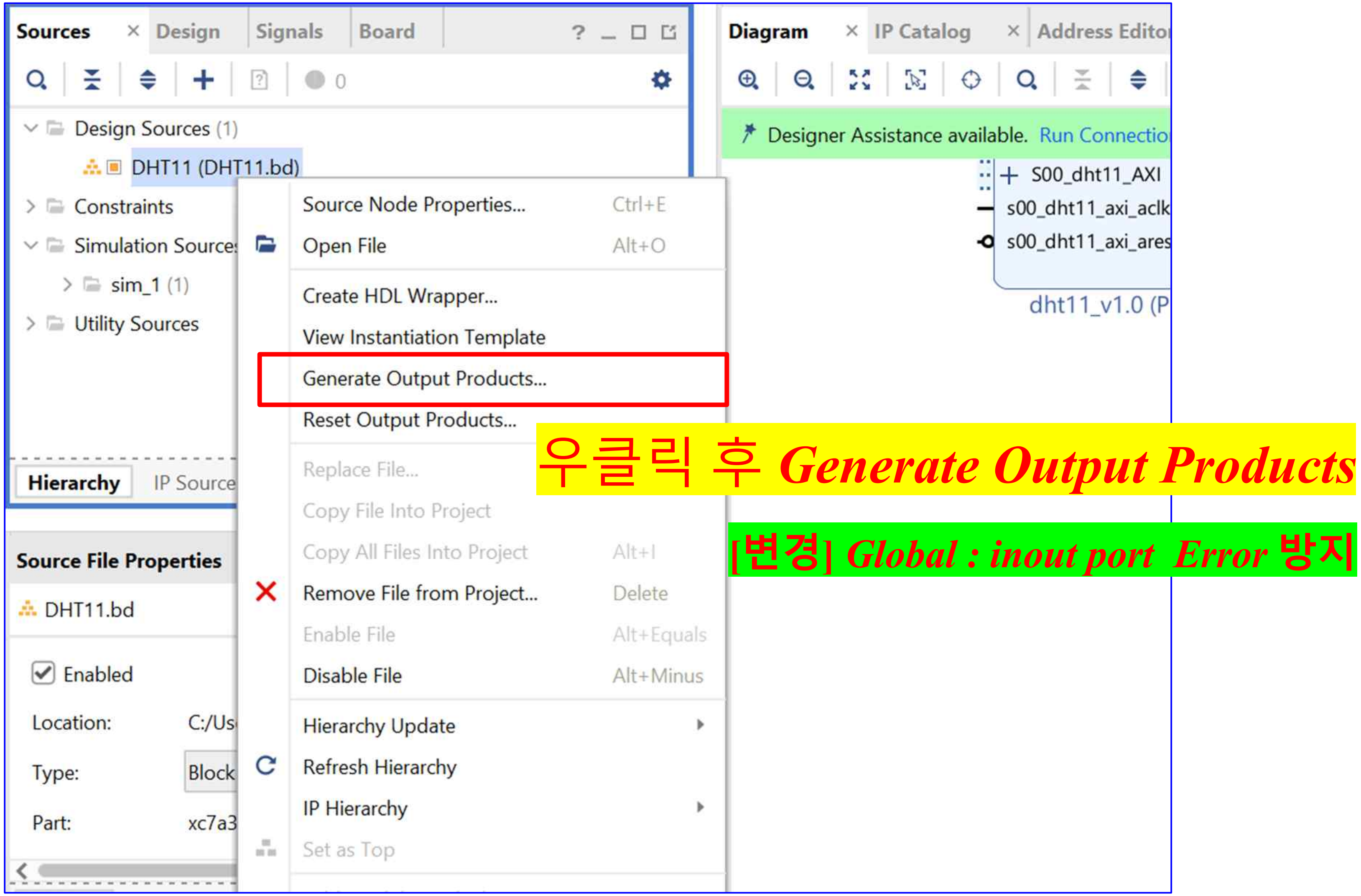
AMBA and AXI Connect [Temperature–Humidity Sensor Module]

Axi_dht11.xpr

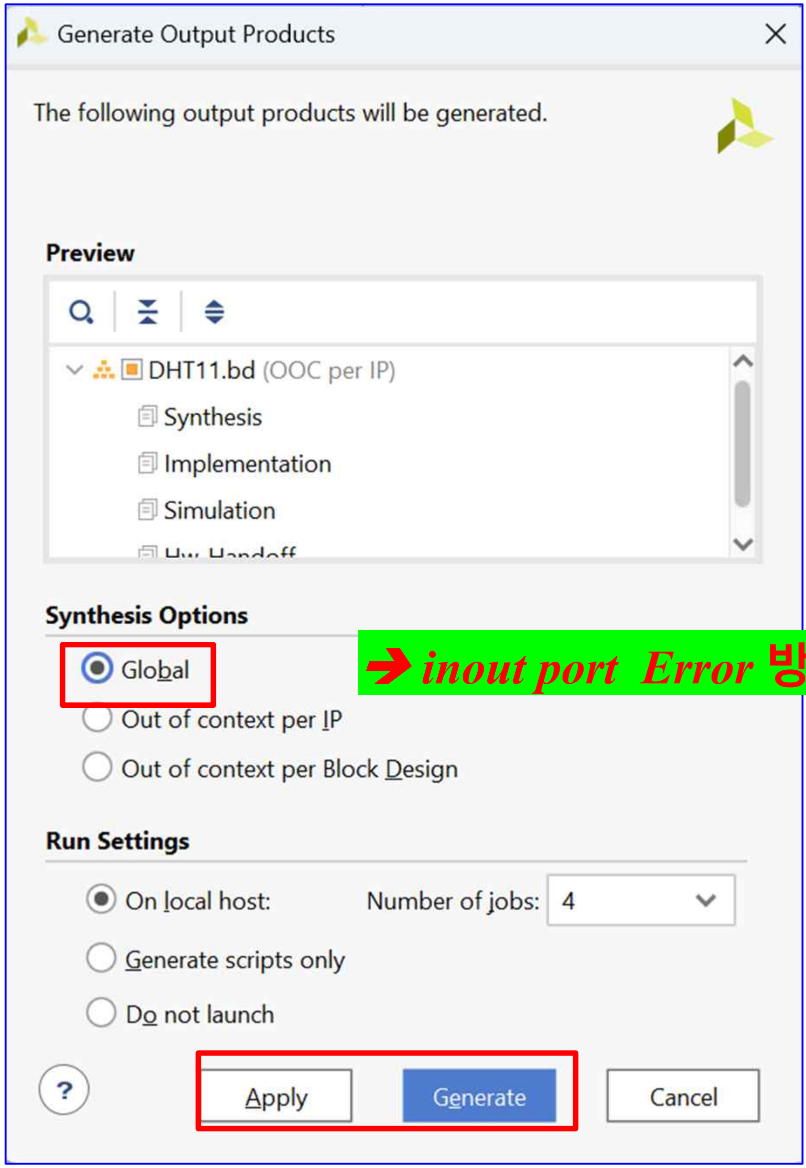
➢ AXI Connect ➔ AXI Template



➢ ***AXI Connect ➔ AXI Template***



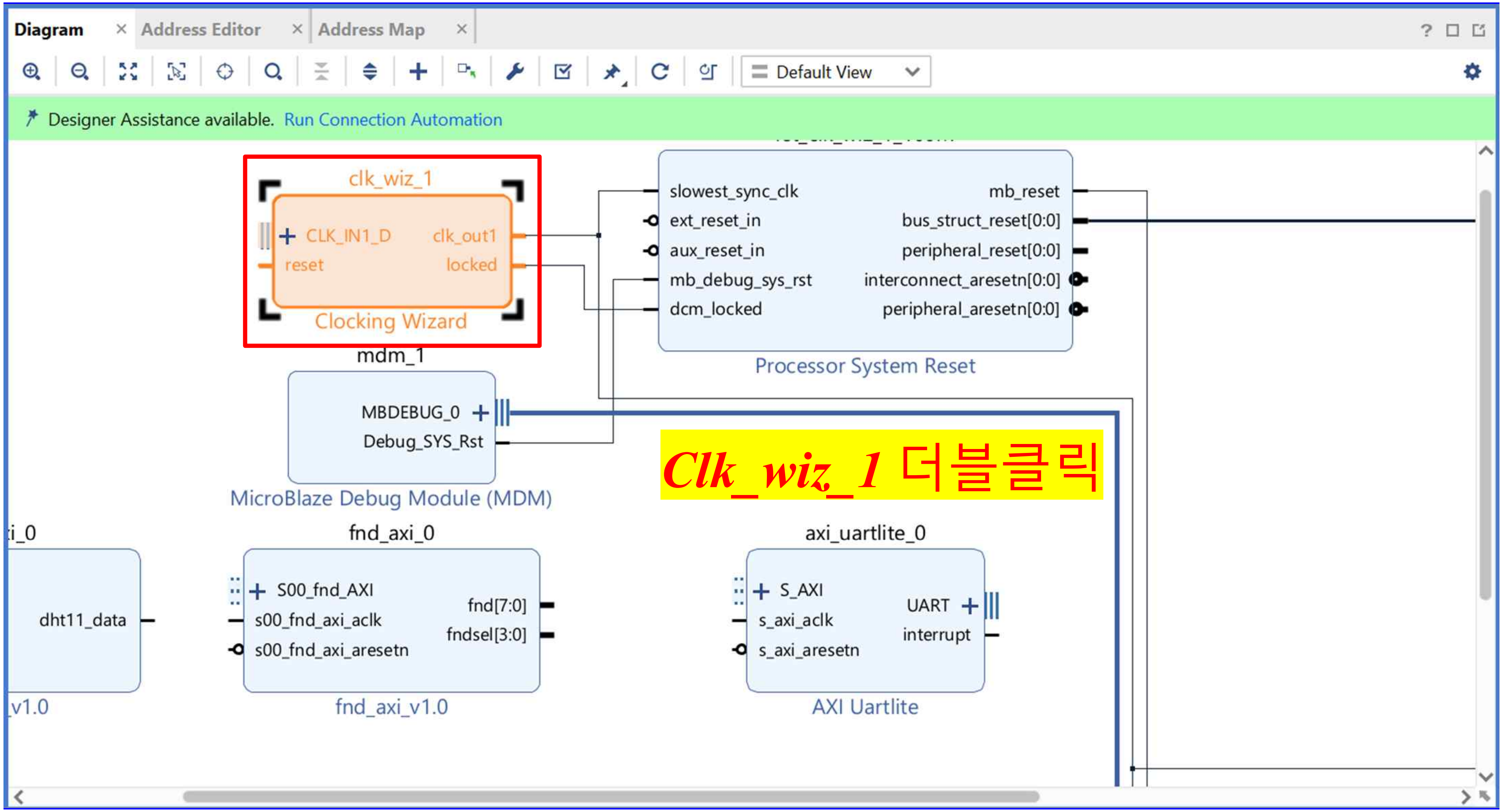
➢ *AXI Connect ➔ AXI Template*



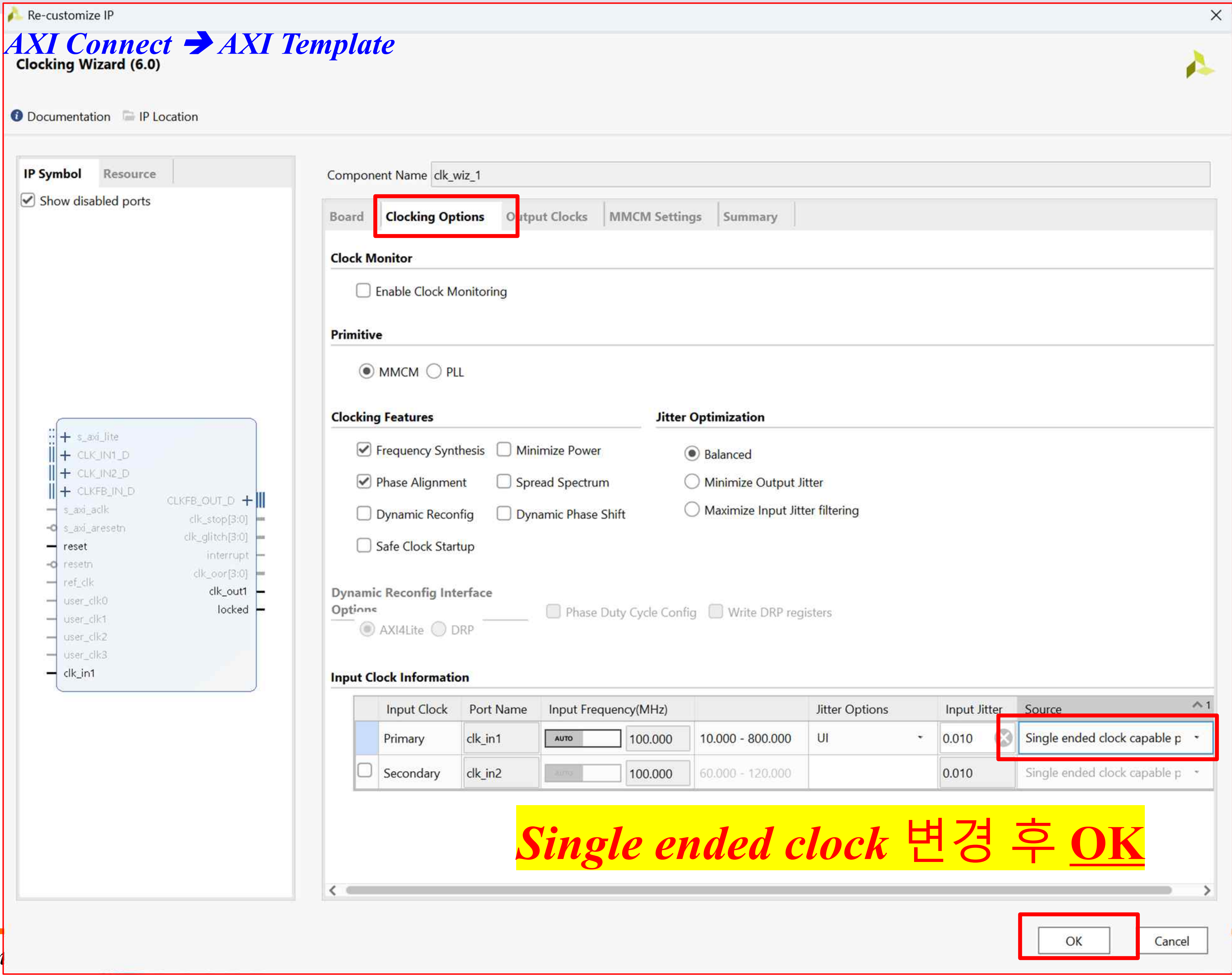
➔ *inout port Error* 방지

*Synthesis Options*를 *Global*로 변경 후 *Generate*

➢ *AXI Connect ➔ AXI Template*



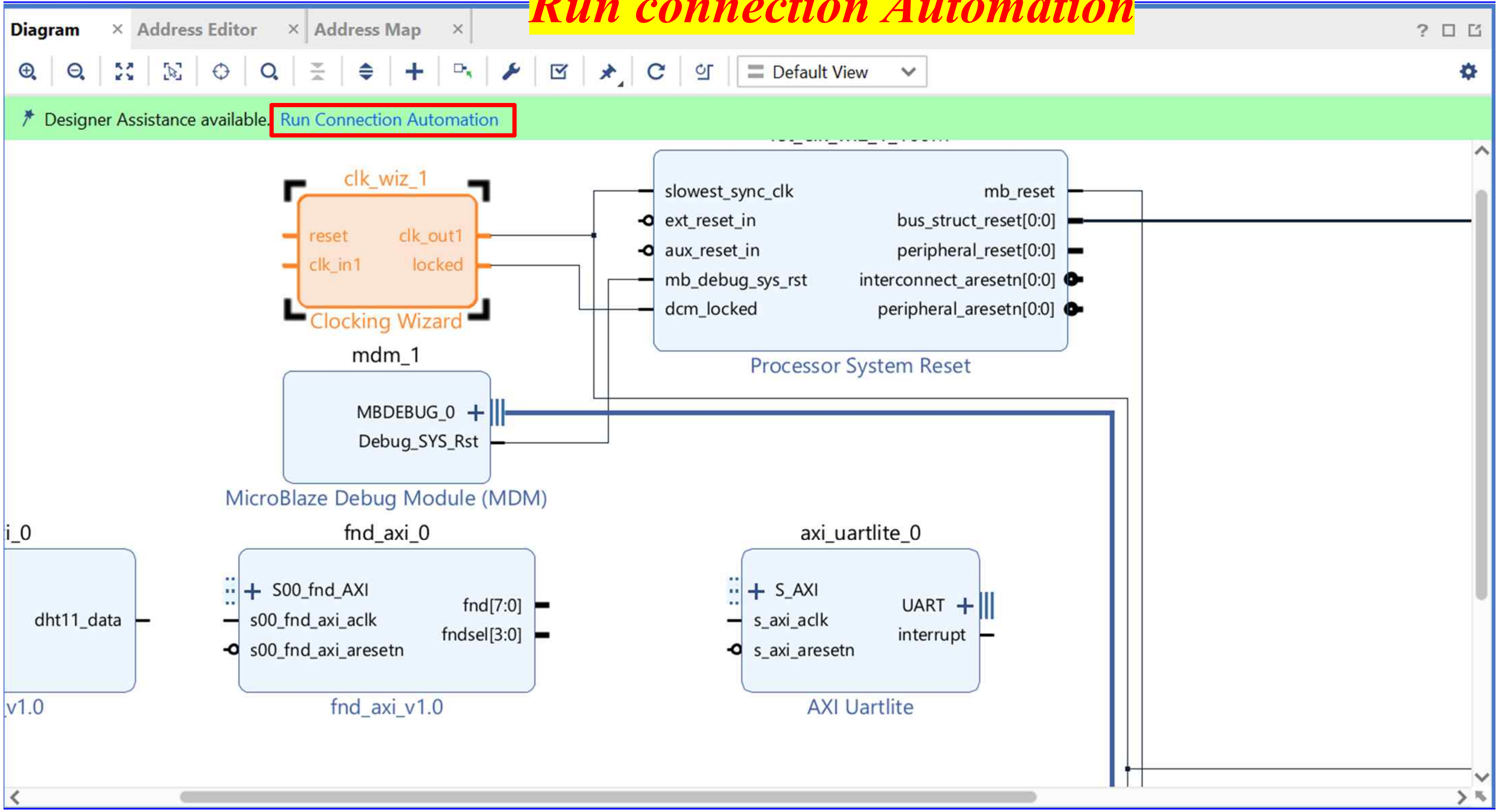
➤ *AXI Connect ➔ AXI Template*



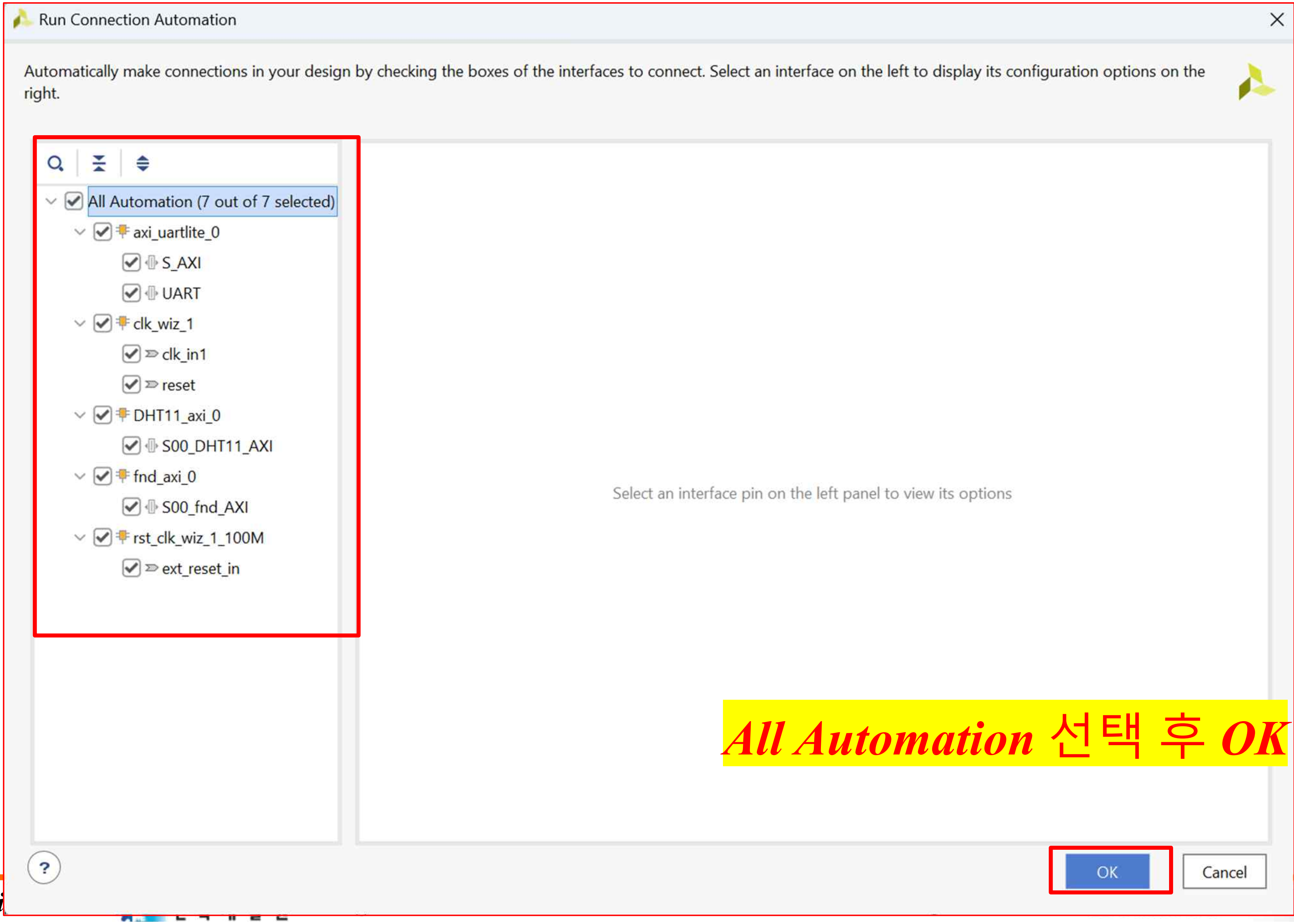
Single ended clock 변경 후 OK

- ***AXI Connect ➔ AXI Template***

Run connection Automation

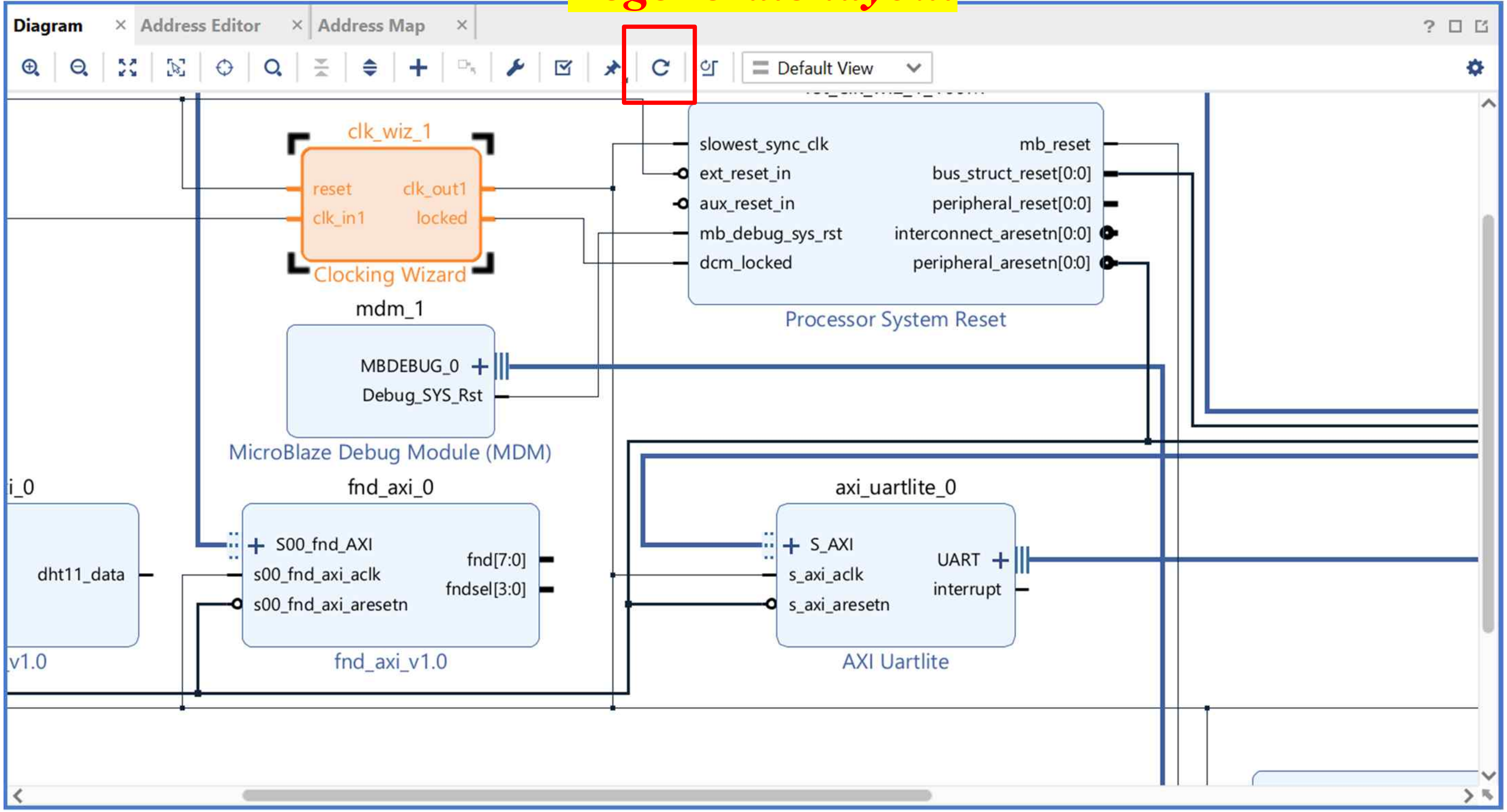


- *AXI Connect ➔ AXI Template*

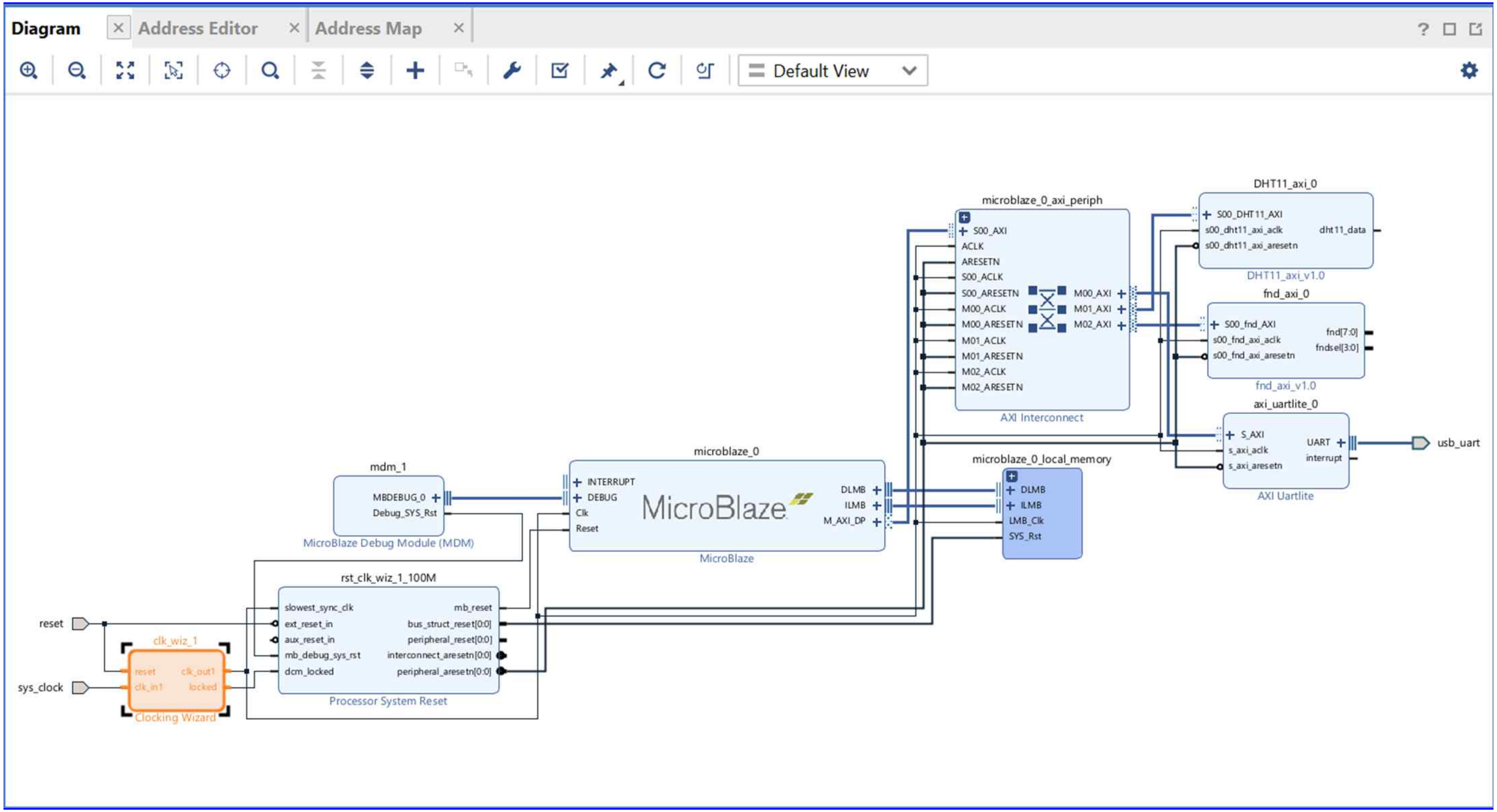


➢ *AXI Connect ➔ AXI Template*

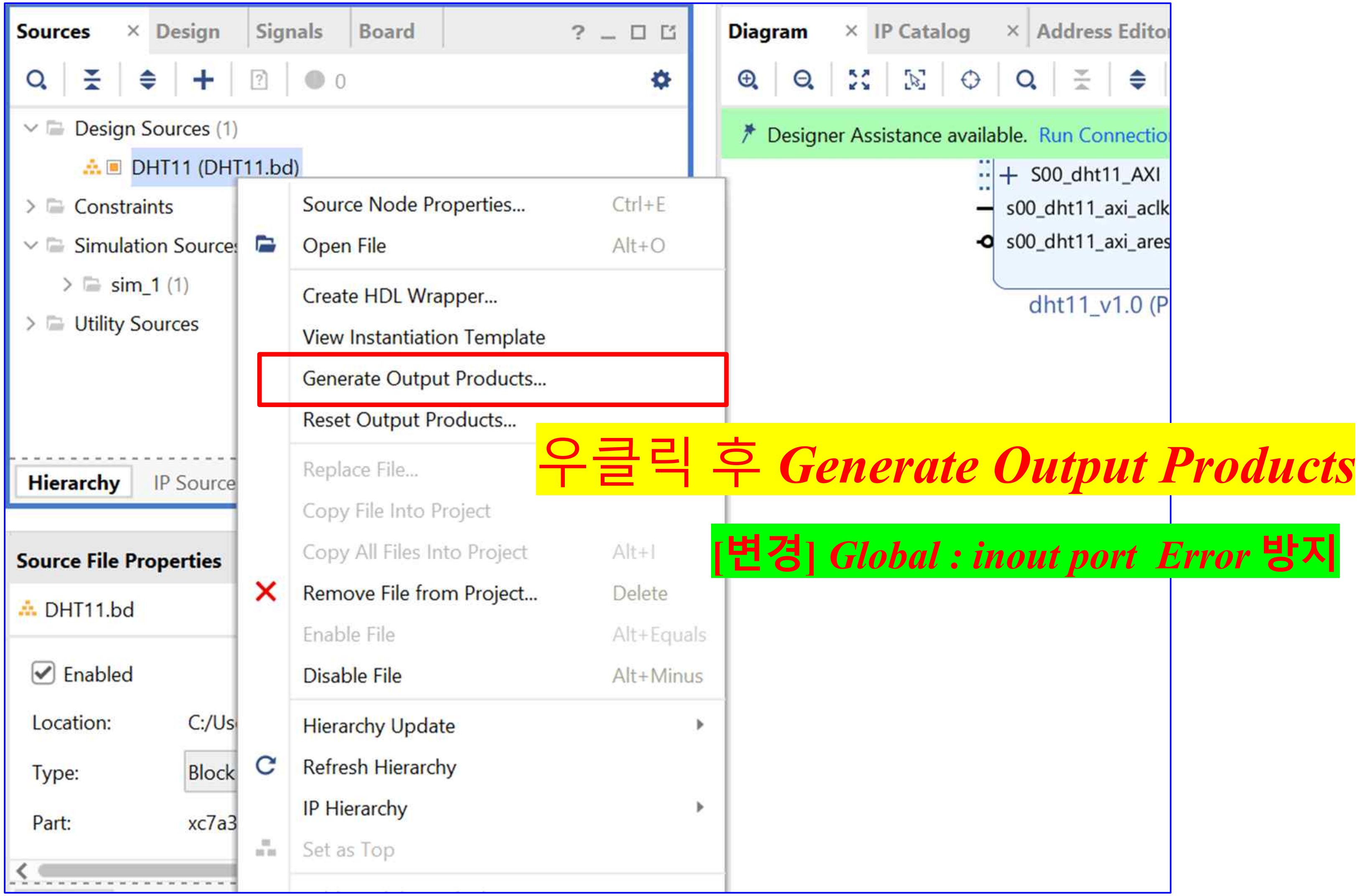
Regenerate layout



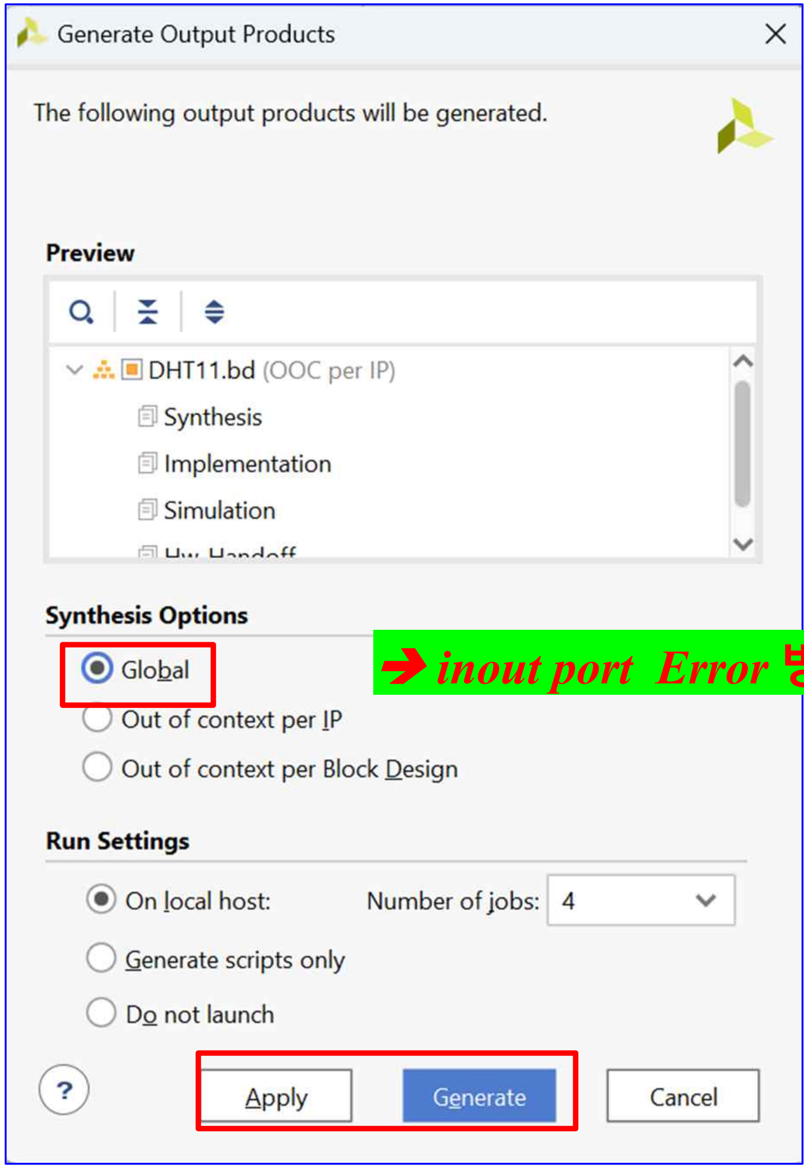
➢ ***AXI Connect → AXI Template***



➢ ***AXI Connect → AXI Template***



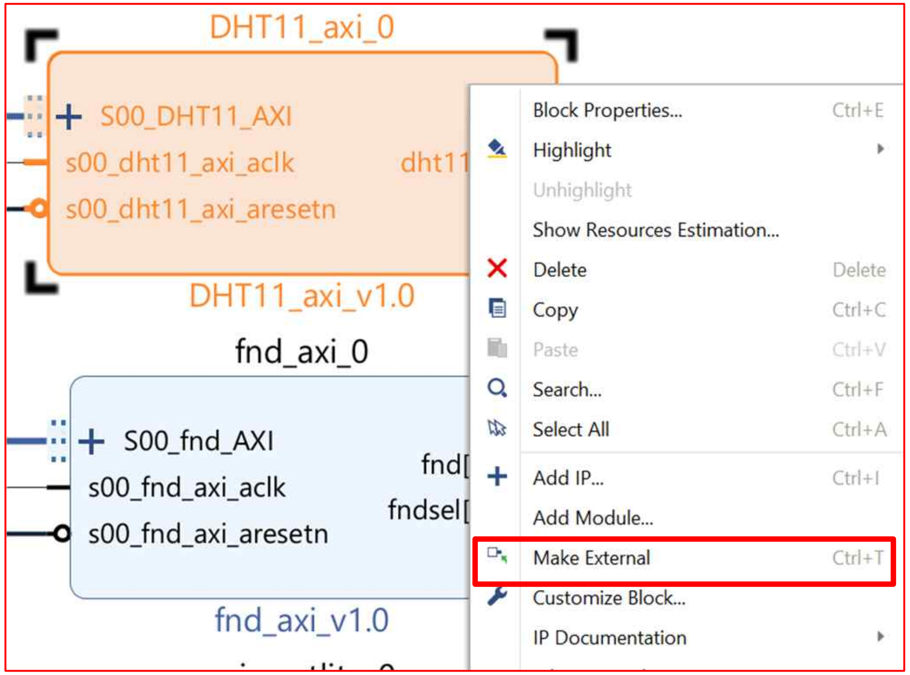
➢ ***AXI Connect ➔ AXI Template***

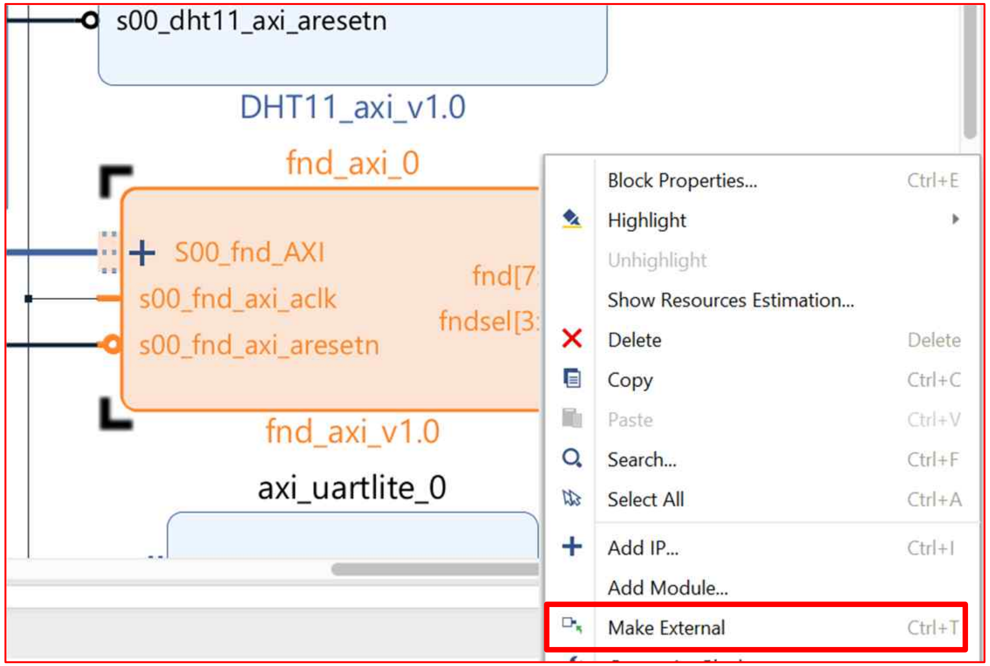


➔ inout port Error 방지

Synthesis Options를 Global로 변경 후 Generate

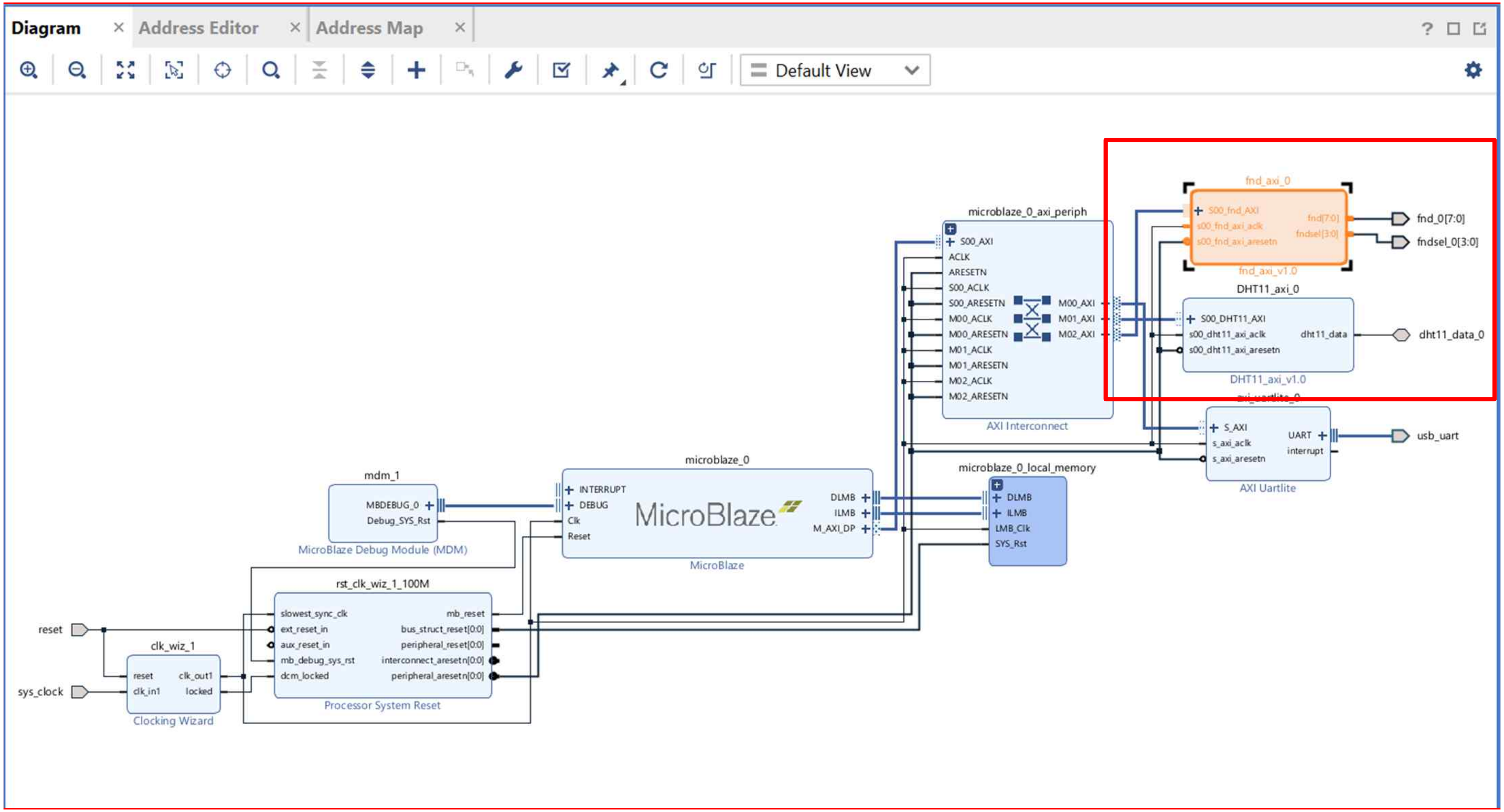
➢ *AXI Connect ➔ AXI Template*



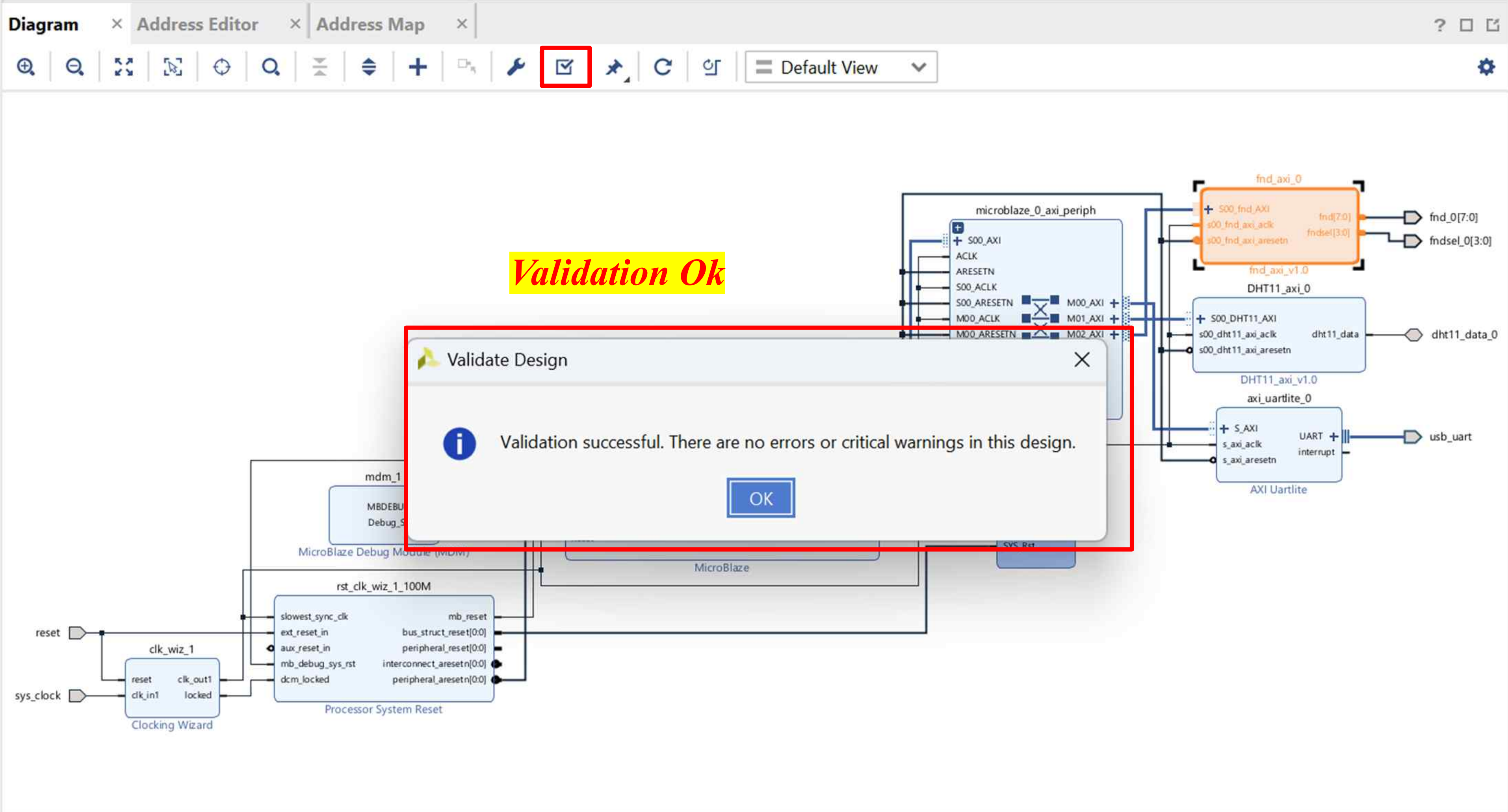


Make External ***Make External***

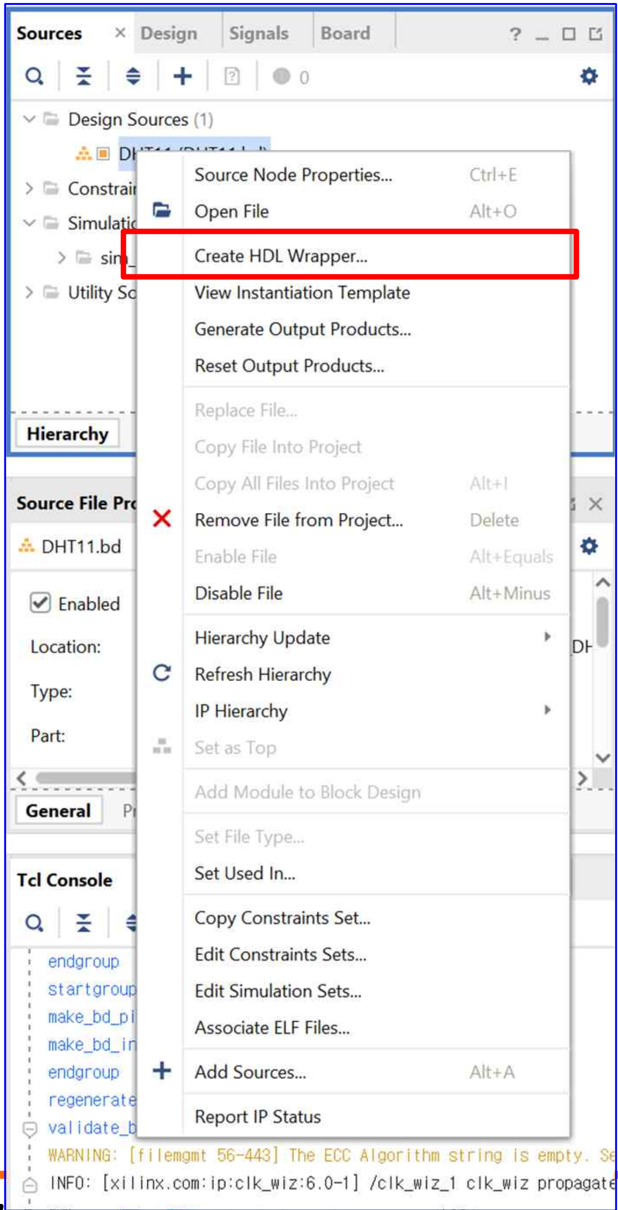
➢ *AXI Connect ➔ AXI Template*

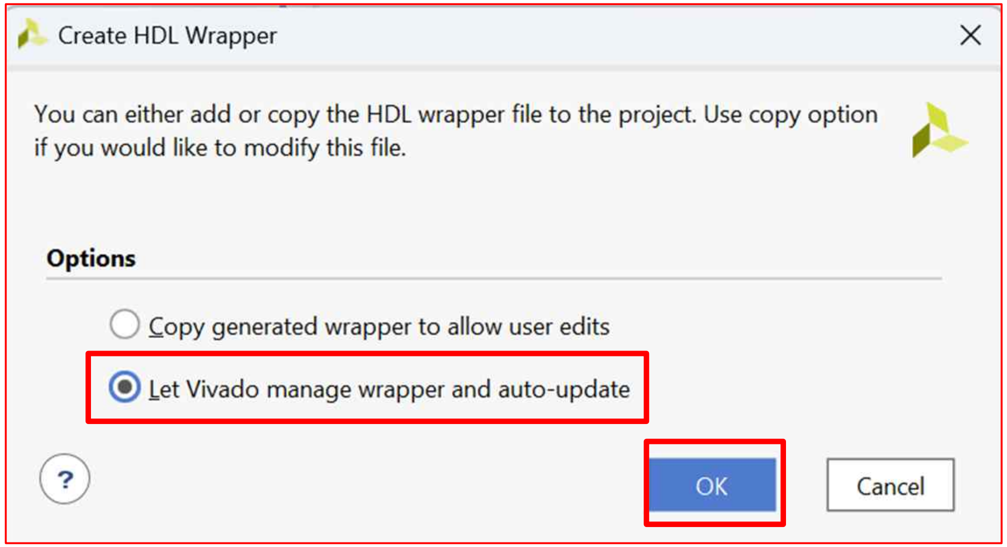


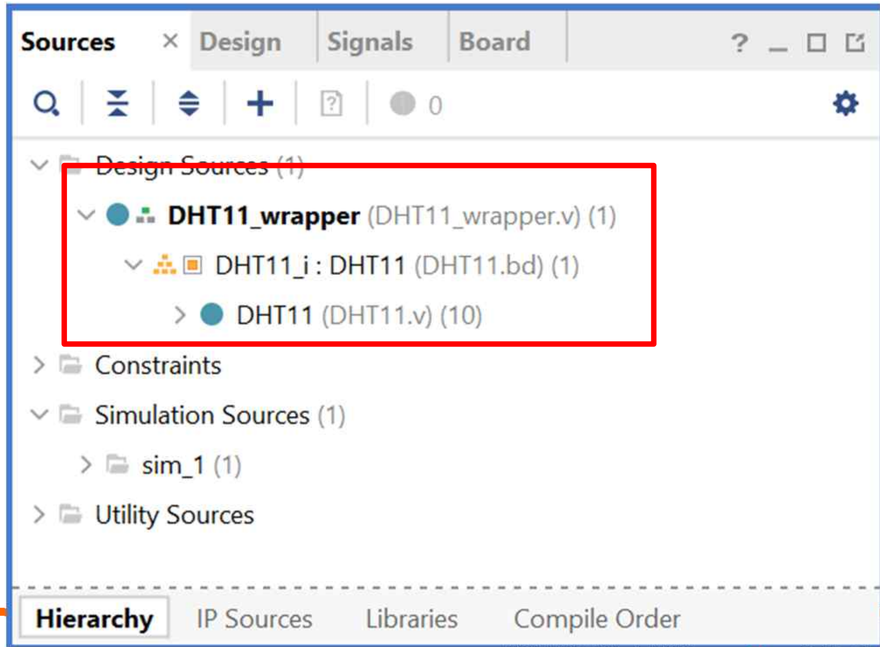
➤ *AXI Connect ➔ AXI Template*



➢ *AXI Connect ➔ AXI Template*

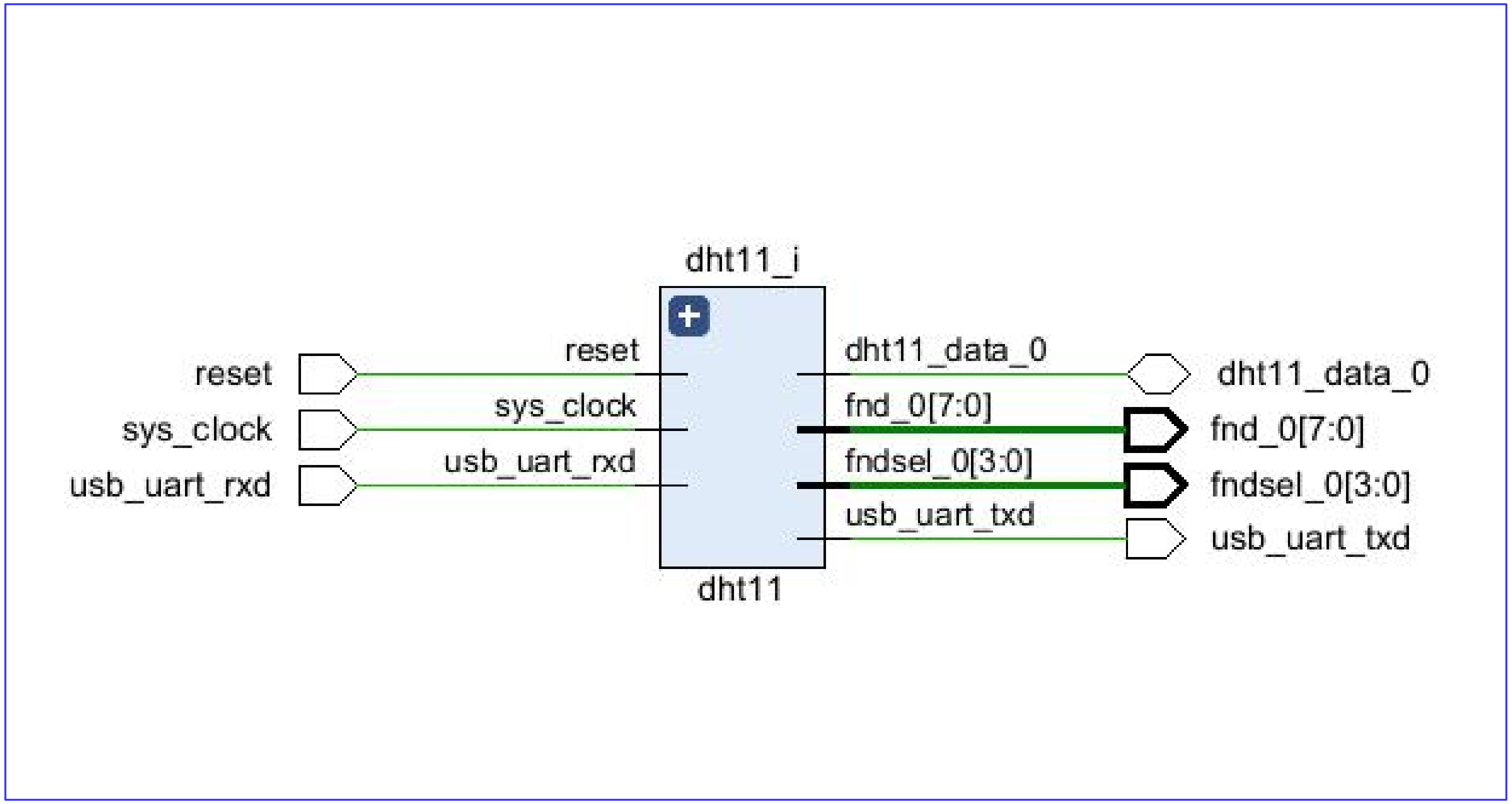






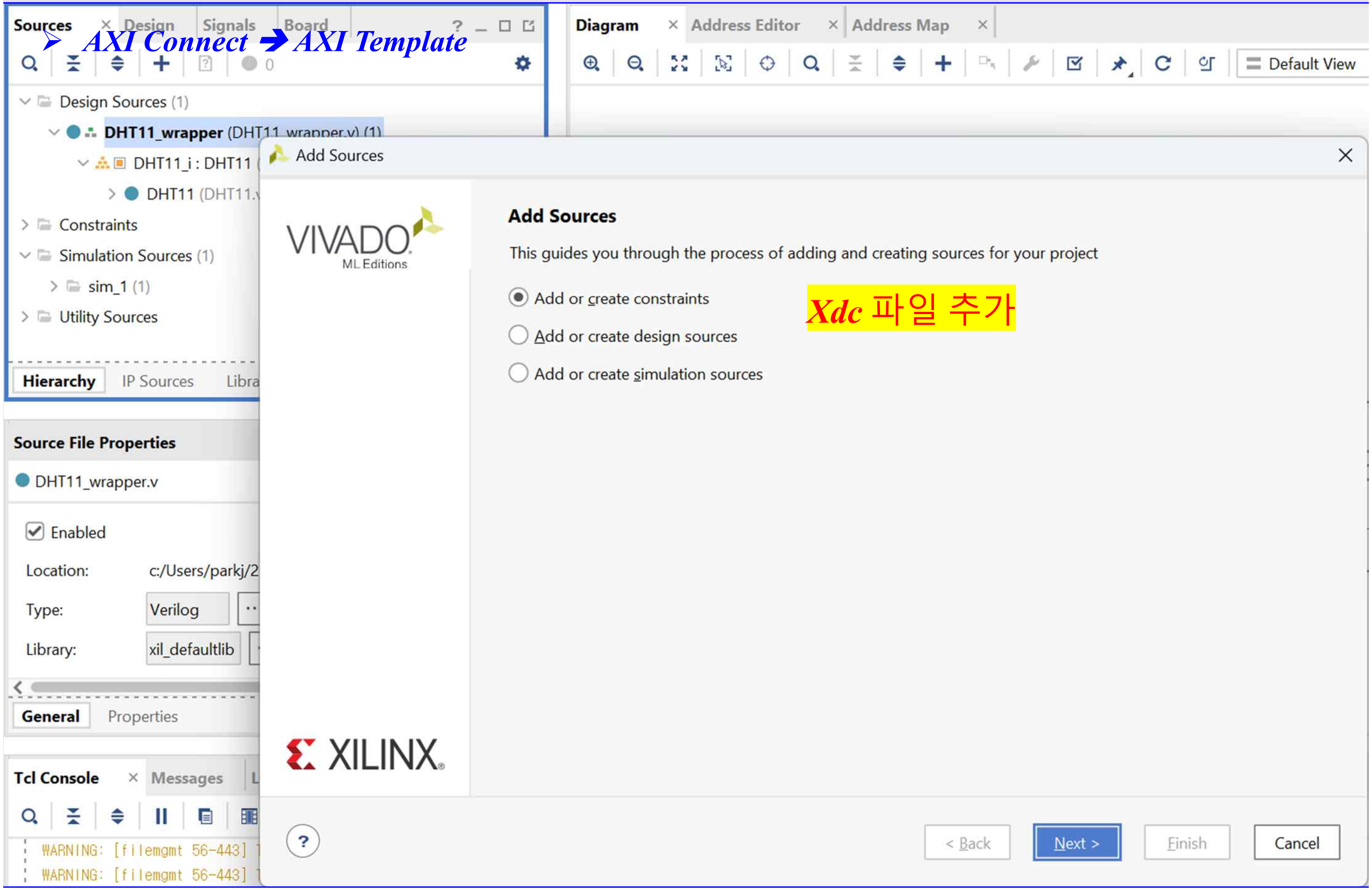
➢ *AXI Connect ➔ AXI Template*

RTL ➔ Schematic



AMBA and AXI Connect [Temperature–Humidity Sensor Module]

Axi_dht11.xpr



Basys-3-Master.xdc

- AXI Connect ➔ AXI Template
  - XDC Constraints

```
set_property -dict { PACKAGE_PIN W5   IOSTANDARD LVCMOS33 } [get_ports sys_clock]
create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports sys_clock]

## Switches
set_property -dict { PACKAGE_PIN R2    IOSTANDARD LVCMOS33 } [get_ports {reset}]

##7 Segment Display
set_property -dict { PACKAGE_PIN W7   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[0]}]
set_property -dict { PACKAGE_PIN W6   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[1]}]
set_property -dict { PACKAGE_PIN U8   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[2]}]
set_property -dict { PACKAGE_PIN V8   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[3]}]
set_property -dict { PACKAGE_PIN U5   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[4]}]
set_property -dict { PACKAGE_PIN V5   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[5]}]
set_property -dict { PACKAGE_PIN U7   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[6]}]
set_property -dict { PACKAGE_PIN V7   IOSTANDARD LVCMOS33 } [get_ports {fnd_0[7]}]

set_property -dict { PACKAGE_PIN U2   IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[0]}]
set_property -dict { PACKAGE_PIN U4   IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[1]}]
set_property -dict { PACKAGE_PIN V4   IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[2]}]
set_property -dict { PACKAGE_PIN W4   IOSTANDARD LVCMOS33 } [get_ports {fndsel_0[3]}]

##Pmod Header JC
set_property -dict { PACKAGE_PIN K17   IOSTANDARD LVCMOS33 } [get_ports {dht11_data_0}];#Sch name = JC1

##USB-RS232 Interface
set_property -dict { PACKAGE_PIN B18   IOSTANDARD LVCMOS33 } [get_ports usb_uart_rxd]
set_property -dict { PACKAGE_PIN A18   IOSTANDARD LVCMOS33 } [get_ports usb_uart_txd]

## Configuration options, can be used for all designs
set_property CONFIG_VOLTAGE 3.3 [current_design]
set_property CFGBVS VCCO [current_design]

## SPI configuration mode options for QSPI boot, can be used for all designs
set_property BITSTREAM.GENERAL.COMPRESS TRUE [current_design]
set_property BITSTREAM.CONFIG.CONFIGRATE 33 [current_design]
set_property CONFIG_MODE SPIx4 [current_design]


## Configuration options, can be used for all designs
set_property CONFIG_VOLTAGE 3.3 [current_design]
set_property CFGBVS VCCO [current_design]

## SPI configuration mode options for QSPI boot, can be used for all designs
set_property BITSTREAM.GENERAL.COMPRESS TRUE [current_design]
set_property BITSTREAM.CONFIG.CONFIGRATE 33 [current_design]
set_property CONFIG_MODE SPIx4 [current_design]
```

- ***AXI Connect → AXI Template***

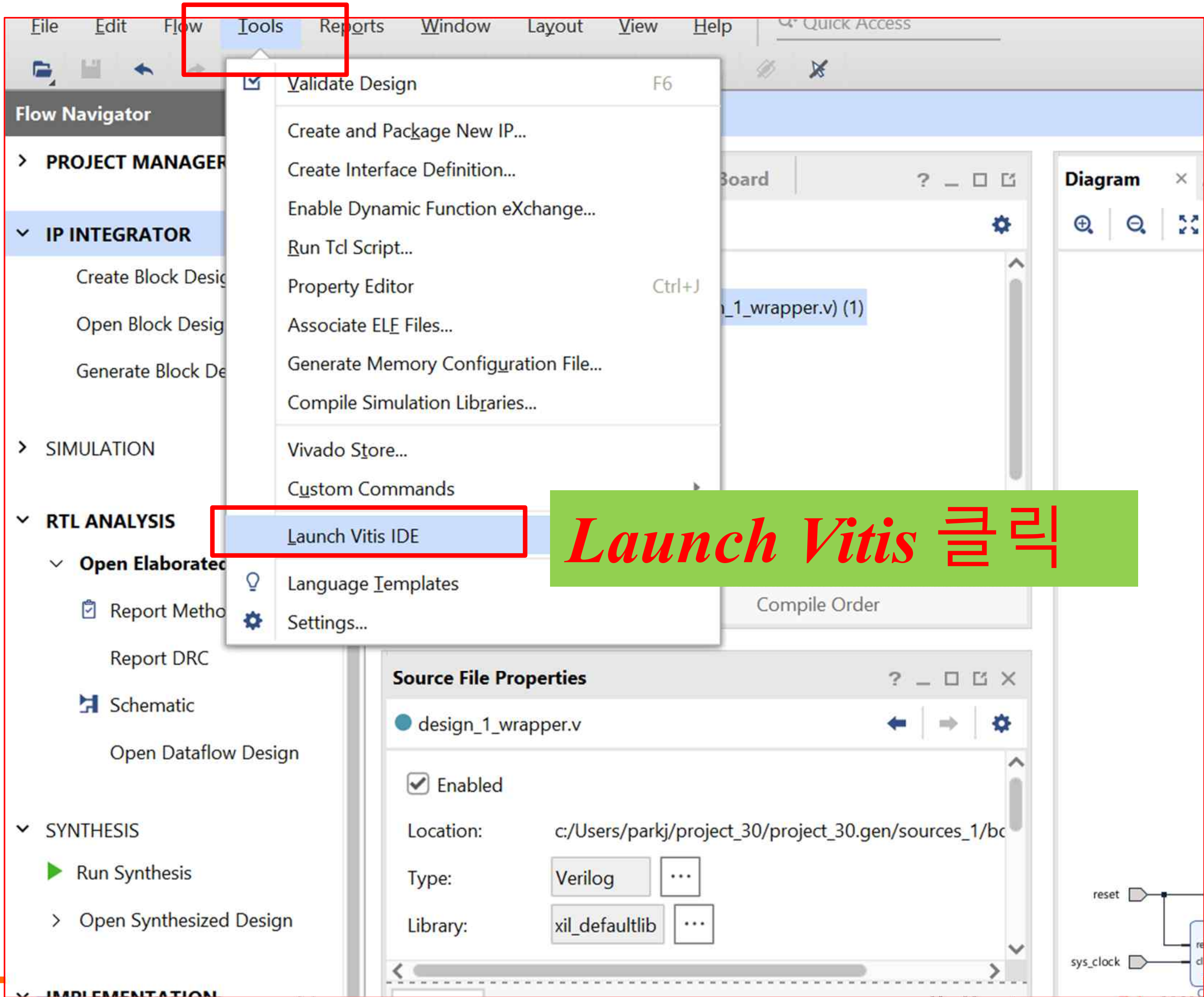
  - ***Create Block Design***
  - ***Blaze Uart & FND*** 추가 및 연결
  - ***sys_clock & reset*** 설정
  - ***FND, FNDsel, Dht11_data External***
  - ***XDC Constraints*** 설정
  - ***Bitstream***
  - ***Export Hardware***

AMBA and AXI Connect

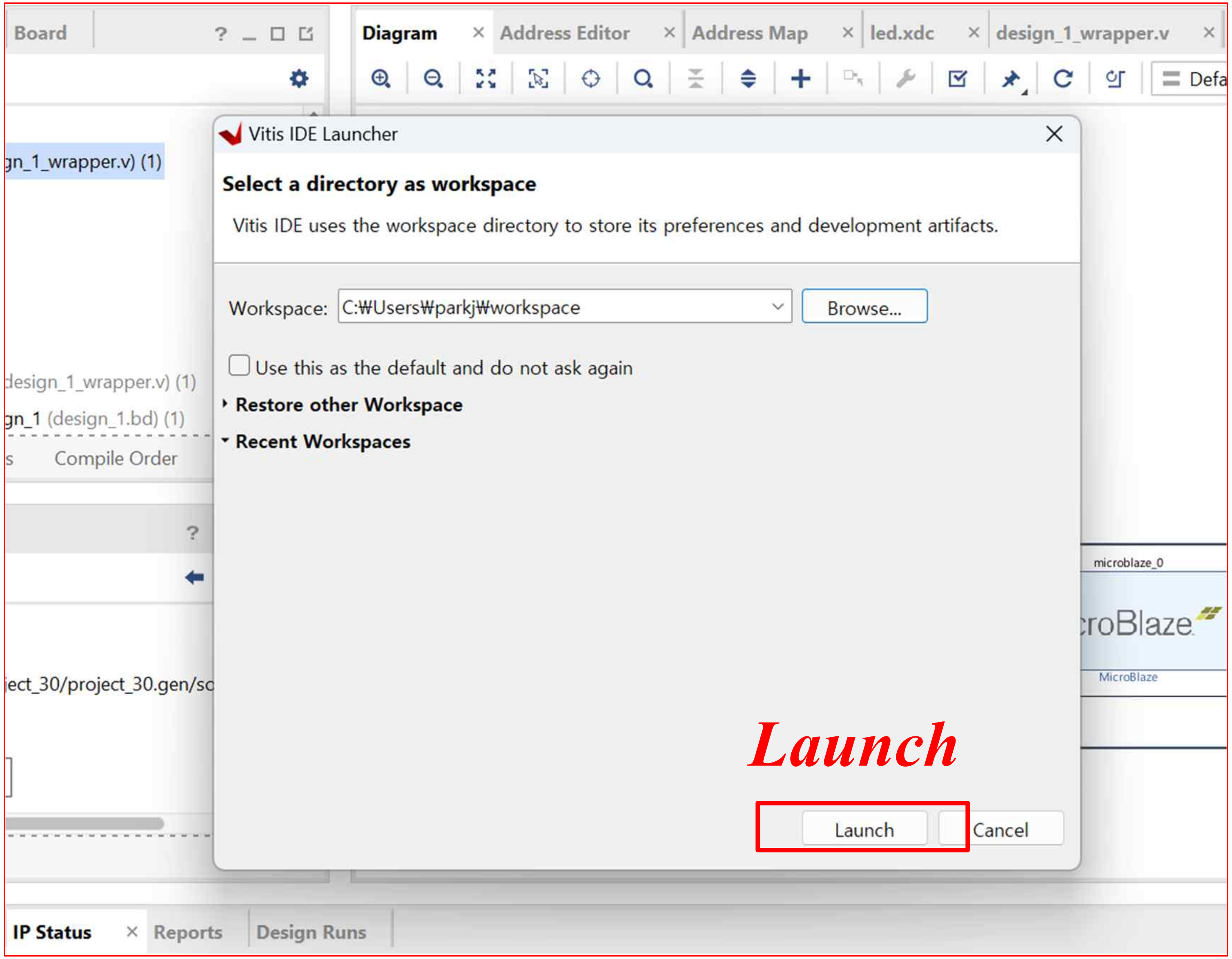
[Temperature–Humidity Sensor Module]

✓ *Application SW(Vitis)*

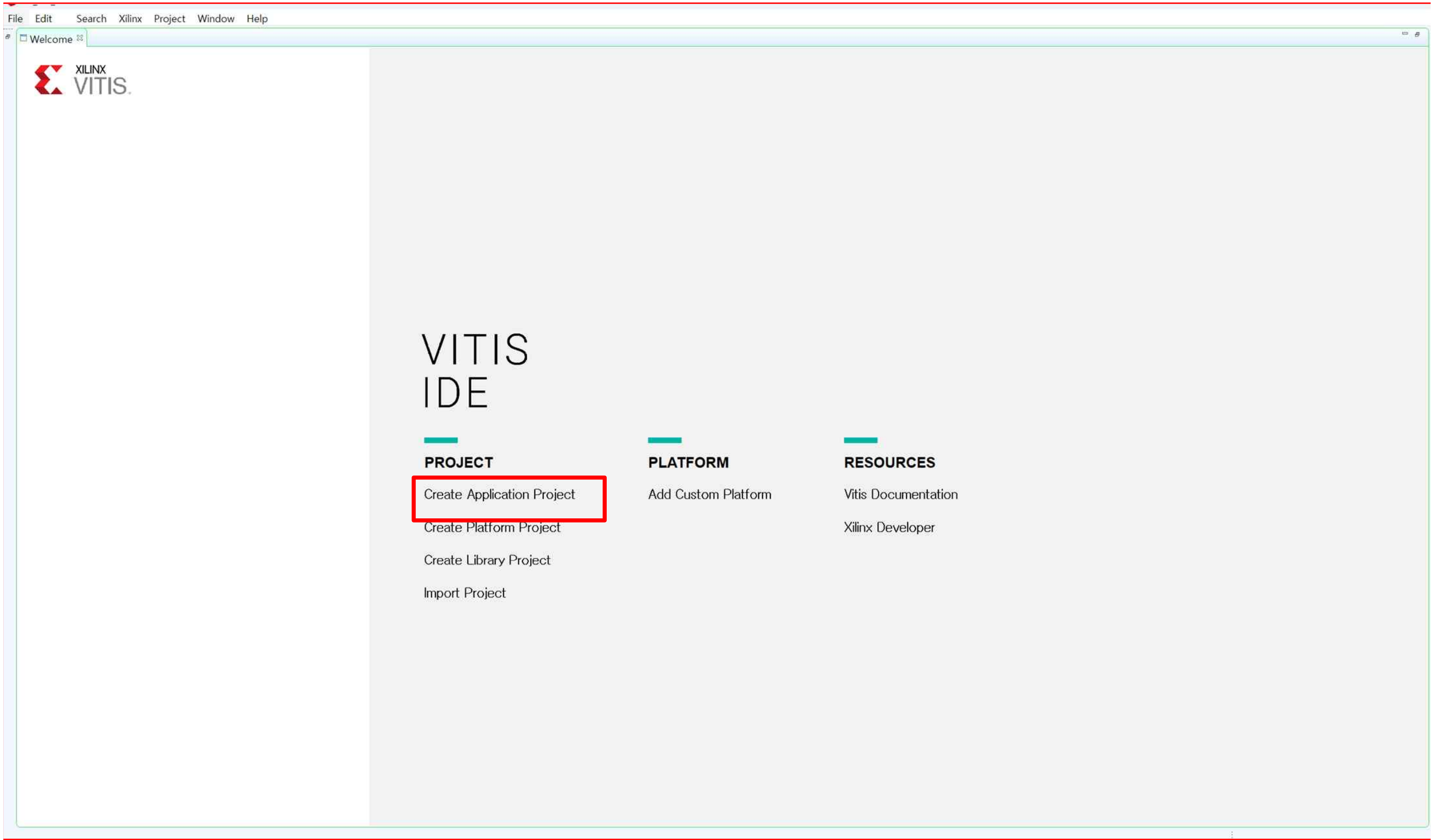
➢ AXI Connect ➔ AXI Template



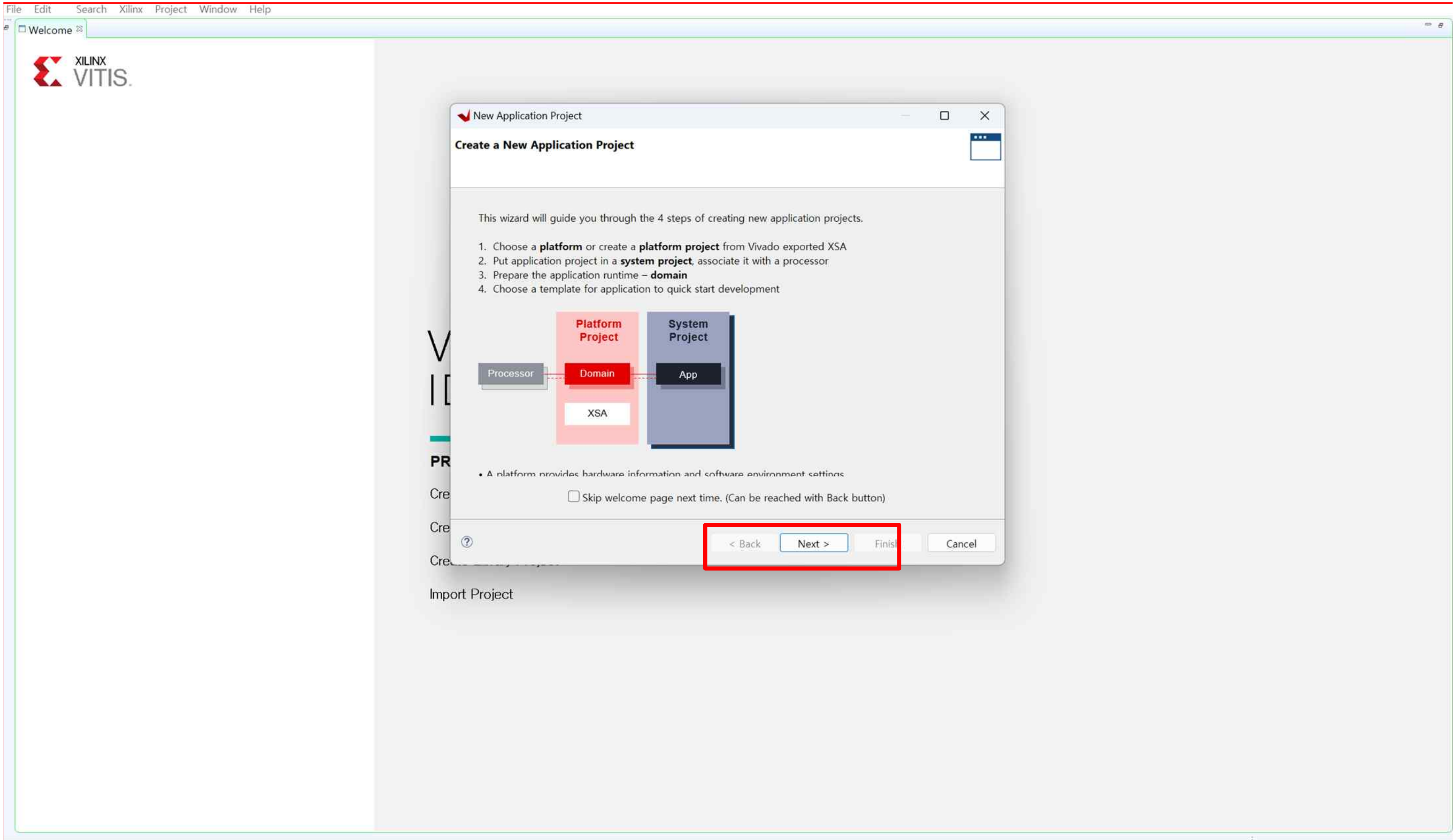
- ***AXI Connect ➔ AXI Template***



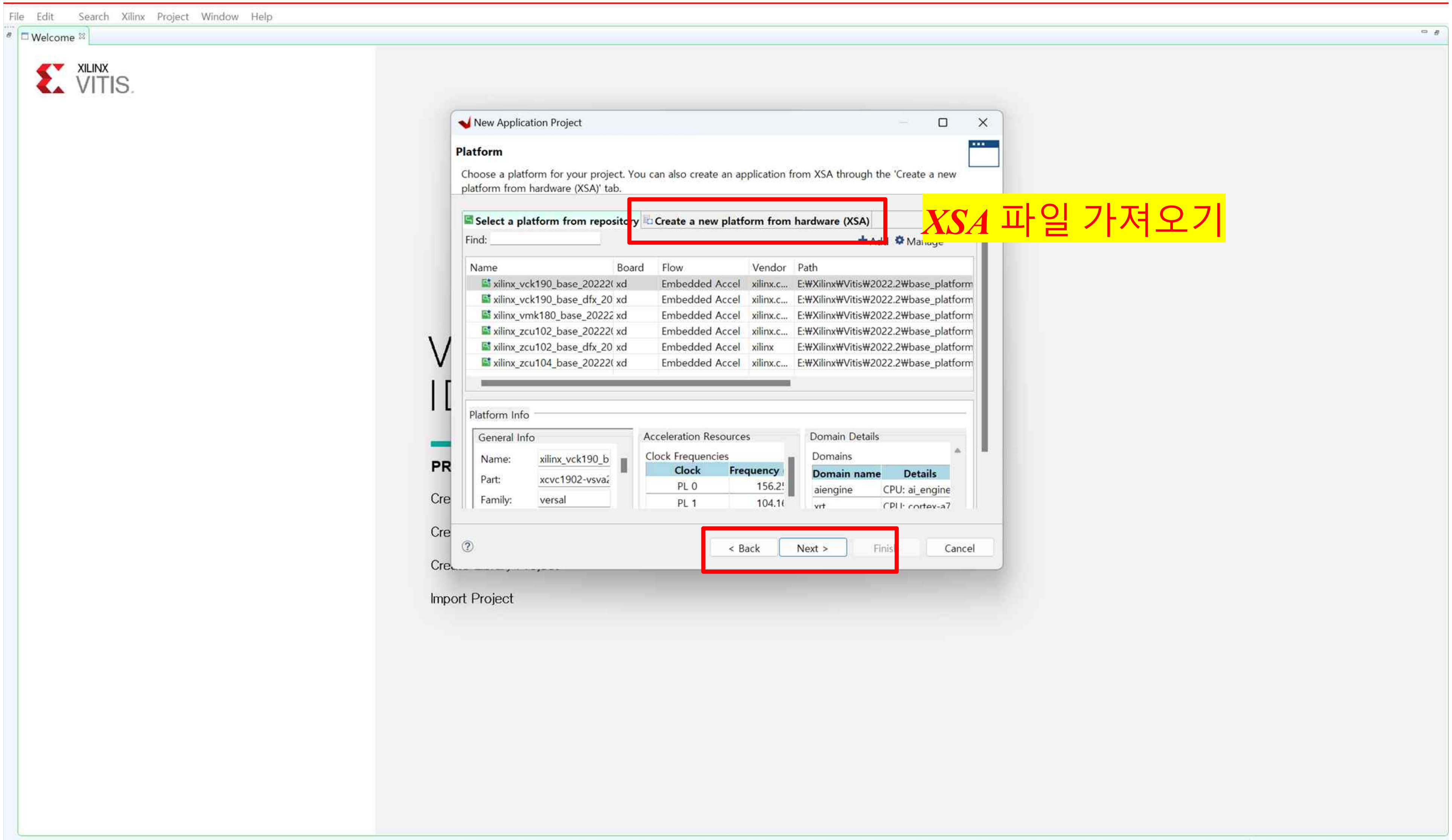
➢ *AXI Connect ➔ AXI Template*



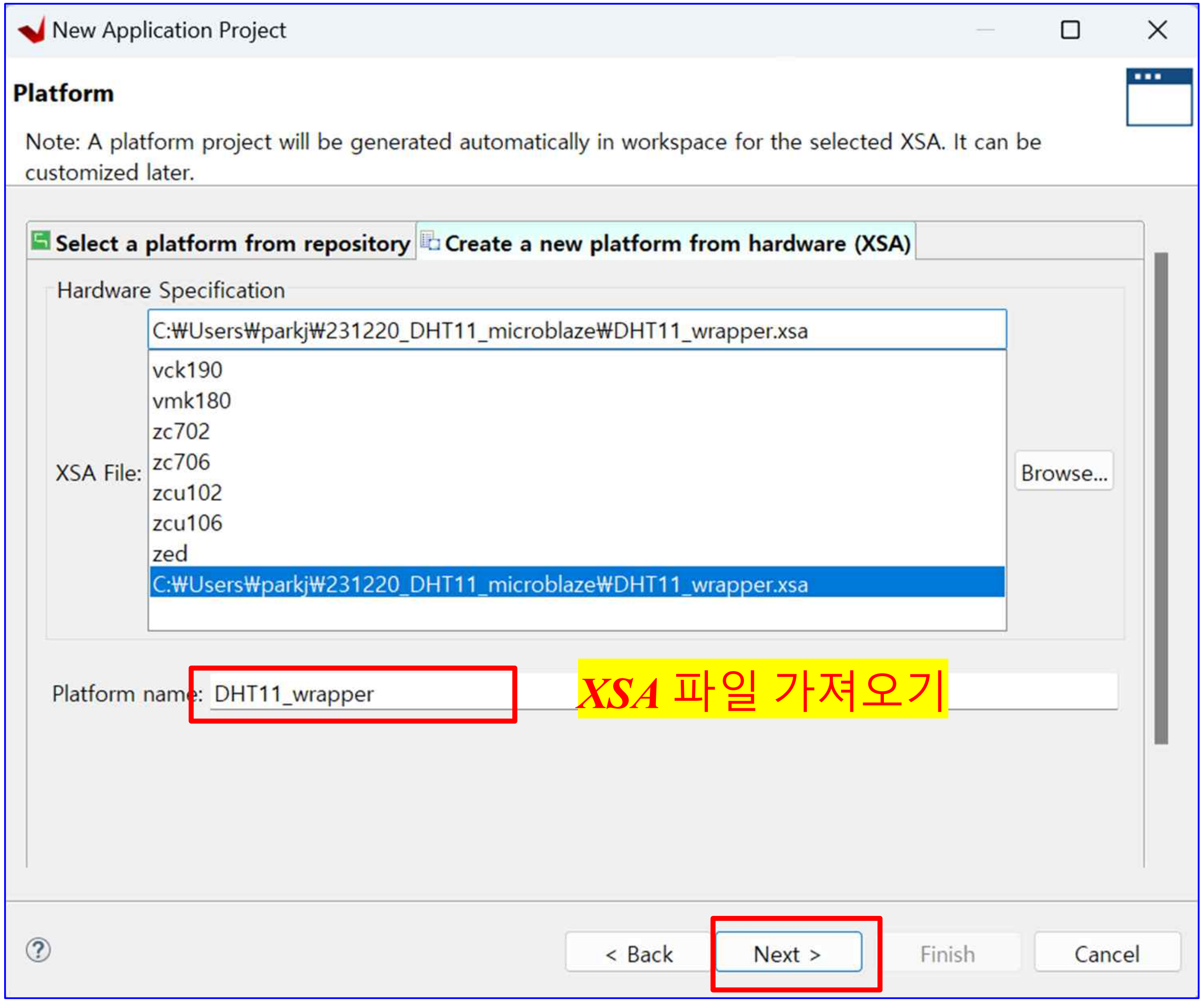
➤ *AXI Connect ➔ AXI Template*



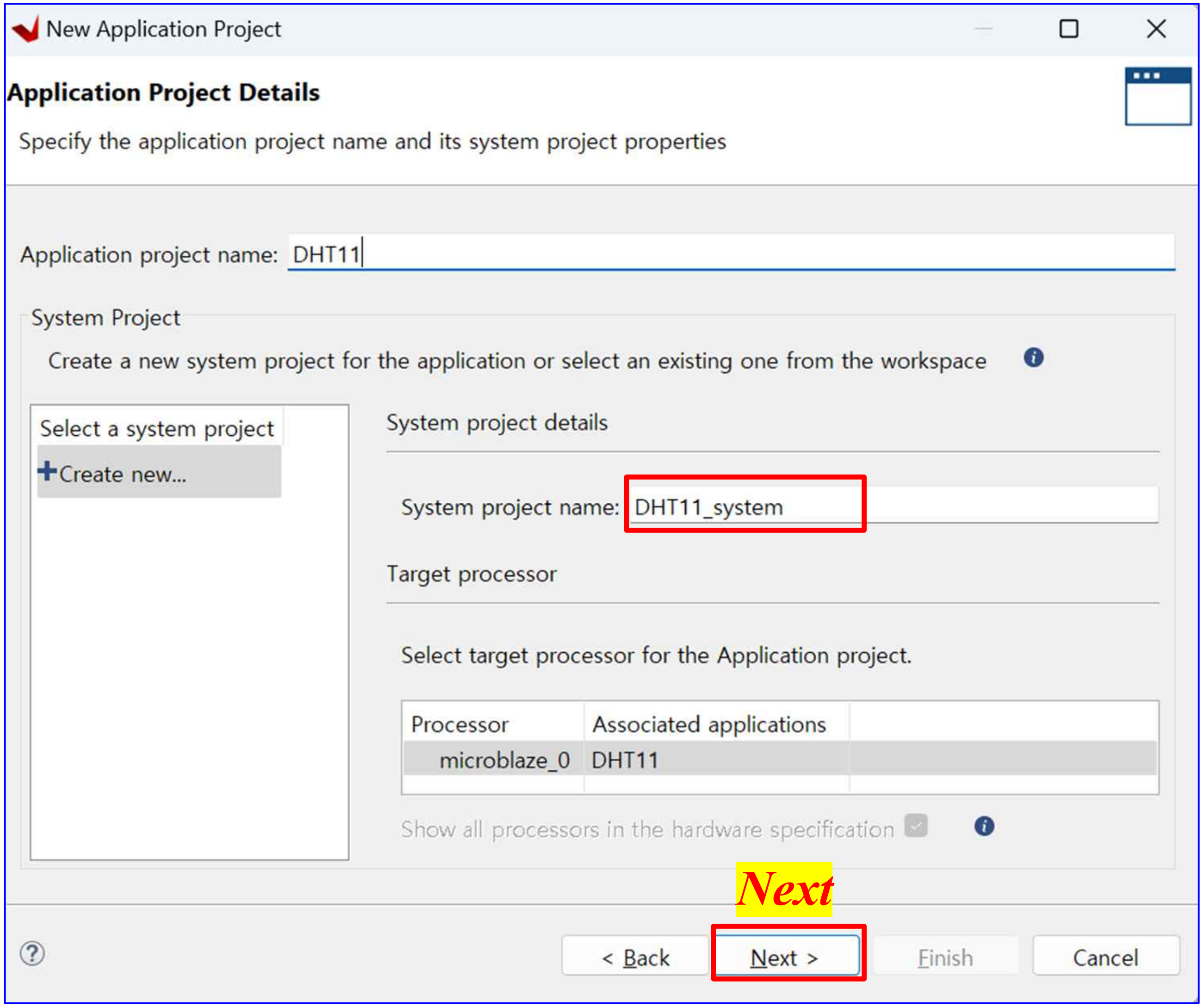
➢ AXI Connect ➔ AXI Template



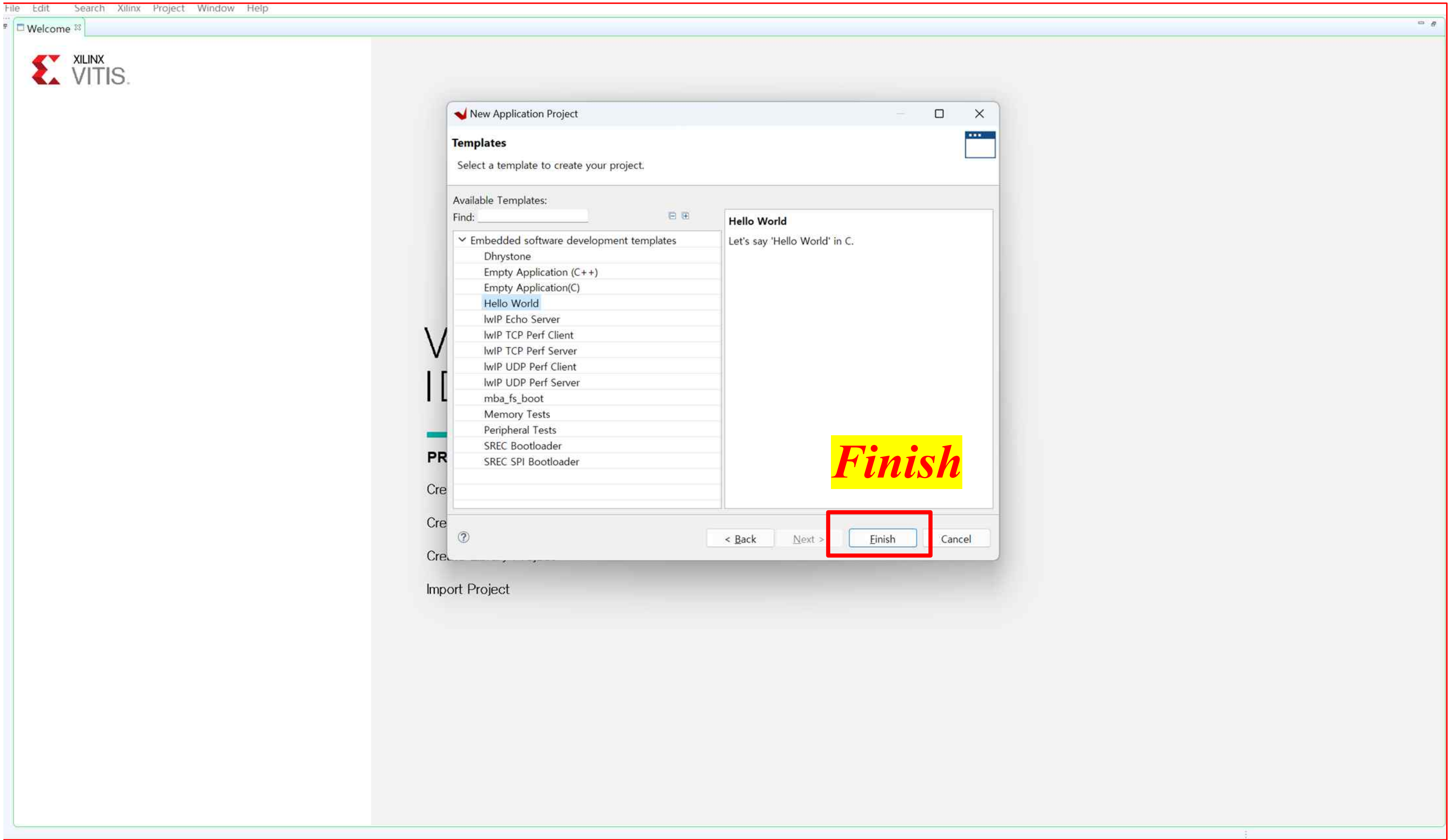
- *AXI Connect ➔ AXI Template*



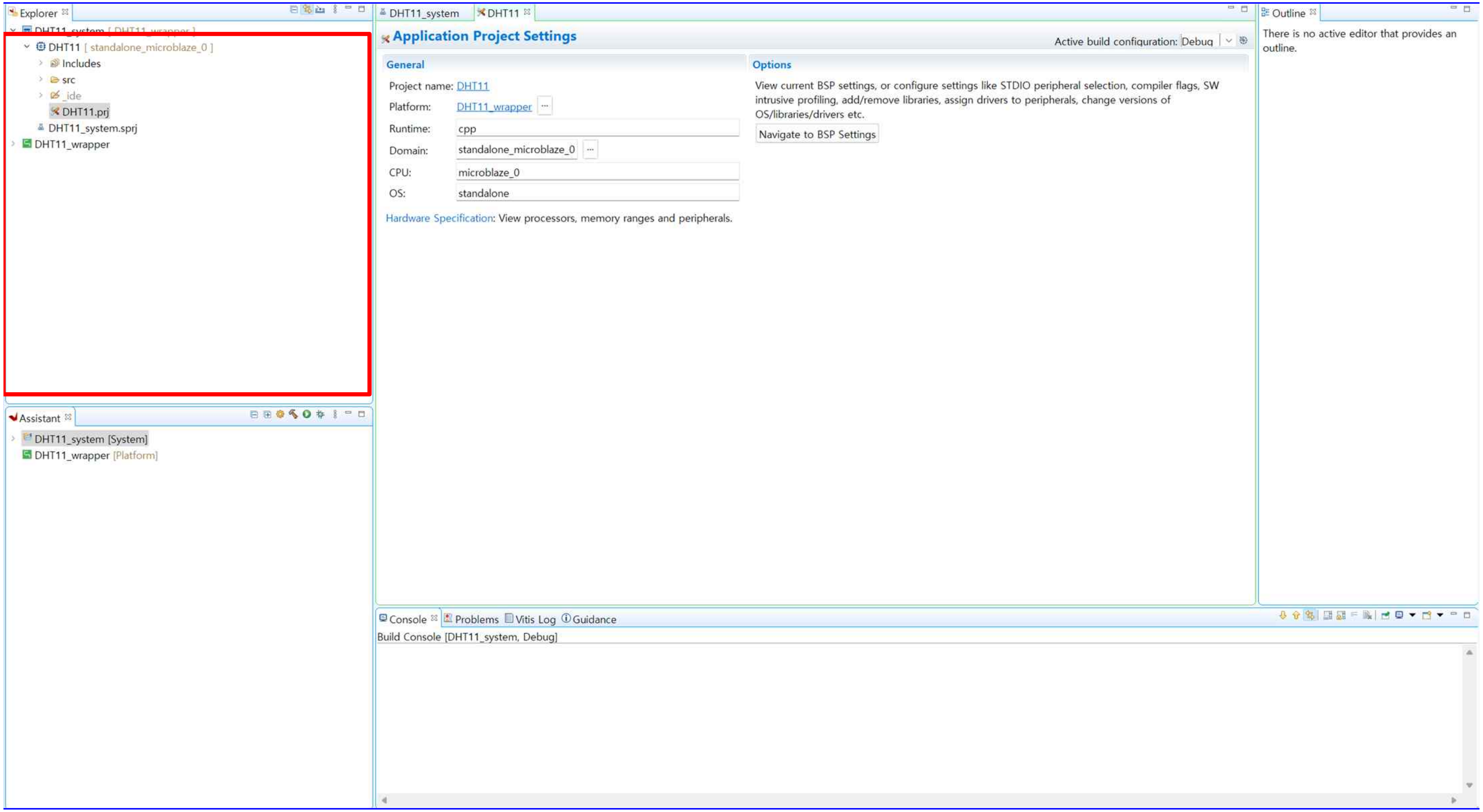
➢ AXI Connect ➔ AXI Template



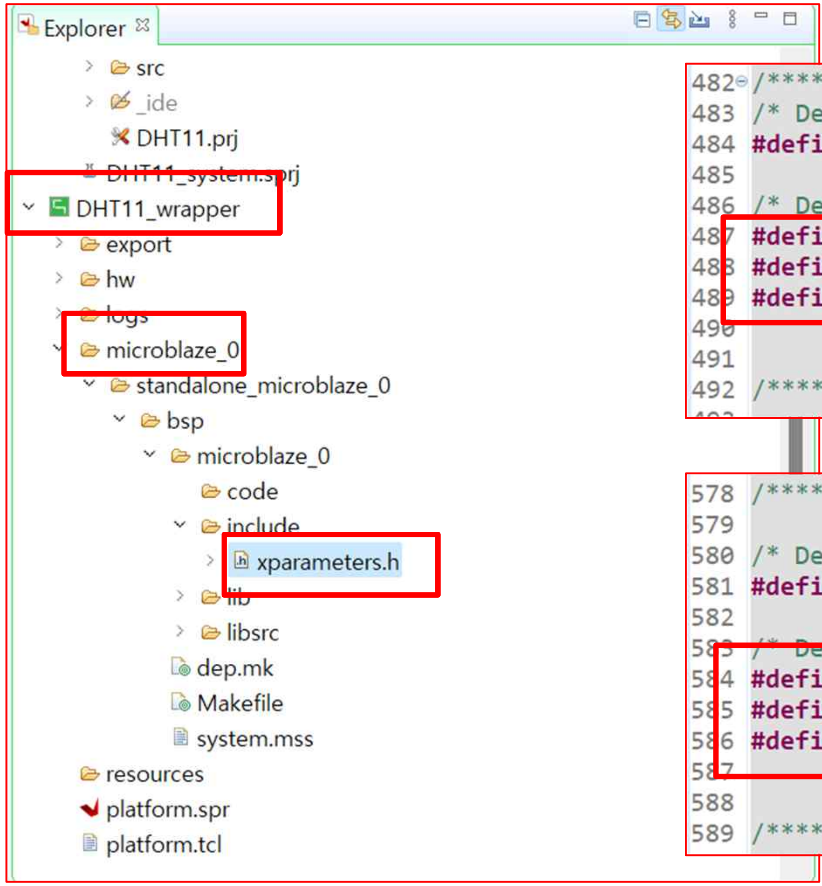
➢ *AXI Connect ➔ AXI Template*



➢ *AXI Connect ➔ AXI Template*



➢ *AXI Connect ➔ AXI Template*

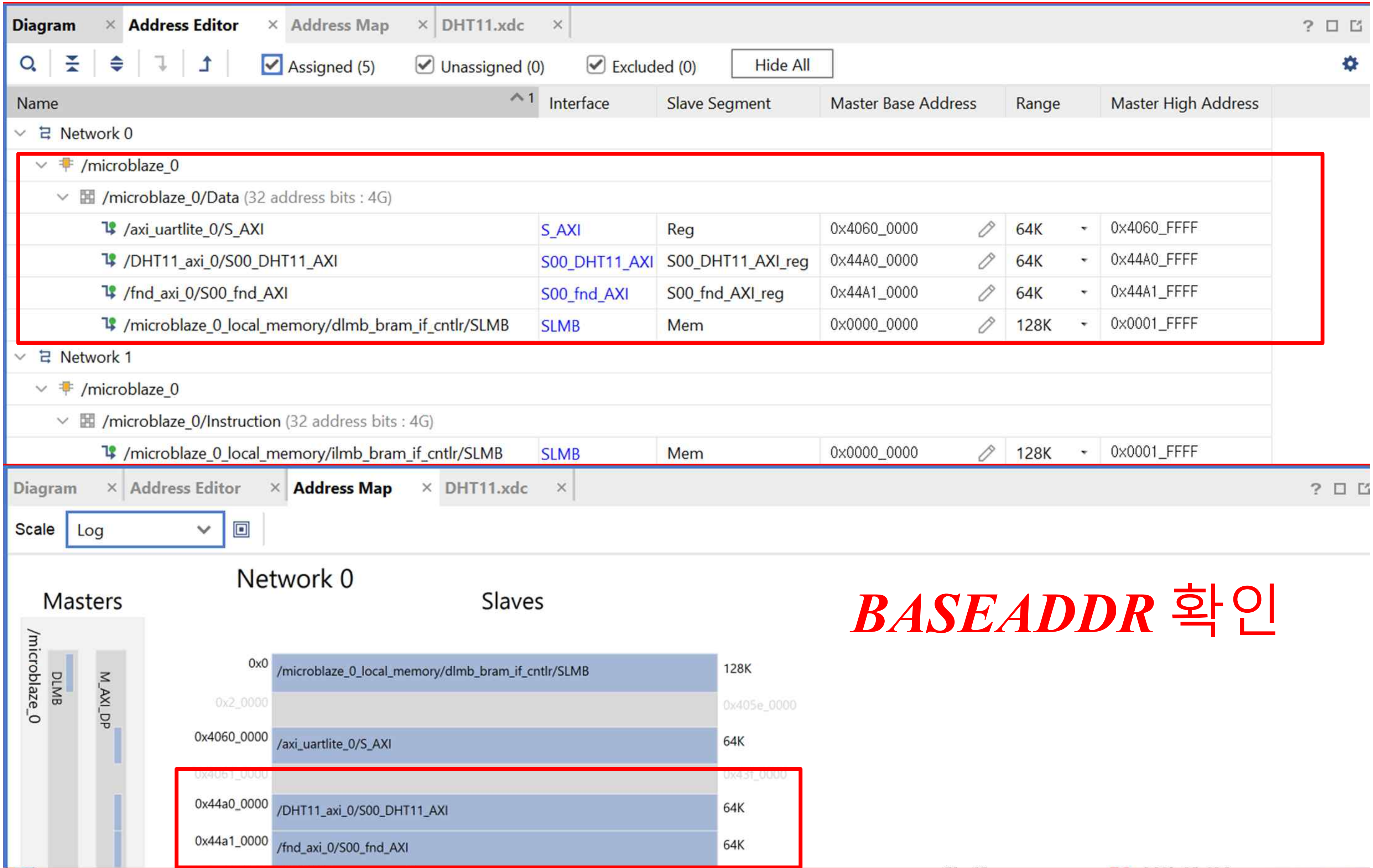



```
/******************************************************************/
/* Definitions for driver DHT11_AXI */
#define XPAR_DHT11_AXI_NUM_INSTANCES 1

/* Definitions for peripheral DHT11_AXI_0 */
#define XPAR_DHT11_AXI_0_DEVICE_ID 0
#define XPAR_DHT11_AXI_0_S00_DHT11_AXI_BASEADDR 0x44A00000
#define XPAR_DHT11_AXI_0_S00_DHT11_AXI_HIGHADDR 0x44A0FFFF


/******************************************************************/
```

```
/******************************************************************/

/* Definitions for driver FND_AXI */
#define XPAR_FND_AXI_NUM_INSTANCES 1

/* Definitions for peripheral FND_AXI_0 */
#define XPAR_FND_AXI_0_DEVICE_ID 0
#define XPAR_FND_AXI_0_S00_FND_AXI_BASEADDR 0x44A10000
#define XPAR_FND_AXI_0_S00_FND_AXI_HIGHADDR 0x44A1FFFF


/******************************************************************/
```

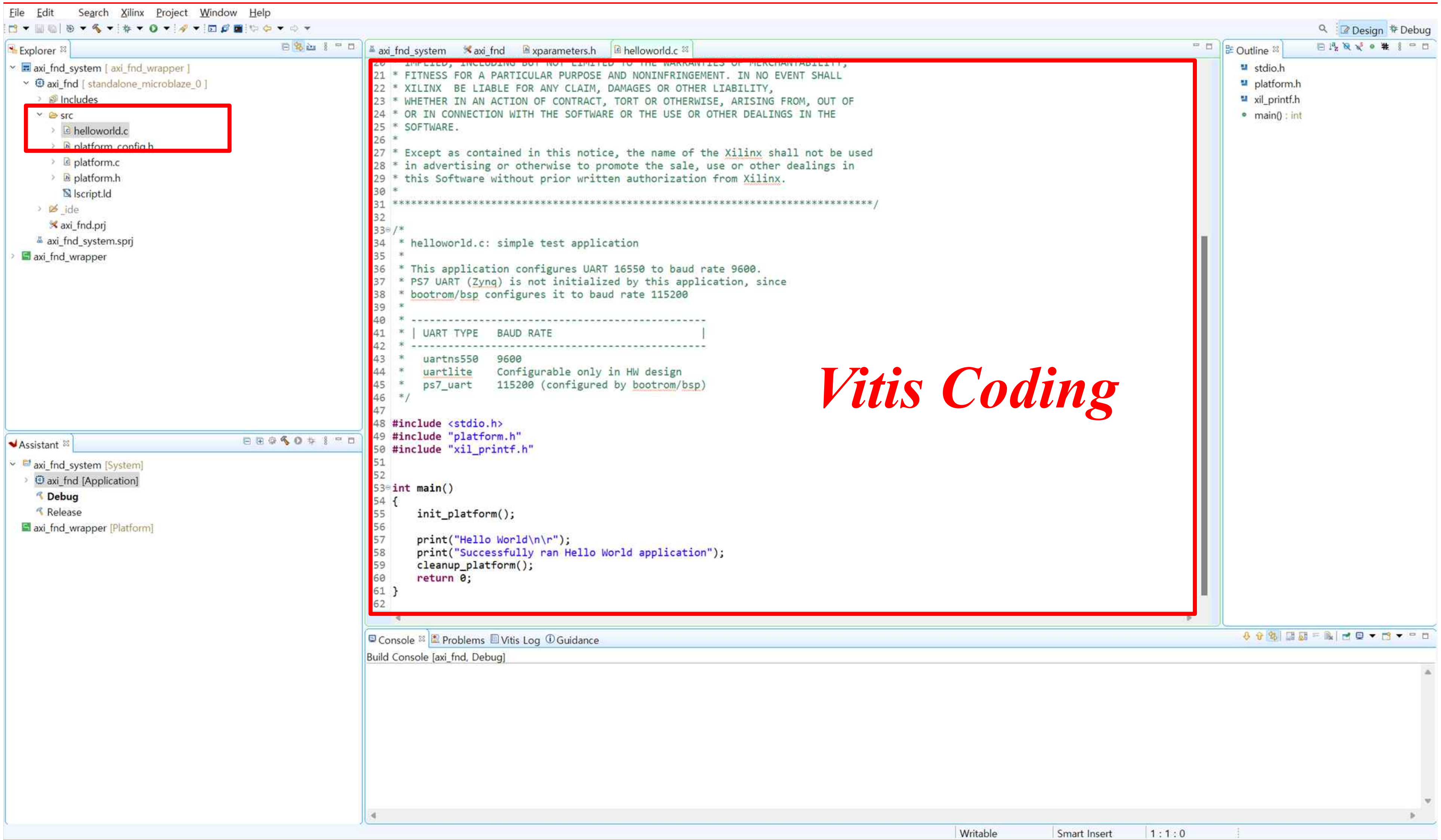
BASEADDR 확인

➢ *AXI Connect ➔ AXI Template*



BASEADDR 확인

➢ AXI Connect ➔ AXI Template



➢ AXI Connect ➔ AXI Template

Helloworld.c

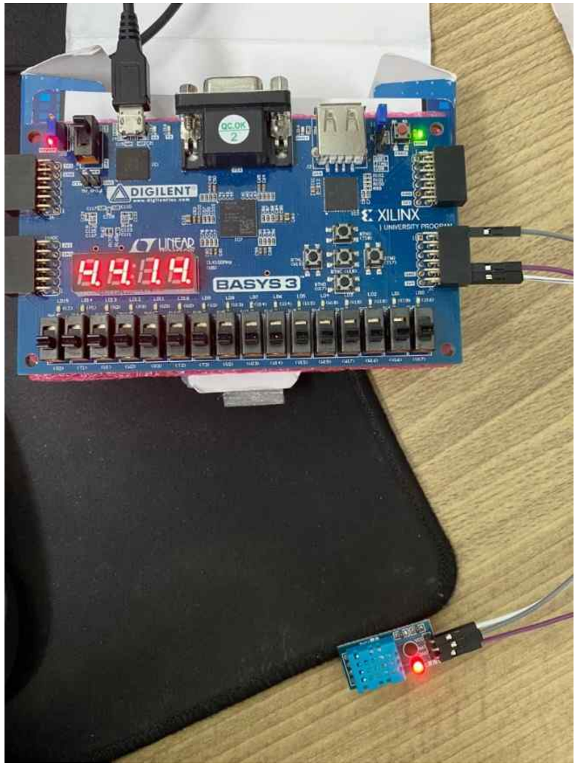
주소 확인

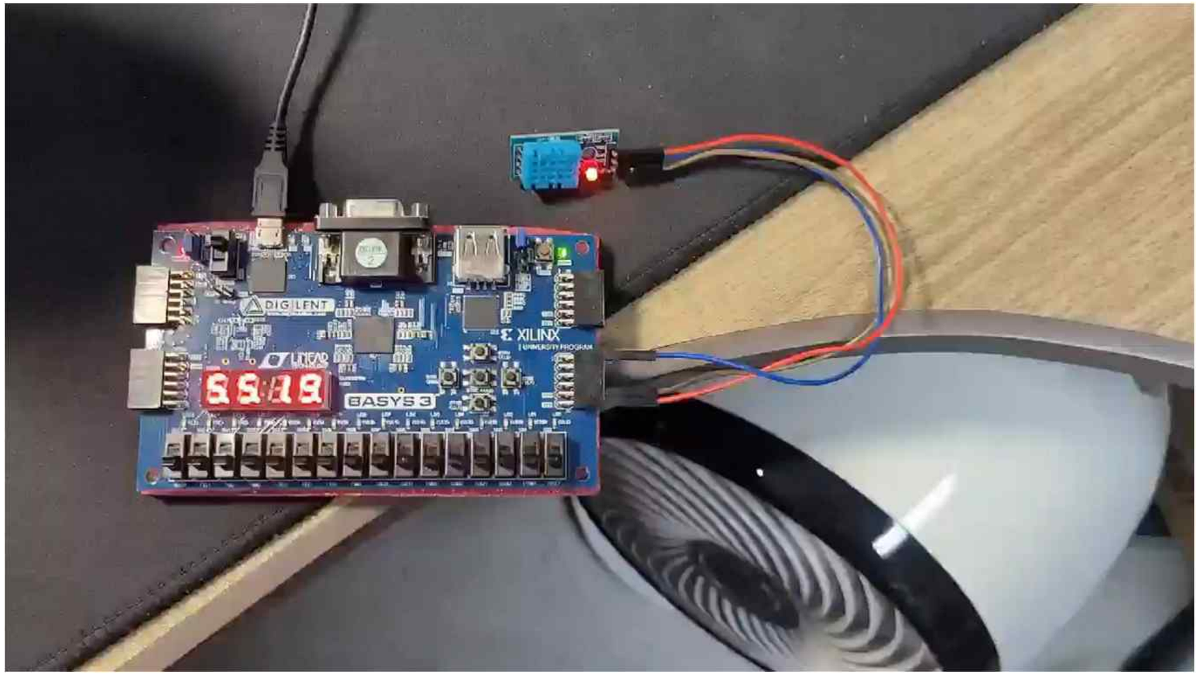
```
#include <stdio.h>
#include "platform.h"
#include "xil_printf.h"
#include "xparameters.h"
#include "sleep.h"
#include "xil_io.h"

#define AXI_FND_BASEADDR  0x44A10000
#define AXI_FND_DIGIT1  AXI_FND_BASEADDR + 0x00
#define AXI_FND_DIGIT2  AXI_FND_BASEADDR + 0x04
#define AXI_FND_DIGIT3  AXI_FND_BASEADDR + 0x08
#define AXI_FND_DIGIT4  AXI_FND_BASEADDR + 0x0c

#define AXI_DHT11_BASEADDR  0x44A00000
#define TEMPERATURE0  AXI_DHT11_BASEADDR + 0x00
#define TEMPERATURE10  AXI_DHT11_BASEADDR + 0x04
#define HUMYDITY0  AXI_DHT11_BASEADDR + 0x08
#define HUMYDITY10  AXI_DHT11_BASEADDR + 0x0c


int main()
{
   init_platform();

   print("Hello World\n\r");
     print("Successfully ran Hello World application");
```

Helloworld.c

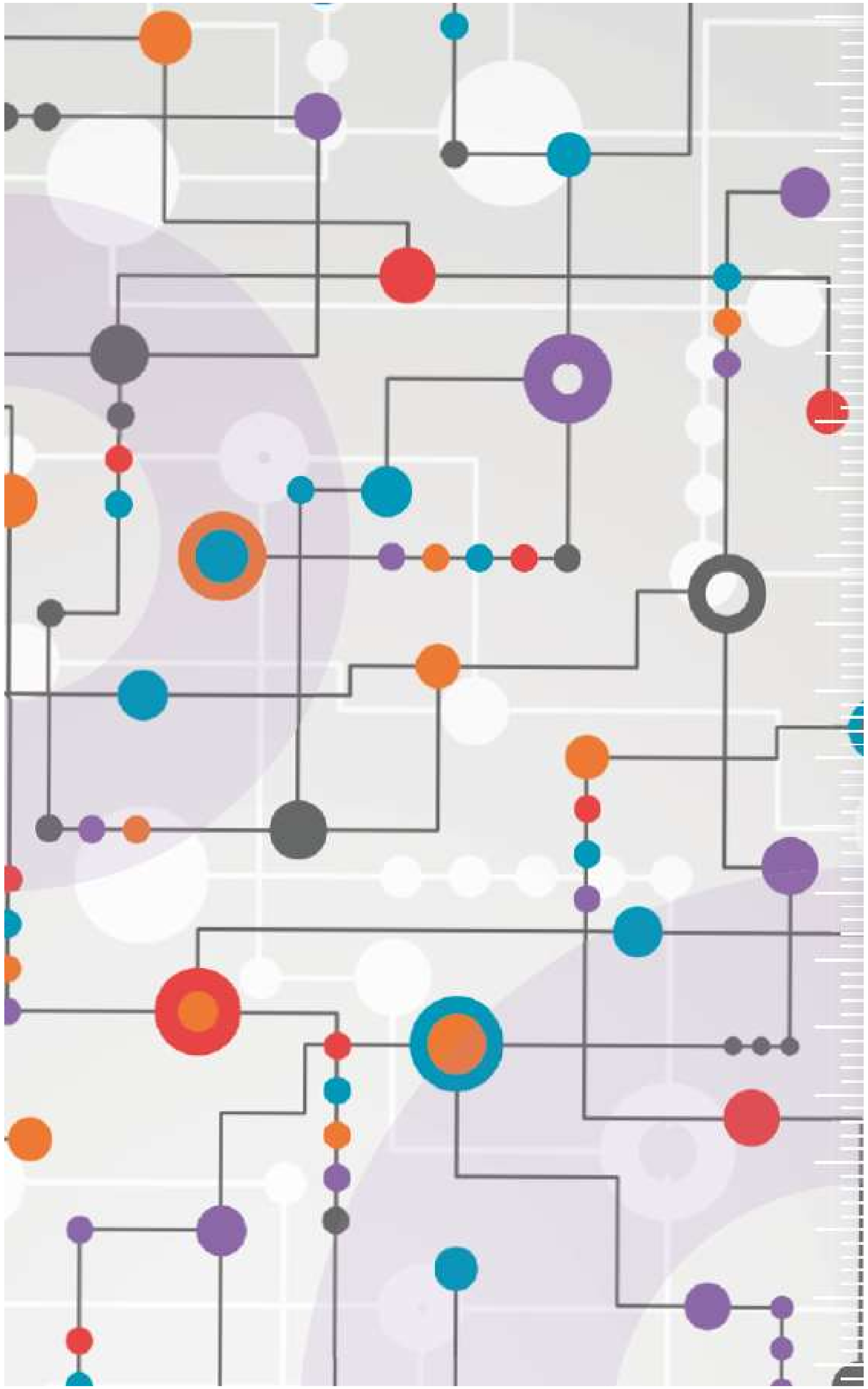
```
    u16 temp0 = 0;
    u16 temp10 = 0;
    u16 hum0 = 0;
    u16 hum10 = 0;


        while(1)
        {
        temp0=Xil_In32(TEMPERATURE0);
temp10=Xil_In32(TEMPERATURE10);
hum0=Xil_In32(HUMYDITY0);
hum10=Xil_In32(HUMYDITY10);

Xil_Out32(AXI_FND_DIGIT1,temp0);
Xil_Out32(AXI_FND_DIGIT2,temp10);
Xil_Out32(AXI_FND_DIGIT3,hum0);
Xil_Out32(AXI_FND_DIGIT4,hum10);

      usleep(500000);

      }
        cleanup_platform();
        return 0;
    }
```

➢ ***Program Device & Check The Result***





현재 온도, 습도가 출력되는 것을 확인



수고하셨습니다.

Inout Port Error

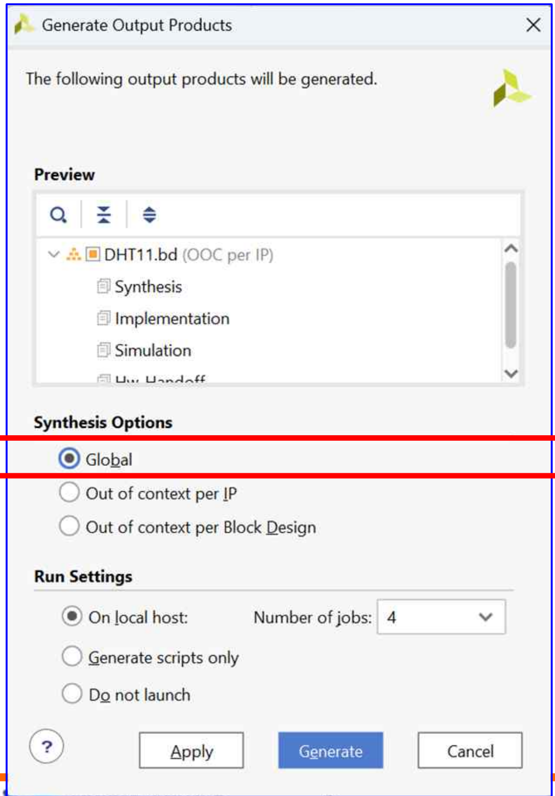
[*Reference*]

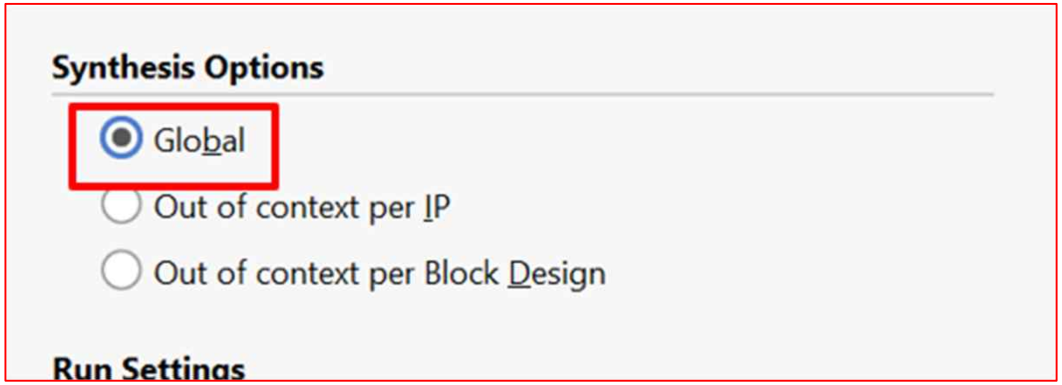
※ 오류해결

https://support.xilinx.com/s/question/0D54U00005ZMzumSAD/how-to-correctly-connect-packaged-ip-to-a-bidirectional-inout-port?language=en_US

- ***CustomIP**에 **inout port** 사용시*
- ***Inout** 포트가 **output** 포트로만 적용됨. (**input x**)*

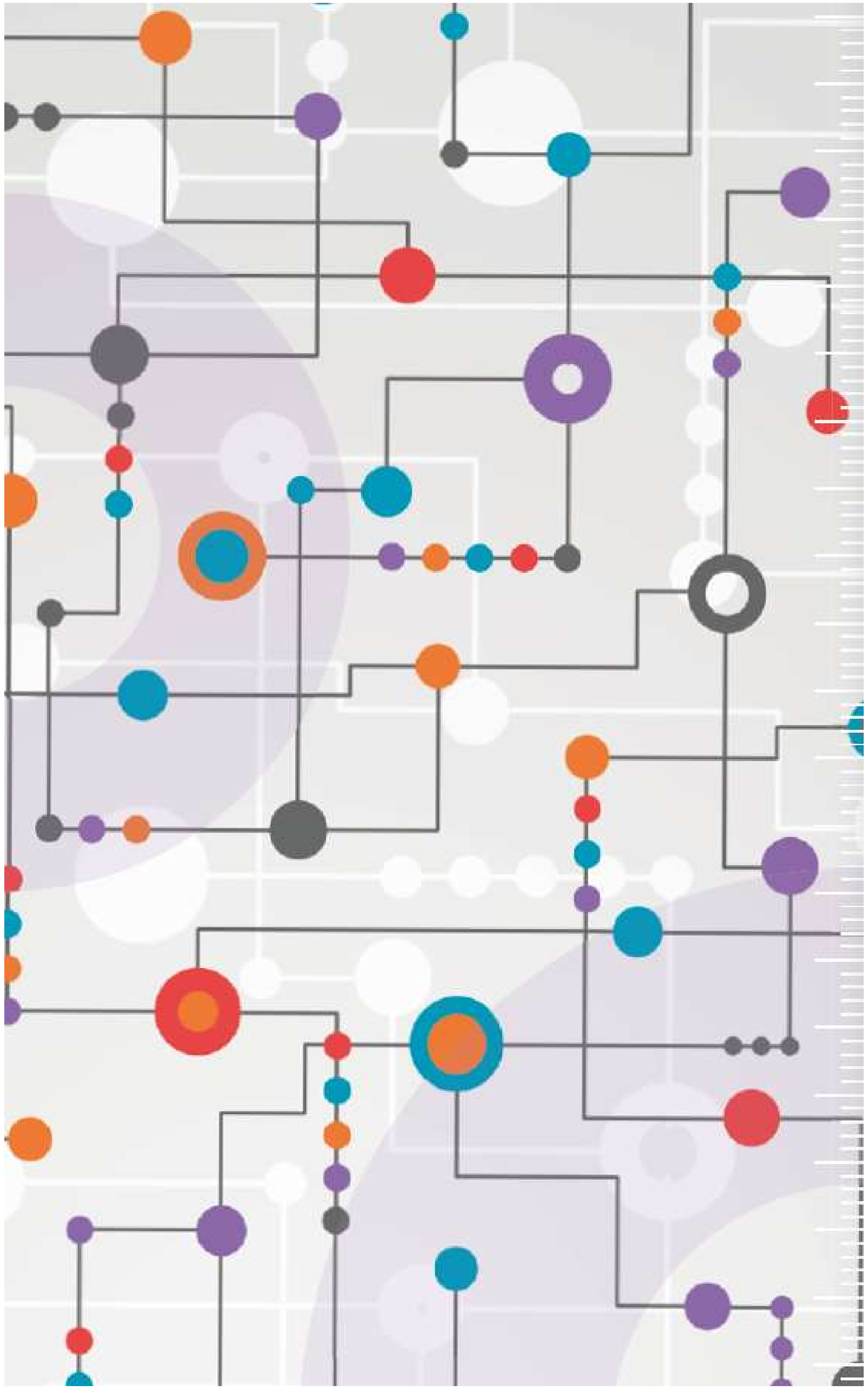
> diagram. This approach did NOT work when synthesized with the default values of "Generate Output Products"... by default that selects Synthesis Option "Out of context per IP". Using these defaults the top-level INOUT port/pin worked as an output only, and always read 0 for input. I had to change the Synthesis Option from the default to "Global" to get things working. No clue why, but that worked.





- 오류 해결을 위하여

  ***Synthesis** 옵션 **Global**로 수정*



수고하셨습니다.